

SmartMaint

Industrial Maintenance & Spare Parts Management System

Complete Source Code Reference

49

FILES

5,400

LINES OF CODE

12

MODULES

Generated 2026-09-03

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vite.config.ts 17 lines
tsconfig.json 8 lines
tsconfig.app.json 31 lines
tsconfig.node.json 23 lines
tailwind.config.js 95 lines
postcss.config.js 7 lines
eslint.config.js 29 lines
index.html 17 lines

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src/main.tsx 11 lines
src/App.tsx 89 lines

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src/types/index.ts 397 lines

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src/store/index.ts 18 lines
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src/components/layout/Sidebar.tsx 116 lines
src/components/layout/TopBar.tsx 149 lines

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src/components/ui/DataTable.tsx 91 lines
src/components/ui/Indicators.tsx 96 lines
src/components/ui/KPICard.tsx 59 lines
src/components/ui/Modal.tsx 50 lines
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STYLES

src/index.css 97 lines
src/vite-env.d.ts 2 lines

```
{
  059669">"name": 059669">"vite-react-typescript-starter",
  059669">"600">private": 600">true,
  059669">"version": 059669">"0.0.0",
  059669">"600">type": 059669">"module",
  059669">"scripts": {
    059669">"dev": 059669">"vite",
    059669">"build": 059669">"vite build",
    059669">"lint": 059669">"eslint .",
    059669">"preview": 059669">"vite preview",
    059669">"typecheck": 059669">"tsc --noEmit -p tsconfig.app.json"
  },
  059669">"dependencies": {
    059669">"@reduxjs/toolkit": 059669">"^2.12.0",
    059669">"@supabase/supabase-js": 059669">"^2.57.4",
    059669">"clsx": 059669">"^2.1.1",
    059669">"date-fns": 059669">"^4.4.0",
    059669">"lucide-react": 059669">"^0.446.0",
    059669">"react": 059669">"^18.3.1",
    059669">"react-dom": 059669">"^18.3.1",
    059669">"react-redux": 059669">"^9.3.0",
    059669">"react-router-dom": 059669">"^7.18.2",
    059669">"recharts": 059669">"^3.10.1"
  },
  059669">"devDependencies": {
    059669">"@eslint/js": 059669">"^9.9.1",
    059669">"@types/react": 059669">"^18.3.5",
    059669">"@types/react-dom": 059669">"^18.3.0",
    059669">"@vitejs/plugin-react": 059669">"^4.3.1",
    059669">"autoprefixer": 059669">"^10.4.18",
    059669">"eslint": 059669">"^9.9.1",
    059669">"eslint-plugin-react-hooks": 059669">"^5.1.0-rc.0",
    059669">"eslint-plugin-react-refresh": 059669">"^0.4.11",
    059669">"globals": 059669">"^15.9.0",
    059669">"postcss": 059669">"^8.4.35",
    059669">"tailwindcss": 059669">"^3.4.1",
    059669">"typescript": 059669">"^5.5.3",
    059669">"typescript-eslint": 059669">"^8.3.0",
    059669">"vite": 059669">"^5.4.2"
  }
}
```

```
import { defineConfig } from 'vite';
import react from '@vitejs/plugin-react';
import { fileURLToPath, URL } from 'node:url';

// https://vitejs.dev/config/
export default defineConfig({
  plugins: [react()],
  resolve: {
    alias: {
      '@': fileURLToPath(new URL('./src', import.meta.url)),
    },
  },
  optimizeDeps: {
    exclude: ['lucide-react'],
  },
});
```

```
{
  059669">"files": [],
  059669">"references": [
    { 059669">"path": 059669">"./tsconfig.app.json" },
    { 059669">"path": 059669">".tsconfig.node.json" }
  ]
}
```

```
{
  059669">"compilerOptions": {
    059669">"target": 059669">"ES2020",
    059669">"useDefineForClassFields": 600">true,
    059669">"lib": [059669">"ES2020", 059669">"DOM", 059669">"DOM.Iterable"],
    059669">"module": 059669">"ESNext",
    059669">"skipLibCheck": 600">true,

    059669">"color:#6b7280;font-style:italic"> /* Bundler mode */
    059669">"moduleResolution": 059669">"bundler",
    059669">"allowImportingTsExtensions": 600">true,
    059669">"isolatedModules": 600">true,
    059669">"moduleDetection": 059669">"force",
    059669">"noEmit": 600">true,
    059669">"jsx": 059669">"react-jsx",

    059669">"color:#6b7280;font-style:italic"> /* Path alias – depth-invariant imports 600">for src(@/foo === src/foo
600">from any file) */
    059669">"baseUrl": 059669">".",
    059669">"paths": {
      059669">"@059669">">/*": [059669">"src/*"]
    },

    /* Linting */
    059669">"strict": 600">true,
    059669">"noUnusedLocals": 600">false,
    059669">"noUnusedParameters": 600">false,
    059669">"noFallthroughCasesInSwitch": 600">true
  },
  059669">"include": [059669">"src"]
}
```

```
{
  059669">"compilerOptions": {
    059669">"target": 059669">"ES2022",
    059669">"lib": [059669">"ES2023"],
    059669">"module": 059669">"ESNext",
    059669">"skipLibCheck": 600">true,

    059669">"color:#6b7280;font-style:italic">/* Bundler mode */
    059669">"moduleResolution": 059669">"bundler",
    059669">"allowImportingTsExtensions": 600">true,
    059669">"isolatedModules": 600">true,
    059669">"moduleDetection": 059669">"force",
    059669">"noEmit": 600">true,

    059669">"color:#6b7280;font-style:italic">/* Linting */
    059669">"strict": 600">true,
    059669">"noUnusedLocals": 600">>false,
    059669">"noUnusedParameters": 600">>false,
    059669">"noFallthroughCasesInSwitch": 600">true
  },
  059669">"include": [059669">"vite.config.ts"]
}
```



```

059669">"color:#6b7280;font-style:italic">/** @600">type {600">import(059669">'tailwindcss').Config} */
600">export 600">default {
  content: [059669">'./index.html', 059669">'./src/**/*.{js,ts,jsx,tsx}'],
  darkMode: 059669">'600">'class',
  theme: {
    extend: {
      colors: {
        primary: {
          50: 059669">'#eef6ff',
          100: 059669">'#d9eaff',
          200: 059669">'#bcd9ff',
          300: 059669">'#8ec0ff',
          400: 059669">'#599cff',
          500: 059669">'#3479ff',
          600: 059669">'#1f5cf5',
          700: 059669">'#1849e1',
          800: 059669">'#1a3db6',
          900: 059669">'#1b378f',
          950: 059669">'#142357',
        },
        accent: {
          50: 059669">'#ecfdf5',
          100: 059669">'#d1fae5',
          200: 059669">'#a7f3d0',
          300: 059669">'#6ee7b7',
          400: 059669">'#34d399',
          500: 059669">'#10b981',
          600: 059669">'#059669',
          700: 059669">'#047857',
          800: 059669">'#065f46',
          900: 059669">'#064e3b',
        },
        warning: {
          50: 059669">'#ffffbe',
          100: 059669">'#fef3c7',
          200: 059669">'#fde68a',
          300: 059669">'#fcd34d',
          400: 059669">'#fbbf24',
          500: 059669">'#f59e0b',
          600: 059669">'#d97706',
          700: 059669">'#b45309',
          800: 059669">'#92400e',
          900: 059669">'#78350f',
        },
        danger: {
          50: 059669">'#fef2f2',
          100: 059669">'#fee2e2',
          200: 059669">'#fecaca',
          300: 059669">'#fca5a5',
          400: 059669">'#f87171',
          500: 059669">'#ef4444',
          600: 059669">'#dc2626',
          700: 059669">'#b91c1c',
          800: 059669">'#991b1b',
          900: 059669">'#7f1d1d',
        },
      },
    },
    fontFamily: {
      sans: [059669">'Inter', 059669">'ui-sans-serif', 059669">'system-ui', 059669">'sans-serif'],
      mono: [059669">'JetBrains Mono', 059669">'ui-monospace', 059669">'monospace'],
    },
    boxShadow: {
      card: 059669">'0 1px 3px 0 rgb(0 0 0 / 0.06), 0 1px 2px -1px rgb(0 0 0 / 0.05)',
      059669">'card-hover': 059669">'0 8px 24px -6px rgb(0 0 0 / 0.12), 0 2px 6px -2px rgb(0 0 0 / 0.06)',
    },
    keyframes: {
      059669">'fade-600">'in': {
        059669">'0%': { opacity: 059669">'0', transform: 059669">'translateY(4px)' },
        059669">'100%': { opacity: 059669">'1', transform: 059669">'translateY(0)' },
      },
    },
  },
}

```

```

    },
    059669">'slide-600">in': {
      059669">'0%': { opacity: 059669">'0', transform: 059669">'translateX(8px)' },
      059669">'100%': { opacity: 059669">'1', transform: 059669">'translateX(0)' },
    },
    059669">'scale-600">in': {
      059669">'0%': { opacity: 059669">'0', transform: 059669">'scale(0.96)' },
      059669">'100%': { opacity: 059669">'1', transform: 059669">'scale(1)' },
    },
    059669">'pulse-ring': {
      059669">'0%': { boxShadow: 059669">'0 0 0 0 rgb(239 68 68 / 0.5)' },
      059669">'70%': { boxShadow: 059669">'0 0 0 8px rgb(239 68 68 / 0)' },
      059669">'100%': { boxShadow: 059669">'0 0 0 0 rgb(239 68 68 / 0)' },
    },
  },
  animation: {
    059669">'fade-600">in': 059669">'fade-600">in 0.25s ease-out',
    059669">'slide-600">in': 059669">'slide-600">in 0.25s ease-out',
    059669">'scale-600">in': 059669">'scale-600">in 0.15s ease-out',
    059669">'pulse-ring': 059669">'pulse-ring 1.8s ease-out infinite',
  },
},
},
plugins: [],
};

```

```
600">export 600">default {  
  plugins: {  
    tailwindcss: {},  
    autoprefixer: {},  
  },  
};
```

```
import js from '@eslint/js';
import globals from 'globals';
import reactHooks from 'eslint-plugin-react-hooks';
import reactRefresh from 'eslint-plugin-react-refresh';
import tseslint from 'typescript-eslint';

export default tseslint.config(
  { ignores: ['dist'] },
  {
    extends: [js.configs.recommended, ...tseslint.configs.recommended],
    files: ['**/*.ts', '**/*.tsx'],
    languageOptions: {
      ecmaVersion: 2020,
      globals: globals.browser,
    },
    plugins: {
      'react-hooks': reactHooks,
      'react-refresh': reactRefresh,
    },
    rules: {
      ...reactHooks.configs.recommended.rules,
      'react-refresh/only-export-components': [
        'warn',
        { allowConstantExport: true },
      ],
    },
  },
);
```

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="/vite.svg" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>SmartMaint – Industrial Maintenance & Spare Parts Management</title>
    <meta property="og:image" content="https://"bolt.new/static/og_default.png" />
    <meta name="twitter:card" content="summary_large_image" />
    <meta name="twitter:image" content="https://"bolt.new/static/og_default.png"
  />
</head>
<body>
  <div id="root"></div>
  <script type="module" src="/src/main.tsx"></script>
</body>
</html>
```

```
import { StrictMode } from 'react';
import { createRoot } from 'react-dom/client';
import App from './App.tsx';
import './index.css';

createRoot(document.getElementById('root')!).render(
  <StrictMode>
    <App />
  </StrictMode>
);
```

```
600">import { BrowserRouter, Routes, Route, Navigate, useLocation } 600">from 059669">'react-router-dom';
600">import { Provider } 600">from 059669">'react-redux';
600">import { store } 600">from 059669">'@/store';
600">import { ThemeProvider } 600">from 059669">'@/components/ThemeProvider';
600">import { AppLayout } 600">from 059669">'@/components/layout/AppLayout';
600">import { LoginPage } 600">from 059669">'@/pages/LoginPage';
600">import { DashboardPage } 600">from 059669">'@/pages/DashboardPage';
600">import { MachinesPage } 600">from 059669">'@/pages/MachinesPage';
600">import { MachineDetailPage } 600">from 059669">'@/pages/MachineDetailPage';
600">import { MaintenancePage } 600">from 059669">'@/pages/MaintenancePage';
600">import { BreakdownsPage } 600">from 059669">'@/pages/BreakdownsPage';
600">import { InventoryPage } 600">from 059669">'@/pages/InventoryPage';
600">import { ForecastingPage } 600">from 059669">'@/pages/ForecastingPage';
600">import { SuppliersPage } 600">from 059669">'@/pages/SuppliersPage';
600">import { WarrantyPage } 600">from 059669">'@/pages/WarrantyPage';
600">import { ServiceHistoryPage } 600">from 059669">'@/pages/ServiceHistoryPage';
600">import { PredictivePage } 600">from 059669">'@/pages/PredictivePage';
600">import { NotificationsPage } 600">from 059669">'@/pages/NotificationsPage';
600">import { AuditPage } 600">from 059669">'@/pages/AuditPage';
600">import { UsersPage } 600">from 059669">'@/pages/UsersPage';
600">import { SettingsPage } 600">from 059669">'@/pages/SettingsPage';
600">import { useAppSelector } 600">from 059669">'@/store';
600">import { canAccessModule } 600">from 059669">'@/lib/rbac';
600">import 600">type { RoleKey } 600">from 059669">'@/types';
600">import 600">type { ReactNode } 600">from 059669">'react';

600">function ProtectedRoute({ children, module }: { children: ReactNode; module?: string }) {
  600">const isAuthenticated = useAppSelector((s) => s.auth.isAuthenticated);
  600">const role = useAppSelector((s) => s.auth.user?.role) 600">as RoleKey | 600">undefined;
  600">const location = useLocation();

  600">if (!isAuthenticated || !role) {
    600">return <Navigate to=059669">"/login" state={{ 600">from: location }} replace />;
  }

  600">if (module && !canAccessModule(role, module)) {
    600">return <Navigate to=059669">"/dashboard" replace />;
  }

  600">return <>{children}</>;
}

600">function AppRoutes() {
  600">return (
    <Routes>
      <Route path=059669">"/login" element={<LoginPage />} />
      <Route
        element={
          <ProtectedRoute>
            <AppLayout />
          </ProtectedRoute>
        }
      >
        <Route path=059669">"/dashboard" element={<ProtectedRoute module=059669">"dashboard"><DashboardPage />
</ProtectedRoute>} />
        <Route path=059669">"/machines" element={<ProtectedRoute module=059669">"machines"><MachinesPage />
</ProtectedRoute>} />
        <Route path=059669">"/machines/:id" element={<ProtectedRoute module=059669">"machines"><MachineDetailPage />
</ProtectedRoute>} />
        <Route path=059669">"/maintenance" element={<ProtectedRoute module=059669">"maintenance"><MaintenancePage />
</ProtectedRoute>} />
        <Route path=059669">"/breakdowns" element={<ProtectedRoute module=059669">"breakdowns"><BreakdownsPage />
</ProtectedRoute>} />
        <Route path=059669">"/inventory" element={<ProtectedRoute module=059669">"inventory"><InventoryPage />
</ProtectedRoute>} />
        <Route path=059669">"/forecasting" element={<ProtectedRoute module=059669">"forecasting"><ForecastingPage />
</ProtectedRoute>} />
        <Route path=059669">"/suppliers" element={<ProtectedRoute module=059669">"suppliers"><SuppliersPage />
</ProtectedRoute>} />
    </Routes>
  )
}
```

```

        <Route path=059669>"/warranty" element={<ProtectedRoute module=059669>"warranty"><WarrantyPage />
</ProtectedRoute>} />
        <Route path=059669>"/service-history" element={<ProtectedRoute module=059669>"service_history">
<ServiceHistoryPage /></ProtectedRoute>} />
        <Route path=059669>"/predictive" element={<ProtectedRoute module=059669>"predictive"><PredictivePage />
</ProtectedRoute>} />
        <Route path=059669>"/notifications" element={<ProtectedRoute module=059669>"notifications">
<NotificationsPage /></ProtectedRoute>} />
        <Route path=059669>"/audit" element={<ProtectedRoute module=059669>"audit"><AuditPage /></ProtectedRoute>}
/>

        <Route path=059669>"/users" element={<ProtectedRoute><UsersPage /></ProtectedRoute>} />
        <Route path=059669>"/settings" element={<ProtectedRoute><SettingsPage /></ProtectedRoute>} />
    </Route>
    <Route path=059669>"/" element={<Navigate to=059669>"/dashboard" replace />} />
    <Route path=059669>"*" element={<Navigate to=059669>"/dashboard" replace />} />
</Routes>
);
}

600">function App() {
  600">return (
    <Provider store={store}>
      <ThemeProvider>
        <BrowserRouter>
          <AppRoutes />
        </BrowserRouter>
      </ThemeProvider>
    </Provider>
  );
}

600">export 600">default App;

```



```
059669">"color:#6b7280;font-style:italic">// Domain types 600">for the Smart Industrial Maintenance & Spare Parts Management System
```

```
600">export 600">type RoleKey =  
  | 059669">'super_admin'  
  | 059669">'factory_manager'  
  | 059669">'maintenance_manager'  
  | 059669">'maintenance_engineer'  
  | 059669">'inventory_manager'  
  | 059669">'procurement_officer'  
  | 059669">'supplier';
```

```
600">export 600">interface Role {  
  key: RoleKey;  
  name: string;  
  description: string;  
  permissions: string[];  
}
```

```
600">export 600">interface User {  
  id: string;  
  name: string;  
  email: string;  
  role: RoleKey;  
  role_name: string;  
  avatar_color: string;  
  factory_id?: string;  
  phone?: string;  
  last_active: string;  
}
```

```
600">export 600">type CriticalityLevel = 059669">'critical' | 059669">'high' | 059669">'medium' | 059669">'low';  
600">export 600">type MachineStatus = 059669">'operational' | 059669">'maintenance' | 059669">'breakdown' |  
059669">'idle' | 059669">'decommissioned';
```

```
600">export 600">interface Factory {  
  id: string;  
  name: string;  
  location: string;  
  country: string;  
  departments: Department[];  
  manager: string;  
  established: string;  
}
```

```
600">export 600">interface Department {  
  id: string;  
  name: string;  
  factory_id: string;  
  production_lines: string[];  
}
```

```
600">export 600">interface MachineComponent {  
  id: string;  
  name: string;  
  part_number: string;  
  category: string;  
  current_status: 059669">'ok' | 059669">'warning' | 059669">'critical' | 059669">'replaced';  
  last_replaced?: string;  
  expected_life_hours?: number;  
}
```

```
600">export 600">interface Machine {  
  id: string;  
  machine_id: string;  
  name: string;  
  factory_id: string;  
  factory_name: string;  
  department_id: string;
```

```

department_name: string;
production_line: string;
600">type: string;
manufacturer: string;
model: string;
serial_number: string;
installation_date: string;
criticality: CriticalityLevel;
status: MachineStatus;
operating_hours: number;
total_downtime_hours: number;
health_score: number;
risk_level: 059669">'low' | 059669">'medium' | 059669">'high' | 059669">'critical';
last_maintenance: string;
next_maintenance: string;
components: MachineComponent[];
mtbf_hours: number;
mttr_hours: number;
failure_count_30d: number;
}

600">export 600">type MaintenanceType = 059669">'preventive' | 059669">'corrective' | 059669">'predictive' |
059669">'inspection';
600">export 600">type MaintenancePriority = 059669">'low' | 059669">'medium' | 059669">'high' | 059669">'urgent';
600">export 600">type MaintenanceStatus = 059669">'scheduled' | 059669">'in_progress' | 059669">'completed' |
059669">'overdue' | 059669">'cancelled';

600">export 600">interface MaintenanceChecklistItem {
  id: string;
  task: string;
  completed: boolean;
}

600">export 600">interface MaintenanceSchedule {
  id: string;
  machine_id: string;
  machine_name: string;
  machine_id_code: string;
  600">type: MaintenanceType;
  priority: MaintenancePriority;
  status: MaintenanceStatus;
  scheduled_date: string;
  completed_date?: string;
  assigned_technician: string;
  technician_id: string;
  description: string;
  checklist: MaintenanceChecklistItem[];
  estimated_hours: number;
  actual_hours?: number;
  cost: number;
  recurring: boolean;
  interval_days: number;
  factory_id: string;
  factory_name: string;
}

600">export 600">interface MaintenanceWorkOrder 600">extends MaintenanceSchedule {
  parts_used: WorkOrderPart[];
  labor_cost: number;
  parts_cost: number;
  total_cost: number;
  notes?: string;
}

600">export 600">interface WorkOrderPart {
  part_id: string;
  part_name: string;
  part_number: string;
  quantity: number;
  unit_cost: number;
}

600">export 600">type BreakdownStatus = 059669">'reported' | 059669">'diagnosing' | 059669">'in_repair' |

```

```

059669">'resolved' | 059669">'closed';
600">export 600">type BreakdownSeverity = 059669">'minor' | 059669">'major' | 059669">'severe' | 059669">'critical';

600">export 600">interface Breakdown {
  id: string;
  breakdown_id: string;
  machine_id: string;
  machine_name: string;
  machine_id_code: string;
  factory_id: string;
  factory_name: string;
  reported_date: string;
  reported_by: string;
  assigned_technician: string;
  status: BreakdownStatus;
  severity: BreakdownSeverity;
  symptoms: string;
  root_cause?: string;
  corrective_action?: string;
  downtime_hours: number;
  parts_replaced: WorkOrderPart[];
  repair_cost: number;
  resolved_date?: string;
  priority: MaintenancePriority;
}

600">export 600">type PartCategory = 059669">'mechanical' | 059669">'electrical' | 059669">'hydraulic' |
059669">'pneumatic' | 059669">'electronic' | 059669">'consumable' | 059669">'lubricant' | 059669">'safety';

600">export 600">interface SparePart {
  id: string;
  part_number: string;
  name: string;
  category: PartCategory;
  description: string;
  compatible_machines: string[];
  current_stock: number;
  reserved_stock: number;
  min_stock: number;
  max_stock: number;
  reorder_point: number;
  unit_cost: number;
  warehouse_id: string;
  warehouse_name: string;
  supplier_id: string;
  supplier_name: string;
  lead_time_days: number;
  last_restocked: string;
  batch_number?: string;
  movement_class: 059669">'fast' | 059669">'medium' | 059669">'slow';
  monthly_consumption: number[];
  stock_status: 059669">'in_stock' | 059669">'low_stock' | 059669">'critical' | 059669">'out_of_stock' |
059669">'overstock';
}

600">export 600">type TransactionType = 059669">'stock_in' | 059669">'stock_out' | 059669">'reserved' |
059669">'returned' | 059669">'adjusted';

600">export 600">interface InventoryTransaction {
  id: string;
  part_id: string;
  part_name: string;
  part_number: string;
  600">type: TransactionType;
  quantity: number;
  balance_after: number;
  reference: string;
  reference_type: 059669">'work_order' | 059669">'purchase_order' | 059669">'adjustment' | 059669">'600">return';
  performed_by: string;
  warehouse_id: string;
  timestamp: string;
  notes?: string;
}

```

```

600">export 600">interface Warehouse {
  id: string;
  name: string;
  location: string;
  factory_id: string;
  capacity: number;
  utilized: number;
  manager: string;
}

600">export 600">type SupplierStatus = 059669">'active' | 059669">'inactive' | 059669">'preferred' |
059669">'blacklisted';

600">export 600">interface Supplier {
  id: string;
  name: string;
  contact_person: string;
  email: string;
  phone: string;
  address: string;
  rating: number;
  status: SupplierStatus;
  lead_time_days: number;
  on_time_delivery_rate: number;
  quality_rate: number;
  parts_supplied: number;
  total_orders: number;
  total_value: number;
  established: string;
}

600">export 600">type PurchaseRequestStatus = 059669">'draft' | 059669">'pending_approval' | 059669">'approved' |
059669">'rejected' | 059669">'ordered' | 059669">'received' | 059669">'cancelled';
600">export 600">type PurchaseOrderStatus = 059669">'draft' | 059669">'sent' | 059669">'confirmed' |
059669">'partial' | 059669">'received' | 059669">'cancelled';

600">export 600">interface PurchaseRequest {
  id: string;
  request_number: string;
  requested_by: string;
  department: string;
  status: PurchaseRequestStatus;
  priority: MaintenancePriority;
  items: PurchaseItem[];
  total_value: number;
  created_date: string;
  required_date: string;
  approved_by?: string;
  notes?: string;
  justification: string;
}

600">export 600">interface PurchaseOrder {
  id: string;
  order_number: string;
  supplier_id: string;
  supplier_name: string;
  status: PurchaseOrderStatus;
  items: PurchaseItem[];
  total_value: number;
  order_date: string;
  expected_delivery: string;
  actual_delivery?: string;
  created_by: string;
  delivery_progress: number;
}

600">export 600">interface PurchaseItem {
  part_id: string;
  part_name: string;
  part_number: string;
  quantity: number;

```

```

    unit_cost: number;
    total: number;
}

600">export 600">type WarrantyStatus = 059669">'active' | 059669">'expiring_soon' | 059669">'expired' |
059669">'claimed';

600">export 600">interface Warranty {
    id: string;
    warranty_number: string;
    machine_id: string;
    machine_name: string;
    machine_id_code: string;
    component_id?: string;
    component_name?: string;
    600">type: 059669">'machine' | 059669">'component';
    start_date: string;
    end_date: string;
    status: WarrantyStatus;
    provider: string;
    coverage: string;
    terms: string;
    days_remaining: number;
}

600">export 600">interface ServiceRecord {
    id: string;
    service_id: string;
    machine_id: string;
    machine_name: string;
    machine_id_code: string;
    date: string;
    technician: string;
    600">type: MaintenanceType;
    problem_description: string;
    diagnosis: string;
    parts_replaced: WorkOrderPart[];
    labor_hours: number;
    labor_cost: number;
    parts_cost: number;
    total_cost: number;
    downtime_hours: number;
    root_cause?: string;
    corrective_action?: string;
    attachments: string[];
    factory_id: string;
}

600">export 600">type NotificationType =
    | 059669">'maintenance_due'
    | 059669">'maintenance_overdue'
    | 059669">'breakdown'
    | 059669">'low_stock'
    | 059669">'critical_stock'
    | 059669">'warranty_expiry'
    | 059669">'purchase_approval'
    | 059669">'supplier_delay'
    | 059669">'system';

600">export 600">type NotificationPriority = 059669">'info' | 059669">'warning' | 059669">'critical';

600">export 600">interface AppNotification {
    id: string;
    600">type: NotificationType;
    priority: NotificationPriority;
    title: string;
    message: string;
    module: string;
    timestamp: string;
    read: boolean;
    action_url?: string;
    target_roles: RoleKey[];
}

```

```
600">export 600">interface AuditLog {
  id: string;
  user: string;
  user_role: RoleKey;
  action: string;
  module: string;
  entity_id?: string;
  entity_type?: string;
  previous_value?: string;
  updated_value?: string;
  timestamp: string;
  ip_address: string;
  device: string;
}

600">export 600">interface KPIData {
  total_machines: number;
  operational_machines: number;
  machines_under_maintenance: number;
  breakdown_count: number;
  total_downtime_hours: number;
  upcoming_maintenance: number;
  overdue_maintenance: number;
  low_stock_parts: number;
  critical_stock_parts: number;
  maintenance_cost_ytd: number;
  inventory_value: number;
  avg_supplier_performance: number;
  open_purchase_orders: number;
  active_warranties: number;
  expiring_warranties: number;
}

600">export 600">interface ForecastData {
  part_id: string;
  part_name: string;
  part_number: string;
  historical_consumption: number[];
  forecasted_consumption: number[];
  recommended_reorder_qty: number;
  safety_stock: number;
  stock_out_risk: number;
  excess_inventory: boolean;
  movement_class: 059669">'fast' | 059669">'medium' | 059669">'slow';
  current_stock: number;
}
```

```
600">import { configureStore } 600">from 059669">'@reduxjs/toolkit';
600">import { useDispatch, useSelector, 600">type TypedUseSelectorHook } 600">from 059669">'react-redux';
600">import authReducer 600">from 059669">'./authSlice';
600">import themeReducer 600">from 059669">'./themeSlice';

600">export 600">const store = configureStore({
  reducer: {
    auth: authReducer,
    theme: themeReducer,
  },
});

600">export 600">type RootState = ReturnType<600">typeof store.getState>;
600">export 600">type AppDispatch = 600">typeof store.dispatch;

600">export 600">const useAppDispatch = () => useDispatch<AppDispatch>();
600">export 600">const useAppSelector: TypedUseSelectorHook<RootState> = useSelector;
```

```
600">import { createSlice, 600">type PayloadAction } 600">from 059669">'@reduxjs/toolkit';
600">import 600">type { RoleKey } 600">from 059669">'@/types';
600">import { users } 600">from 059669">'@/data/seed';

600">export 600">interface AuthState {
  user: 600">typeof users[number] | 600">>null;
  isAuthenticated: boolean;
}

600">const initialState: AuthState = {
  user: 600">>null,
  isAuthenticated: 600">>false,
};

600">const authSlice = createSlice({
  name: 059669">'auth',
  initialState,
  reducers: {
    login(state, action: PayloadAction<RoleKey>) {
      600">const user = users.find((u) => u.role === action.payload);
      600">if (user) {
        state.user = user;
        state.isAuthenticated = 600">>true;
      }
    },
    logout(state) {
      state.user = 600">>null;
      state.isAuthenticated = 600">>false;
    },
  },
});

600">export 600">const { login, logout } = authSlice.actions;
600">export 600">default authSlice.reducer;
```



```
import { createSlice, type PayloadAction } from '@reduxjs/toolkit';

export type Theme = 'light' | 'dark';

interface ThemeState {
  theme: Theme;
}

function getInitialTheme(): Theme {
  if (typeof window !== 'undefined') {
    const stored = localStorage.getItem('theme');
    if (stored === 'light' || stored === 'dark') return stored;
    if (window.matchMedia('(prefers-color-scheme: dark)').matches) return 'dark';
  }
  return 'light';
}

const initialState: ThemeState = {
  theme: getInitialTheme(),
};

const themeSlice = createSlice({
  name: 'theme',
  initialState,
  reducers: {
    toggleTheme(state) {
      state.theme = state.theme === 'light' ? 'dark' : 'light';
    },
    setTheme(state, action: PayloadAction<Theme>) {
      state.theme = action.payload;
    },
  },
});

export const { toggleTheme, setTheme } = themeSlice.actions;
export default themeSlice.reducer;
```

```

600">import {
  factories, departments, machines, maintenanceSchedules, breakdowns,
  spareParts, inventoryTransactions, warehouses, suppliers,
  purchaseRequests, purchaseOrders, warranties, serviceRecords,
  notifications, auditLogs, users, kpiData, forecastData,
  downtimeTrendData, maintenanceCostTrendData, breakdownFrequencyData,
  sparePartConsumptionData, inventoryTurnoverData, preventiveVsCorrectiveData,
  machineReliabilityData,
} 600">from 059669">'@/data/seed';

600">export 600">interface PaginatedResult<T> {
  data: T[];
  total: number;
  page: number;
  pageSize: number;
  totalPages: number;
}

600">export 600">interface QueryParams {
  page?: number;
  pageSize?: number;
  search?: string;
  sortBy?: string;
  sortDir?: 059669">'asc' | 059669">'desc';
  filters?: Record<string, string | 600">undefined>;
}

600">function paginate<T>(items: T[], params: QueryParams): PaginatedResult<T> {
  600">const page = params.page || 1;
  600">const pageSize = params.pageSize || 10;
  600">const total = items.length;
  600">const totalPages = Math.ceil(total / pageSize);
  600">const start = (page - 1) * pageSize;
  600">return {
    data: items.slice(start, start + pageSize),
    total,
    page,
    pageSize,
    totalPages,
  };
}

600">function sortItems<T>(items: T[], sortBy?: string, sortDir?: 059669">'asc' | 059669">'desc'): T[] {
  600">if (!sortBy) 600">return items;
  600">const dir = sortDir === 059669">'desc' ? -1 : 1;
  600">return [...items].sort((a, b) => {
    600">const av = (a 600">as Record<string, unknown>)[sortBy];
    600">const bv = (b 600">as Record<string, unknown>)[sortBy];
    600">if (av == 600">null && bv == 600">null) 600">return 0;
    600">if (av == 600">null) 600">return 1;
    600">if (bv == 600">null) 600">return -1;
    600">if (600">typeof av === 059669">'number' && 600">typeof bv === 059669">'number') 600">return (av - bv) * dir;
    600">return String(av).localeCompare(String(bv)) * dir;
  });
}

600">function searchItems<T>(items: T[], search: string, fields: string[]): T[] {
  600">if (!search) 600">return items;
  600">const q = search.toLowerCase();
  600">return items.filter((item) => {
    600">const record = item 600">as Record<string, unknown>;
    600">return fields.some((f) => String(record[f] ?? 059669">'').toLowerCase().includes(q));
  });
}

600">function filterItems<T>(items: T[], filters: Record<string, string | 600">undefined>): T[] {
  600">const activeFilters = Object.entries(filters).filter(([, v]) => v != 600">null && v != 059669">' ' && v != 059669">'all');
  600">if (activeFilters.length === 0) 600">return items;

```

```

600">return items.filter((item) => {
600">const record = item 600">as Record<string, unknown>;
600">return activeFilters.every(([key, val]) => String(record[key] ?? 059669">'').toLowerCase() ===
String(val).toLowerCase());
});
}

```

```

600">function query<T>(items: T[], params: QueryParams, searchFields: string[]): PaginatedResult<T> {
600">let result = searchItems(items, params.search || 059669">', searchFields);
result = filterItems(result, params.filters || {});
result = sortItems(result, params.sortBy, params.sortDir);
600">return paginate(result, params);
}

```

```

600">const delay = (ms: number) => 600">new Promise((r) => setTimeout(r, ms));

```

```

059669">"color:#6b7280;font-style:italic">/// — API

```

```

600">export 600">const api = {
600">async getMachines(params: QueryParams = {}) {
600">await delay(150);
600">return query(machines, params, [059669">'name', 059669">'machine_id', 059669">'manufacturer',
059669">'model', 059669">'serial_number', 059669">'600">type']);
},
600">async getMachine(id: string) {
600">await delay(100);
600">return machines.find((m) => m.id === id) || 600">>null;
},
600">async getMaintenanceSchedules(params: QueryParams = {}) {
600">await delay(150);
600">return query(maintenanceSchedules, params, [059669">'machine_name', 059669">'machine_id_code',
059669">'assigned_technician', 059669">'description']);
},
600">async getBreakdowns(params: QueryParams = {}) {
600">await delay(150);
600">return query(breakdowns, params, [059669">'machine_name', 059669">'machine_id_code', 059669">'symptoms',
059669">'assigned_technician', 059669">'breakdown_id']);
},
600">async getSpareParts(params: QueryParams = {}) {
600">await delay(150);
600">return query(spareParts, params, [059669">'name', 059669">'part_number', 059669">'supplier_name',
059669">'warehouse_name']);
},
600">async getInventoryTransactions(params: QueryParams = {}) {
600">await delay(150);
600">return query(inventoryTransactions, params, [059669">'part_name', 059669">'part_number',
059669">'reference', 059669">'performed_by']);
},
600">async getSuppliers(params: QueryParams = {}) {
600">await delay(150);
600">return query(suppliers, params, [059669">'name', 059669">'contact_person', 059669">'email',
059669">'address']);
},
600">async getPurchaseRequests(params: QueryParams = {}) {
600">await delay(150);
600">return query(purchaseRequests, params, [059669">'request_number', 059669">'requested_by',
059669">'department']);
},
600">async getPurchaseOrders(params: QueryParams = {}) {
600">await delay(150);
600">return query(purchaseOrders, params, [059669">'order_number', 059669">'supplier_name',
059669">'created_by']);
},
600">async getWarranties(params: QueryParams = {}) {
600">await delay(150);
600">return query(warranties, params, [059669">'warranty_number', 059669">'machine_name',
059669">'machine_id_code', 059669">'provider', 059669">'component_name']);
},
600">async getServiceRecords(params: QueryParams = {}) {
600">await delay(150);
600">return query(serviceRecords, params, [059669">'service_id', 059669">'machine_name',
059669">'machine_id_code', 059669">'technician', 059669">'problem_description']);
},
}

```

```

600">async getNotifications(params: QueryParams = {}) {
600">await delay(100);
600">return query(notifications, params, [059669">'title', 059669">'message', 059669">'module']);
},
600">async getAuditLogs(params: QueryParams = {}) {
600">await delay(150);
600">return query(auditLogs, params, [059669">'user', 059669">'action', 059669">'module', 059669">'entity_id']);
},
600">async getFactories() {
600">await delay(100);
600">return factories;
},
600">async getDepartments() {
600">await delay(100);
600">return departments;
},
600">async getWarehouses() {
600">await delay(100);
600">return warehouses;
},
600">async getUsers() {
600">await delay(100);
600">return users;
},
600">async getKPIs() {
600">await delay(100);
600">return kpiData;
},
600">async getForecastData() {
600">await delay(200);
600">return forecastData;
},
600">async getDowntimeTrend() {
600">await delay(100);
600">return downtimeTrendData;
},
600">async getMaintenanceCostTrend() {
600">await delay(100);
600">return maintenanceCostTrendData;
},
600">async getBreakdownFrequency() {
600">await delay(100);
600">return breakdownFrequencyData;
},
600">async getSparePartConsumption() {
600">await delay(100);
600">return sparePartConsumptionData;
},
600">async getInventoryTurnover() {
600">await delay(100);
600">return inventoryTurnoverData;
},
600">async getPreventiveVsCorrective() {
600">await delay(100);
600">return preventiveVsCorrectiveData;
},
600">async getMachineReliability() {
600">await delay(100);
600">return machineReliabilityData;
},
};

```

```
600">import 600">type { RoleKey } 600">from 059669">'@/types';
600">import {
  LayoutDashboard, Cpu, Wrench, AlertTriangle, Package,
  TrendingUp, Truck, ShieldCheck, ClipboardList, Activity,
  Bell, FileClock, Settings, Users,
} 600">from 059669">'lucide-react';
600">import 600">type { ComponentType } 600">from 059669">'react';

600">export 600">interface NavItem {
  key: string;
  label: string;
  icon: ComponentType<{ className?: string }>;
  path: string;
  permission: string;
}

600">export 600">interface NavSection {
  label: string;
  items: NavItem[];
}

600">export 600">const navSections: NavSection[] = [
  {
    label: 059669">'Overview',
    items: [
      { key: 059669">'dashboard', label: 059669">'Dashboard', icon: LayoutDashboard, path: 059669">' /dashboard',
        permission: 059669">'dashboard:view' },
    ],
  },
  {
    label: 059669">'Asset Management',
    items: [
      { key: 059669">'machines', label: 059669">'Machines', icon: Cpu, path: 059669">' /machines', permission:
059669">'machines:view' },
      { key: 059669">'maintenance', label: 059669">'Maintenance', icon: Wrench, path: 059669">' /maintenance',
        permission: 059669">'maintenance:view' },
      { key: 059669">'breakdowns', label: 059669">'Breakdowns', icon: AlertTriangle, path: 059669">' /breakdowns',
        permission: 059669">'breakdowns:view' },
      { key: 059669">'service_history', label: 059669">'Service History', icon: ClipboardList, path:
059669">' /service-history', permission: 059669">'service_history:view' },
    ],
  },
  {
    label: 059669">'Inventory & Procurement',
    items: [
      { key: 059669">'inventory', label: 059669">'Spare Parts', icon: Package, path: 059669">' /inventory',
        permission: 059669">'inventory:view' },
      { key: 059669">'forecasting', label: 059669">'Forecasting', icon: TrendingUp, path: 059669">' /forecasting',
        permission: 059669">'forecasting:view' },
      { key: 059669">'suppliers', label: 059669">'Suppliers & Procurement', icon: Truck, path: 059669">' /suppliers',
        permission: 059669">'suppliers:view' },
    ],
  },
  {
    label: 059669">'Analytics & Compliance',
    items: [
      { key: 059669">'predictive', label: 059669">'Predictive Maintenance', icon: Activity, path:
059669">' /predictive', permission: 059669">'predictive:view' },
      { key: 059669">'warranty', label: 059669">'Warranty', icon: ShieldCheck, path: 059669">' /warranty', permission:
059669">'warranty:view' },
      { key: 059669">'notifications', label: 059669">'Notifications', icon: Bell, path: 059669">' /notifications',
        permission: 059669">'notifications:view' },
      { key: 059669">'audit', label: 059669">'Audit Log', icon: FileClock, path: 059669">' /audit', permission:
059669">'audit:view' },
    ],
  },
];

600">export 600">const adminOnlyItems: NavItem[] = [
```

```

    { key: 059669">'users', label: 059669">'User Management', icon: Users, path: 059669">' /users', permission:
059669">'users:view' },
    { key: 059669">'settings', label: 059669">'Settings', icon: Settings, path: 059669">' /settings', permission:
059669">'settings:view' },
  ];

600">export 600">function getNavItemsForRole(role: RoleKey): NavSection[] {
  600">const sections = navSections.map((section) => ({
    ...section,
    items: section.items.filter((item) => hasPermission(role, item.permission)),
  })).filter((section) => section.items.length > 0);

  600">if (role === 059669">'super_admin') {
    sections.push({ label: 059669">'Administration', items: adminOnlyItems });
  }
  600">return sections;
}

600">import { hasPermission } 600">from 059669">'@/lib/rbac';

```

```

600">import 600">type { Role, RoleKey } 600">from 059669">'@/types';

600">export 600">const ROLE_CATALOG: Role[] = [
  {
    key: 059669">'super_admin',
    name: 059669">'Super Admin',
    description: 059669">'Full system access, user management, and global configuration',
    permissions: [059669">'*'],
  },
  {
    key: 059669">'factory_manager',
    name: 059669">'Factory Manager',
    description: 059669">'Oversees factory operations, reviews analytics and KPIs',
    permissions: [
      059669">'dashboard:view',
      059669">'machines:view',
      059669">'maintenance:view',
      059669">'breakdowns:view',
      059669">'inventory:view',
      059669">'forecasting:view',
      059669">'suppliers:view',
      059669">'warranty:view',
      059669">'service_history:view',
      059669">'predictive:view',
      059669">'notifications:view',
      059669">'audit:view',
    ],
  },
  {
    key: 059669">'maintenance_manager',
    name: 059669">'Maintenance Manager',
    description: 059669">'Plans and manages maintenance schedules and work orders',
    permissions: [
      059669">'dashboard:view',
      059669">'machines:view',
      059669">'maintenance:view',
      059669">'maintenance:edit',
      059669">'breakdowns:view',
      059669">'breakdowns:edit',
      059669">'inventory:view',
      059669">'forecasting:view',
      059669">'warranty:view',
      059669">'service_history:view',
      059669">'service_history:edit',
      059669">'predictive:view',
      059669">'notifications:view',
    ],
  },
  {
    key: 059669">'maintenance_engineer',
    name: 059669">'Maintenance Engineer',
    description: 059669">'Executes maintenance tasks and reports breakdowns',
    permissions: [
      059669">'dashboard:view',
      059669">'machines:view',
      059669">'maintenance:view',
      059669">'maintenance:edit',
      059669">'breakdowns:view',
      059669">'breakdowns:edit',
      059669">'service_history:view',
      059669">'service_history:edit',
      059669">'notifications:view',
    ],
  },
  {
    key: 059669">'inventory_manager',
    name: 059669">'Inventory Manager',
    description: 059669">'Manages spare parts inventory, warehouses, and stock transactions',
    permissions: [

```

```

        059669">'dashboard:view',
        059669">'machines:view',
        059669">'inventory:view',
        059669">'inventory:edit',
        059669">'forecasting:view',
        059669">'suppliers:view',
        059669">'notifications:view',
    ],
},
{
    key: 059669">'procurement_officer',
    name: 059669">'Procurement Officer',
    description: 059669">'Handles purchase requests, orders, and supplier relationships',
    permissions: [
        059669">'dashboard:view',
        059669">'inventory:view',
        059669">'suppliers:view',
        059669">'suppliers:edit',
        059669">'notifications:view',
    ],
},
{
    key: 059669">'supplier',
    name: 059669">'Supplier',
    description: 059669">'Views purchase orders and updates delivery status',
    permissions: [
        059669">'dashboard:view',
        059669">'suppliers:view',
        059669">'notifications:view',
    ],
},
];

600">export 600">const ROLE_MAP: Record<RoleKey, Role> = Object.fromEntries(
    ROLE_CATALOG.map((r) => [r.key, r]),
) 600">as Record<RoleKey, Role>;

600">export 600">function hasPermission(role: RoleKey, permission: string): boolean {
    600">const r = ROLE_MAP[role];
    600">if (!r) 600">return 600">>false;
    600">return r.permissions.includes(059669">'*') || r.permissions.includes(permission);
}

600">export 600">function canAccessModule(role: RoleKey, moduleKey: string): boolean {
    600">const modulePermissions: Record<string, string> = {
        dashboard: 059669">'dashboard:view',
        machines: 059669">'machines:view',
        maintenance: 059669">'maintenance:view',
        breakdowns: 059669">'breakdowns:view',
        inventory: 059669">'inventory:view',
        forecasting: 059669">'forecasting:view',
        suppliers: 059669">'suppliers:view',
        warranty: 059669">'warranty:view',
        service_history: 059669">'service_history:view',
        predictive: 059669">'predictive:view',
        notifications: 059669">'notifications:view',
        audit: 059669">'audit:view',
    };
    600">const perm = modulePermissions[moduleKey];
    600">if (!perm) 600">return 600">>false;
    600">return hasPermission(role, perm);
}

```



```

600">import { useState, useEffect, useCallback } 600">from 059669">'react';
600">import { api, 600">type QueryParams, 600">type PaginatedResult } 600">from 059669">'@/lib/api';

600">export 600">function usePaginatedData<T 600">extends { id: string }>(
  fetcher: (params: QueryParams) => Promise<PaginatedResult<T>>,
  searchFields: string[],
  initialFilters: Record<string, string> = {},
) {
  600">const [data, setData] = useState<T[]>([]);
  600">const [total, setTotal] = useState(0);
  600">const [page, setPage] = useState(1);
  600">const [pageSize, setPageSize] = useState(10);
  600">const [totalPages, setTotalPages] = useState(0);
  600">const [search, setSearch] = useState(059669">'');
  600">const [sortBy, setSortBy] = useState<string | 600">undefined>(600">undefined);
  600">const [sortDir, setSortDir] = useState<059669">'asc' | 059669">'desc'>(059669">'asc');
  600">const [filters, setFilters] = useState<Record<string, string>>(initialFilters);
  600">const [isLoading, setIsLoading] = useState(600">true);

  600">const fetchData = useCallback(600">async () => {
    setIsLoading(600">true);
    600">const params: QueryParams = { page, pageSize, search, sortBy, sortDir, filters };
    600">const result = 600">await fetcher(params);
    setData(result.data);
    setTotal(result.total);
    setTotalPages(result.totalPages);
    setIsLoading(600">false);
  }, [fetcher, page, pageSize, search, sortBy, sortDir, filters]);

  useEffect(() => { fetchData(); }, [fetchData]);

  600">const handleSort = (key: string) => {
    600">if (sortBy === key) {
      setSortDir(sortDir === 059669">'asc' ? 059669">'desc' : 059669">'asc');
    } 600">else {
      setSortBy(key);
      setSortDir(059669">'asc');
    }
  };

  600">const handleSearch = (value: string) => {
    setSearch(value);
    setPage(1);
  };

  600">const handleFilter = (key: string, value: string) => {
    setFilters((prev) => ({ ...prev, [key]: value }));
    setPage(1);
  };

  600">return {
    data, total, page, pageSize, totalPages, search, sortBy, sortDir, filters, isLoading,
    setPage, setPageSize, setSearch: handleSearch, setSortBy: handleSort, setFilters: handleFilter,
  };
}

```

```

600">import 600">type {
    Factory,
    Department,
    Machine,
    MachineComponent,
    MaintenanceSchedule,
    Breakdown,
    SparePart,
    InventoryTransaction,
    Warehouse,
    Supplier,
    PurchaseRequest,
    PurchaseOrder,
    Warranty,
    ServiceRecord,
    AppNotification,
    AuditLog,
    User,
    KPIData,
    ForecastData,
    MachineStatus,
    CriticalityLevel,
    MaintenanceType,
    MaintenancePriority,
    MaintenanceStatus,
    BreakdownStatus,
    BreakdownSeverity,
    PartCategory,
    TransactionType,
    WarrantyStatus,
    SupplierStatus,
    PurchaseRequestStatus,
    PurchaseOrderStatus,
    WorkOrderPart,
} 600">from 059669">'@/types';

059669">"color:#6b7280;font-style:italic">// Deterministic pseudo-random generator 600">for reproducible seed data
600">let seed = 42;
600">function rand(): number {
    seed = (seed * 9301 + 49297) % 233280;
    600">return seed / 233280;
}
600">function randInt(min: number, max: number): number {
    600">return Math.floor(rand() * (max - min + 1)) + min;
}
600">function randFloat(min: number, max: number, decimals = 1): number {
    600">return parseFloat((rand() * (max - min) + min).toFixed(decimals));
}
600">function pick<T>(arr: T[]): T {
    600">return arr[randInt(0, arr.length - 1)];
}
600">function pickN<T>(arr: T[], n: number): T[] {
    600">const copy = [...arr];
    600">const result: T[] = [];
    600">for (600">let i = 0; i < n && copy.length > 0; i++) {
        600">const idx = randInt(0, copy.length - 1);
        result.push(copy.splice(idx, 1)[0]);
    }
    600">return result;
}

600">const today = 600">new Date(059669">'2026-08-28');
600">function daysFromNow(days: number): string {
    600">const d = 600">new Date(today);
    d.setDate(d.getDate() + days);
    600">return d.toISOString().split(059669">'T')[0];
}
600">function daysAgo(days: number): string {
    600">return daysFromNow(-days);
}

```

```
}
```

```
059669">"color:#6b7280;font-style:italic">// — Factories & Departments
```

```
600">export 600">const factories: Factory[] = [
  {
    id: 059669">'fac-001',
    name: 059669">'Detroit Assembly Plant',
    location: 059669">'Detroit, Michigan',
    country: 059669">'USA',
    manager: 059669">'Robert Chen',
    established: 059669">'1998-03-15',
    departments: [],
  },
  {
    id: 059669">'fac-002',
    name: 059669">'Stuttgart Precision Works',
    location: 059669">'Stuttgart, Baden-Württemberg',
    country: 059669">'Germany',
    manager: 059669">'Hans Mueller',
    established: 059669">'1985-07-01',
    departments: [],
  },
  {
    id: 059669">'fac-003',
    name: 059669">'Osaka Electronics Facility',
    location: 059669">'Osaka, Kansai',
    country: 059669">'Japan',
    manager: 059669">'Takeshi Yamamoto',
    established: 059669">'2005-11-20',
    departments: [],
  },
];

600">export 600">const departments: Department[] = [
  { id: 059669">'dep-001', name: 059669">'Engine Assembly', factory_id: 059669">'fac-001', production_lines:
[059669">'Line A', 059669">'Line B', 059669">'Line C'] },
  { id: 059669">'dep-002', name: 059669">'Body Welding', factory_id: 059669">'fac-001', production_lines:
[059669">'Line A', 059669">'Line B'] },
  { id: 059669">'dep-003', name: 059669">'Paint Shop', factory_id: 059669">'fac-001', production_lines:
[059669">'Line A'] },
  { id: 059669">'dep-004', name: 059669">'Final Assembly', factory_id: 059669">'fac-001', production_lines:
[059669">'Line A', 059669">'Line B'] },
  { id: 059669">'dep-005', name: 059669">'CNC Machining', factory_id: 059669">'fac-002', production_lines:
[059669">'Line A', 059669">'Line B', 059669">'Line C'] },
  { id: 059669">'dep-006', name: 059669">'Stamping', factory_id: 059669">'fac-002', production_lines: [059669">'Line
A'] },
  { id: 059669">'dep-007', name: 059669">'Quality Control', factory_id: 059669">'fac-002', production_lines:
[059669">'Line A', 059669">'Line B'] },
  { id: 059669">'dep-008', name: 059669">'PCB Assembly', factory_id: 059669">'fac-003', production_lines:
[059669">'Line A', 059669">'Line B'] },
  { id: 059669">'dep-009', name: 059669">'Surface Mount', factory_id: 059669">'fac-003', production_lines:
[059669">'Line A'] },
  { id: 059669">'dep-010', name: 059669">'Testing & Packaging', factory_id: 059669">'fac-003', production_lines:
[059669">'Line A', 059669">'Line B'] },
];

factories.forEach((f) => {
  f.departments = departments.filter((d) => d.factory_id === f.id);
});
```

```
059669">"color:#6b7280;font-style:italic">// — Users
```

```
600">export 600">const users: User[] = [
  { id: 059669">'usr-001', name: 059669">'Sarah Mitchell', email: 059669">'sarah.mitchell@smartmaint.com', role:
059669">'super_admin', role_name: 059669">'Super Admin', avatar_color: 059669">'#1f5cf5', last_active: daysAgo(0) },
  { id: 059669">'usr-002', name: 059669">'Robert Chen', email: 059669">'robert.chen@smartmaint.com', role:
059669">'factory_manager', role_name: 059669">'Factory Manager', avatar_color: 059669">'#059669', factory_id:
059669">'fac-001', last_active: daysAgo(0) },
  { id: 059669">'usr-003', name: 059669">'Hans Mueller', email: 059669">'hans.mueller@smartmaint.com', role:
059669">'factory_manager', role_name: 059669">'Factory Manager', avatar_color: 059669">'#d97706', factory_id:
059669">'fac-002', last_active: daysAgo(1) },
  { id: 059669">'usr-004', name: 059669">'James Patterson', email: 059669">'james.patterson@smartmaint.com', role:
```

```

059669">'maintenance_manager', role_name: 059669">'Maintenance Manager', avatar_color: 059669">'#dc2626', factory_id:
059669">'fac-001', last_active: daysAgo(0) },
  { id: 059669">'usr-005', name: 059669">'Maria Garcia', email: 059669">'maria.garcia@smartmaint.com', role:
059669">'maintenance_manager', role_name: 059669">'Maintenance Manager', avatar_color: 059669">'#7c3aed', factory_id:
059669">'fac-002', last_active: daysAgo(0) },
  { id: 059669">'usr-006', name: 059669">'David Kim', email: 059669">'david.kim@smartmaint.com', role:
059669">'maintenance_engineer', role_name: 059669">'Maintenance Engineer', avatar_color: 059669">'#0891b2',
factory_id: 059669">'fac-001', last_active: daysAgo(0) },
  { id: 059669">'usr-007', name: 059669">'Lisa Anderson', email: 059669">'lisa.anderson@smartmaint.com', role:
059669">'maintenance_engineer', role_name: 059669">'Maintenance Engineer', avatar_color: 059669">'#be185d',
factory_id: 059669">'fac-001', last_active: daysAgo(2) },
  { id: 059669">'usr-008', name: 059669">'Klaus Weber', email: 059669">'klaus.weber@smartmaint.com', role:
059669">'maintenance_engineer', role_name: 059669">'Maintenance Engineer', avatar_color: 059669">'#0369a1',
factory_id: 059669">'fac-002', last_active: daysAgo(0) },
  { id: 059669">'usr-009', name: 059669">'Yuki Tanaka', email: 059669">'yuki.tanaka@smartmaint.com', role:
059669">'maintenance_engineer', role_name: 059669">'Maintenance Engineer', avatar_color: 059669">'#15803d',
factory_id: 059669">'fac-003', last_active: daysAgo(1) },
  { id: 059669">'usr-010', name: 059669">'Michael Brown', email: 059669">'michael.brown@smartmaint.com', role:
059669">'inventory_manager', role_name: 059669">'Inventory Manager', avatar_color: 059669">'#9333ea', factory_id:
059669">'fac-001', last_active: daysAgo(0) },
  { id: 059669">'usr-011', name: 059669">'Emma Schmidt', email: 059669">'emma.schmidt@smartmaint.com', role:
059669">'inventory_manager', role_name: 059669">'Inventory Manager', avatar_color: 059669">'#e11d48', factory_id:
059669">'fac-002', last_active: daysAgo(0) },
  { id: 059669">'usr-012', name: 059669">'Daniel Foster', email: 059669">'daniel.foster@smartmaint.com', role:
059669">'procurement_officer', role_name: 059669">'Procurement Officer', avatar_color: 059669">'#0d9488',
last_active: daysAgo(0) },
  { id: 059669">'usr-013', name: 059669">'Jennifer Walsh', email: 059669">'jennifer.walsh@smartmaint.com', role:
059669">'procurement_officer', role_name: 059669">'Procurement Officer', avatar_color: 059669">'#ca8a04',
last_active: daysAgo(1) },
  { id: 059669">'usr-014', name: 059669">'Bosch Industrial', email: 059669">'sales@bosch-industrial.com', role:
059669">'supplier', role_name: 059669">'Supplier', avatar_color: 059669">'#475569', last_active: daysAgo(3) },
  { id: 059669">'usr-015', name: 059669">'SKF Bearings', email: 059669">'orders@skf.com', role: 059669">'supplier',
role_name: 059669">'Supplier', avatar_color: 059669">'#64748b', last_active: daysAgo(5) },
];

```

059669">"color:#6b7280;font-style:italic">// — Machine Components

```

600">const componentTemplates: Omit<MachineComponent, 059669">'id'>[] = [
  { name: 059669">'Main Drive Motor', part_number: 059669">'MTR-2200', category: 059669">'electrical',
current_status: 059669">'ok', expected_life_hours: 20000 },
  { name: 059669">'Hydraulic Pump', part_number: 059669">'HYP-1500', category: 059669">'hydraulic', current_status:
059669">'ok', expected_life_hours: 15000 },
  { name: 059669">'Spindle Bearing Set', part_number: 059669">'BRG-008', category: 059669">'mechanical',
current_status: 059669">'warning', expected_life_hours: 12000 },
  { name: 059669">'Servo Controller', part_number: 059669">'SVC-300', category: 059669">'electronic', current_status:
059669">'ok', expected_life_hours: 25000 },
  { name: 059669">'Pneumatic Cylinder', part_number: 059669">'PNC-450', category: 059669">'pneumatic',
current_status: 059669">'ok', expected_life_hours: 18000 },
  { name: 059669">'Coolant Pump', part_number: 059669">'CLP-200', category: 059669">'mechanical', current_status:
059669">'critical', expected_life_hours: 8000 },
  { name: 059669">'PLC Module', part_number: 059669">'PLC-900', category: 059669">'electronic', current_status:
059669">'ok', expected_life_hours: 30000 },
  { name: 059669">'Gearbox Assembly', part_number: 059669">'GBX-500', category: 059669">'mechanical', current_status:
059669">'ok', expected_life_hours: 22000 },
  { name: 059669">'Linear Guide Rail', part_number: 059669">'LGR-120', category: 059669">'mechanical',
current_status: 059669">'warning', expected_life_hours: 16000 },
  { name: 059669">'Safety Light Curtain', part_number: 059669">'SLC-100', category: 059669">'safety', current_status:
059669">'ok', expected_life_hours: 50000 },
  { name: 059669">'VFD Drive', part_number: 059669">'VFD-750', category: 059669">'electrical', current_status:
059669">'ok', expected_life_hours: 20000 },
  { name: 059669">'Lubrication Unit', part_number: 059669">'LUB-050', category: 059669">'lubricant', current_status:
059669">'ok', expected_life_hours: 10000 },
];

```

```

600">function genComponents(machineIdx: number): MachineComponent[] {
  600">const count = randInt(5, 9);
  600">const selected = pickN(componentTemplates, count);
  600">return selected.map((c, i) => ({
    ...c,
    id: 059669">`cmp-${machineIdx}-${i}`,
    last_replaced: daysAgo(randInt(30, 800)),
    current_status: rand() > 0.75 ? (rand() > 0.5 ? 059669">'warning' : 059669">'critical') : 059669">'ok',
  }));
}

```

```

}

059669">"color:#6b7280;font-style:italic">// — Machines

600">const machineTypes = [059669">'CNC Milling Center', 059669">'Robotic Welder', 059669">'Conveyor System',
059669">'Hydraulic Press', 059669">'Injection Molder', 059669">'Laser Cutter', 059669">'Assembly Robot',
059669">'Stamping Press', 059669">'Paint Robot', 059669">'Testing Station'];
600">const manufacturers = [059669">'Siemens', 059669">'Bosch Rexroth', 059669">'Mitsubishi Electric',
059669">'Fanuc', 059669">'ABB', 059669">'Kuka', 059669">'DMG Mori', 059669">'Trumpf', 059669">'Yaskawa',
059669">'Okuma'];
600">const machineNames: Record<string, string[]> = {
  059669">'CNC Milling Center': [059669">'DMC-630 V', 059669">'VMC-1000', 059669">'NXV-700A'],
  059669">'Robotic Welder': [059669">'KR-210 R2700', 059669">'ARC-1500', 059669">'MA-1440'],
  059669">'Conveyor System': [059669">'CVX-2000', 059669">'BeltPro-500', 059669">'RollerCon-300'],
  059669">'Hydraulic Press': [059669">'HPS-400', 059669">'PressMax-800', 059669">'HPU-250'],
  059669">'Injection Molder': [059669">'IM-300T', 059669">'EngelVictory-200', 059669">'Arburg-470'],
  059669">'Laser Cutter': [059669">'TruLaser-3030', 059669">'LC-3015', 059669">'Bystronic-3015'],
  059669">'Assembly Robot': [059669">'IRB-6700', 059669">'M-710ic', 059669">'RS-010N'],
  059669">'Stamping Press': [059669">'SP-600', 059669">'StampPro-400', 059669">'SPS-800'],
  059669">'Paint Robot': [059669">'PR-200', 059669">'PaintMate-10', 059669">'EPX-1250'],
  059669">'Testing Station': [059669">'TS-500', 059669">'TestPro-200', 059669">'TS-1000'],
};

600">const statuses: MachineStatus[] = [059669">'operational', 059669">'operational', 059669">'operational',
059669">'operational', 059669">'maintenance', 059669">'breakdown', 059669">'idle'];
600">const criticalities: CriticalityLevel[] = [059669">'critical', 059669">'high', 059669">'medium', 059669">'low'];

600">export 600">const machines: Machine[] = [];
600">let machineCounter = 0;
factories.forEach((factory) => {
  600">const facDepts = departments.filter((d) => d.factory_id === factory.id);
  facDepts.forEach((dept) => {
    600">const machineCount = randInt(3, 6);
    600">for (600">let i = 0; i < machineCount; i++) {
      machineCounter++;
      600">const 600">type = pick(machineTypes);
      600">const manufacturer = pick(manufacturers);
      600">const model = pick(machineNames[600">type] || [600">type.split(059669">' ')[0] + 059669">'-' + 100]);
      600">const status = pick(statuses);
      600">const operating_hours = randInt(2000, 45000);
      600">const total_downtime_hours = randInt(8, 520);
      600">const failure_count = randInt(0, 6);
      600">const mtbf = failure_count > 0 ? Math.round(operating_hours / failure_count) : randInt(8000, 20000);
      600">const mttr = randInt(2, 48);
      600">const health_score = status === 059669">'breakdown' ? randInt(15, 45) : status === 059669">'maintenance' ?
randInt(50, 75) : randInt(65, 98);
      600">const risk_level = health_score >= 80 ? 059669">'low' : health_score >= 60 ? 059669">'medium' :
health_score >= 40 ? 059669">'high' : 059669">'critical';

      machines.push({
        id: 059669">'mch-${String(machineCounter).padStart(3, '0')}',
        machine_id: 059669">'M-${String(machineCounter).padStart(4, '0')}',
        name: 059669">'${model} #${machineCounter}',
        factory_id: factory.id,
        factory_name: factory.name,
        department_id: dept.id,
        department_name: dept.name,
        production_line: pick(dept.production_lines),
        600">type,
        manufacturer,
        model,
        serial_number: 059669">'`SN${randInt(100000, 999999)}'`,
        installation_date: daysAgo(randInt(180, 3600)),
        criticality: pick(criticalities),
        status,
        operating_hours,
        total_downtime_hours,
        health_score,
        risk_level,
        last_maintenance: daysAgo(randInt(5, 120)),
        next_maintenance: daysFromNow(randInt(-10, 30)),
        components: genComponents(machineCounter),
        mtbf_hours: mtbf,
      });
    }
  });
});

```

```
        mtrr_hours: mtrr,
        failure_count_30d: randInt(0, 3),
    });
}
});
});
```

059669">"color:#6b7280;font-style:italic">// — Warehouses

```
600">export 600">const warehouses: Warehouse[] = [
    { id: 059669">'wh-001', name: 059669">'Central Warehouse Detroit', location: 059669">'Detroit, MI', factory_id:
059669">'fac-001', capacity: 50000, utilized: 38500, manager: 059669">'Michael Brown' },
    { id: 059669">'wh-002', name: 059669">'Stuttgart Parts Depot', location: 059669">'Stuttgart, DE', factory_id:
059669">'fac-002', capacity: 40000, utilized: 31200, manager: 059669">'Emma Schmidt' },
    { id: 059669">'wh-003', name: 059669">'Osaka Component Store', location: 059669">'Osaka, JP', factory_id:
059669">'fac-003', capacity: 30000, utilized: 22800, manager: 059669">'Yuki Tanaka' },
    { id: 059669">'wh-004', name: 059669">'Overflow Storage Detroit', location: 059669">'Detroit, MI', factory_id:
059669">'fac-001', capacity: 20000, utilized: 8200, manager: 059669">'Michael Brown' },
];
```

059669">"color:#6b7280;font-style:italic">// — Suppliers

```
600">export 600">const suppliers: Supplier[] = [
    { id: 059669">'sup-001', name: 059669">'Bosch Industrial Supply', contact_person: 059669">'Markus Bauer', email:
059669">'sales@bosch-industrial.com', phone: 059669">'+49 711 400 0', address: 059669">'Stuttgart, Germany', rating:
4.8, status: 059669">'preferred', lead_time_days: 7, on_time_delivery_rate: 96, quality_rate: 99, parts_supplied:
142, total_orders: 318, total_value: 2840000, established: 059669">'2010-01-15' },
    { id: 059669">'sup-002', name: 059669">'SKF Bearing Co.', contact_person: 059669">'Erik Lindqvist', email:
059669">'orders@skf.com', phone: 059669">'+46 31 337 0', address: 059669">'Gothenburg, Sweden', rating: 4.6, status:
059669">'active', lead_time_days: 10, on_time_delivery_rate: 92, quality_rate: 97, parts_supplied: 89, total_orders:
215, total_value: 1450000, established: 059669">'2012-06-01' },
    { id: 059669">'sup-003', name: 059669">'Siemens Automation', contact_person: 059669">'Thomas Wagner', email:
059669">'parts@siemens-auto.com', phone: 059669">'+49 89 636 0', address: 059669">'Munich, Germany', rating: 4.9,
status: 059669">'preferred', lead_time_days: 5, on_time_delivery_rate: 98, quality_rate: 99, parts_supplied: 76,
total_orders: 402, total_value: 3650000, established: 059669">'2008-03-20' },
    { id: 059669">'sup-004', name: 059669">'Parker Hannifin', contact_person: 059669">'Jennifer Park', email:
059669">'hydraulics@parker.com', phone: 059669">'+1 216 896 3000', address: 059669">'Cleveland, OH, USA', rating:
4.3, status: 059669">'active', lead_time_days: 14, on_time_delivery_rate: 88, quality_rate: 95, parts_supplied: 54,
total_orders: 167, total_value: 890000, established: 059669">'2015-09-10' },
    { id: 059669">'sup-005', name: 059669">'SMC Pneumatics', contact_person: 059669">'Hiroshi Sato', email:
059669">'orders@smc-pneumatics.com', phone: 059669">'+81 3 5207 8100', address: 059669">'Tokyo, Japan', rating: 4.5,
status: 059669">'active', lead_time_days: 12, on_time_delivery_rate: 90, quality_rate: 96, parts_supplied: 67,
total_orders: 198, total_value: 1120000, established: 059669">'2013-04-05' },
    { id: 059669">'sup-006', name: 059669">'Omron Electronics', contact_person: 059669">'Chen Wei', email:
059669">'supply@omron-electronics.com', phone: 059669">'+86 755 8373 8888', address: 059669">'Shenzhen, China',
rating: 3.9, status: 059669">'active', lead_time_days: 21, on_time_delivery_rate: 82, quality_rate: 91,
parts_supplied: 43, total_orders: 134, total_value: 560000, established: 059669">'2017-11-12' },
    { id: 059669">'sup-007', name: 059669">'ABB Robotics Parts', contact_person: 059669">'Anna Bergstrom', email:
059669">'parts@abb-robotics.com', phone: 059669">'+41 43 317 7111', address: 059669">'Zurich, Switzerland', rating:
4.7, status: 059669">'preferred', lead_time_days: 8, on_time_delivery_rate: 95, quality_rate: 98, parts_supplied: 38,
total_orders: 156, total_value: 1980000, established: 059669">'2011-02-18' },
    { id: 059669">'sup-008', name: 059669">'DiscountParts Inc.', contact_person: 059669">'Tom Harris', email:
059669">'sales@discountparts.com', phone: 059669">'+1 312 555 0144', address: 059669">'Chicago, IL, USA', rating:
2.8, status: 059669">'blacklisted', lead_time_days: 30, on_time_delivery_rate: 65, quality_rate: 78, parts_supplied:
12, total_orders: 45, total_value: 89000, established: 059669">'2019-08-01' },
];
```

059669">"color:#6b7280;font-style:italic">// — Spare Parts

```
600">const partCategories: PartCategory[] = [059669">'mechanical', 059669">'electrical', 059669">'hydraulic',
059669">'pneumatic', 059669">'electronic', 059669">'consumable', 059669">'lubricant', 059669">'safety'];
600">const partNames: Record<PartCategory, string[]> = {
    mechanical: [059669">'Precision Bearing Set', 059669">'Spindle Assembly', 059669">'Gearbox Unit', 059669">'Linear
Guide Rail', 059669">'Coupling Set', 059669">'Drive Shaft', 059669">'Ball Screw', 059669">'Linear Actuator'],
    electrical: [059669">'Main Drive Motor 15kW', 059669">'Servo Motor 7.5kW', 059669">'VFD Drive 22kW', 059669">'Power
Supply Unit 24V', 059669">'Contactor 40A', 059669">'Circuit Breaker 63A', 059669">'Motor Starter',
059669">'Transformer 5kVA'],
    hydraulic: [059669">'Hydraulic Pump A10VSO', 059669">'Hydraulic Valve 4WE6', 059669">'Hydraulic Cylinder 100mm',
059669">'Pressure Relief Valve', 059669">'Hydraulic Hose 1/2"', 059669">'Accumulator 10L', 059669">'Flow Control
Valve', 059669">'Check Valve'],
    pneumatic: [059669">'Pneumatic Cylinder 50mm', 059669">'Solenoid Valve 1/4"', 059669">'Air Filter Regulator',
059669">'Pneumatic Fitting Set', 059669">'Silencer M5', 059669">'Pneumatic Hose 8mm', 059669">'Pressure Switch',
059669">'Quick Connect Coupling'],
```



```

    electronic: [059669">'PLC CPU Module', 059669">'Servo Controller 3kW', 059669">'HMI Touch Panel 7"', 059669">'I/O
Module 16DI', 059669">'Communication Module', 059669">'Encoder Cable 5m', 059669">'Safety Relay', 059669">'Signal
Converter']],
    consumable: [059669">'Cutting Tool Set HSS', 059669">'Welding Wire 1.2mm', 059669">'Lubricant Grease 1kg',
059669">'Sandpaper Belt 100grit', 059669">'Coolant Concentrate 20L', 059669">'Cleaning Solvent 5L', 059669">'Adhesive
50ml', 059669">'Protective Film'],
    lubricant: [059669">'Hydraulic Oil ISO 46 20L', 059669">'Gear Oil 220 20L', 059669">'Way Lube 68 5L',
059669">'Grease NLGI-2 1kg', 059669">'Cutting Fluid 10L', 059669">'Compressor Oil 20L', 059669">'Chain Lube 5L',
059669">'Anti-seize Compound'],
    safety: [059669">'Safety Light Curtain 300mm', 059669">'Emergency Stop Button', 059669">'Safety Door Switch',
059669">'Pressure Mat 600x400', 059669">'Safety Relay Module', 059669">'Warning Light Amber', 059669">'Lockout Tagout
Kit', 059669">'Guard Interlock'],
  };

```

```

600">export 600">const spareParts: SparePart[] = [];
600">let partCounter = 0;
suppliers.forEach((supplier) => {
  600">if (supplier.status === 059669">'blacklisted') 600">return;
  600">const partCount = randInt(8, 16);
  600">for (600">let i = 0; i < partCount; i++) {
    partCounter++;
    600">const category = pick(partCategories);
    600">const name = pick(partNames[category]);
    600">const min_stock = randInt(5, 30);
    600">const max_stock = min_stock * randInt(3, 6);
    600">const reorder_point = min_stock + randInt(5, 15);
    600">const current_stock = randInt(0, max_stock);
    600">const reserved_stock = randInt(0, Math.floor(current_stock * 0.3));
    600">const warehouse = pick(warehouses);
    600">const movement = rand() > 0.65 ? 059669">'fast' : rand() > 0.4 ? 059669">'medium' : 059669">'slow';
    600">const monthlyConsumption = Array.600">from({ length: 12 }, () => randInt(0, movement === 059669">'fast' ? 80
: movement === 059669">'medium' ? 30 : 10));
    600">const stock_status =
      current_stock === 0 ? 059669">'out_of_stock' :
      current_stock <= min_stock ? 059669">'critical' :
      current_stock <= reorder_point ? 059669">'low_stock' :
      current_stock > max_stock * 1.2 ? 059669">'overstock' : 059669">'in_stock';

    spareParts.push({
      id: 059669">`prt-${String(partCounter).padStart(3, '0')}`,
      part_number: 059669">`${category.substring(0, 3).toUpperCase()}-${String(partCounter).padStart(4, '0')}`,
      name,
      category,
      description: 059669">`${name} – compatible with industrial machinery. OEM-grade quality.`,
      compatible_machines: pickN(machines, randInt(2, 6)).map((m) => m.machine_id),
      current_stock,
      reserved_stock,
      min_stock,
      max_stock,
      reorder_point,
      unit_cost: randFloat(5, 3500, 2),
      warehouse_id: warehouse.id,
      warehouse_name: warehouse.name,
      supplier_id: supplier.id,
      supplier_name: supplier.name,
      lead_time_days: supplier.lead_time_days,
      last_restocked: daysAgo(randInt(1, 60)),
      batch_number: 059669">`B${randInt(20240000, 20249999)}`,
      movement_class: movement,
      monthly_consumption: monthlyConsumption,
      stock_status,
    });
  }
});

```

059669">"color:#6b7280;font-style:italic">/// — Maintenance Schedules

```

600">const maintenanceTypes: MaintenanceType[] = [059669">'preventive', 059669">'preventive', 059669">'preventive',
059669">'corrective', 059669">'predictive', 059669">'inspection'];
600">const priorities: MaintenancePriority[] = [059669">'low', 059669">'medium', 059669">'high', 059669">'urgent'];
600">const maintenanceStatuses: MaintenanceStatus[] = [059669">'scheduled', 059669">'scheduled',
059669">'in_progress', 059669">'completed', 059669">'completed', 059669">'overdue'];
600">const technicians = users.filter((u) => u.role === 059669">'maintenance_engineer');

```

```

600">const checklistTemplates = [
  [059669">'Inspect drive belts 600">'for wear', 059669">'Check oil levels', 059669">'Test emergency stops',
059669">'Calibrate sensors', 059669">'Clean filters'],
  [059669">'Lubricate all moving parts', 059669">'Check hydraulic pressure', 059669">'Inspect electrical
connections', 059669">'Test safety circuits', 059669">'Verify alignment'],
  [059669">'Check bearing temperature', 059669">'Inspect coolant system', 059669">'Test PLC programs', 059669">'Clean
optical sensors', 059669">'Check pneumatic seals'],
  [059669">'Verify spindle runout', 059669">'Inspect way covers', 059669">'Check lubrication system', 059669">'Test
axis movement', 059669">'Inspect cable carriers'],
];

600">export 600">const maintenanceSchedules: MaintenanceSchedule[] = [];
machines.forEach((machine, idx) => {
  600">const scheduleCount = randInt(2, 5);
  600">for (600">let i = 0; i < scheduleCount; i++) {
    600">const 600">type = pick(maintenanceTypes);
    600">const status = pick(maintenanceStatuses);
    600">const tech = pick(technicians);
    600">const offset = randInt(-25, 35);
    600">const scheduledDate = daysFromNow(offset);
    600">const isCompleted = status === 059669">'completed';
    600">const checklistItems = pick(checklistTemplates);
    600">const estHours = randFloat(2, 16, 1);

    maintenanceSchedules.push({
      id: 059669">`sch-${String(idx * 5 + i + 1).padStart(4, '0')}`,
      machine_id: machine.id,
      machine_name: machine.name,
      machine_id_code: machine.machine_id,
      600">type,
      priority: pick(priorities),
      status,
      scheduled_date: scheduledDate,
      completed_date: isCompleted ? daysAgo(randInt(1, 30)) : 600">undefined,
      assigned_technician: tech.name,
      technician_id: tech.id,
      description: 059669">`${600">type.charAt(0).toUpperCase() + 600">type.slice(1)} maintenance 600">for
${machine.name} - ${machine.600">type}`,
      checklist: checklistItems.map((task, ci) => ({ id: 059669">`chk-${idx}-${i}-${ci}`, task, completed:
isCompleted || rand() > 0.5 })),
      estimated_hours: estHours,
      actual_hours: isCompleted ? randFloat(estHours * 0.8, estHours * 1.3, 1) : 600">undefined,
      cost: randFloat(200, 8000, 2),
      recurring: 600">type === 059669">'preventive',
      interval_days: 600">type === 059669">'preventive' ? pick([7, 14, 30, 60, 90, 180]) : 0,
      factory_id: machine.factory_id,
      factory_name: machine.factory_name,
    });
  }
});

```

059669">"color:#6b7280;font-style:italic">/// — Breakdowns

```

600">const breakdownStatuses: BreakdownStatus[] = [059669">'reported', 059669">'diagnosing', 059669">'in_repair',
059669">'resolved', 059669">'resolved', 059669">'closed'];
600">const severities: BreakdownSeverity[] = [059669">'minor', 059669">'major', 059669">'severe',
059669">'critical'];
600">const symptomsList = [
  059669">'Unusual vibration detected on spindle axis',
  059669">'Hydraulic pressure drop below threshold',
  059669">'Motor overheating – temperature exceeded 85°C',
  059669">'PLC communication error on axis 3',
  059669">'Pneumatic cylinder failed to extend',
  059669">'Bearing noise increasing – abnormal sound',
  059669">'Coolant leak detected under machine base',
  059669">'Servo drive fault code E-7301',
  059669">'Encoder signal loss on rotary axis',
  059669">'Emergency stop circuit intermittent failure',
];
600">const rootCauses = [
  059669">'Bearing wear due to insufficient lubrication',
  059669">'Hydraulic seal degradation 600">from age',
  059669">'Motor winding insulation breakdown',

```



```

059669">'PLC firmware bug 600">in motion control module',
059669">'Pneumatic seal failure 600">from contamination',
059669">'Bearing race spalling – end 600">of service life',
059669">'Coolant pump impeller erosion',
059669">'Servo drive power stage failure',
059669">'Encoder cable chafing at cable carrier',
059669">'Safety relay contact oxidation',
];
600">const correctiveActions = [
059669">'Replaced bearing set and re-lubricated assembly',
059669">'Replaced hydraulic seals and flushed system',
059669">'Rewound motor and replaced insulation',
059669">'Updated PLC firmware to v4.2.1',
059669">'Replaced pneumatic cylinder and installed filter',
059669">'Replaced bearing set and adjusted preload',
059669">'Replaced coolant pump and sealed connections',
059669">'Replaced servo drive unit',
059669">'Replaced encoder cable and secured routing',
059669">'Replaced safety relay and cleaned contacts',
];

600">export 600">const breakdowns: Breakdown[] = [];
machines.forEach((machine, idx) => {
600">const breakdownCount = randInt(0, 4);
600">for (600">let i = 0; i < breakdownCount; i++) {
600">const status = pick(breakdownStatuses);
600">const isResolved = status === 059669">'resolved' || status === 059669">'closed';
600">const symptomIdx = randInt(0, symptomsList.length - 1);
600">const partsReplaced: WorkOrderPart[] = rand() > 0.4 ? pickN(spareParts, randInt(1, 3)).map((p) => ({
part_id: p.id, part_name: p.name, part_number: p.part_number, quantity: randInt(1, 4), unit_cost: p.unit_cost,
})) : [];

breakdowns.push({
id: 059669">'brk-${String(idx * 4 + i + 1).padStart(4, '0')}`,
breakdown_id: 059669">'BD-${String(idx * 4 + i + 1).padStart(5, '0')}`,
machine_id: machine.id,
machine_name: machine.name,
machine_id_code: machine.machine_id,
factory_id: machine.factory_id,
factory_name: machine.factory_name,
reported_date: daysAgo(randInt(1, 180)),
reported_by: pick(users.filter((u) => u.role === 059669">'maintenance_engineer' || u.role ===
059669">'maintenance_manager')).name,
assigned_technician: pick(technicians).name,
status,
severity: pick(severities),
symptoms: symptomsList[symptomIdx],
root_cause: isResolved ? rootCauses[symptomIdx] : 600">undefined,
corrective_action: isResolved ? correctiveActions[symptomIdx] : 600">undefined,
downtime_hours: randFloat(1, 72, 1),
parts_replaced: partsReplaced,
repair_cost: randFloat(200, 15000, 2),
resolved_date: isResolved ? daysAgo(randInt(0, 30)) : 600">undefined,
priority: pick(priorities),
});
}
});

059669">"color:#6b7280;font-style:italic">{// — Inventory Transactions

600">const transactionTypes: TransactionType[] = [059669">'stock_in', 059669">'stock_out', 059669">'stock_out',
059669">'reserved', 059669">'returned', 059669">'adjusted'];
600">export 600">const inventoryTransactions: InventoryTransaction[] = [];
spareParts.forEach((part, idx) => {
600">const txCount = randInt(5, 15);
600">for (600">let i = 0; i < txCount; i++) {
600">const 600">type = pick(transactionTypes);
600">const qty = randInt(1, 50);
inventoryTransactions.push({
id: 059669">'txn-${String(idx * 15 + i + 1).padStart(5, '0')}`,
part_id: part.id,
part_name: part.name,
part_number: part.part_number,

```

```

        600">type,
        quantity: qty,
        balance_after: Math.max(0, part.current_stock + randInt(-20, 20)),
        reference: 600">type === 059669">'stock_out' ? 059669">'W0-${randInt(10000, 19999)}\` : 600">type ===
059669">'stock_in' ? 059669">'PO-${randInt(20000, 29999)}\` : 059669">'ADJ-${randInt(30000, 39999)}\`,
        reference_type: 600">type === 059669">'stock_out' ? 059669">'work_order' : 600">type === 059669">'stock_in' ?
059669">'purchase_order' : 600">type === 059669">'returned' ? 059669">'600">return' : 059669">'adjustment',
        performed_by: pick(users.filter((u) => u.role === 059669">'inventory_manager' || u.role ===
059669">'maintenance_engineer')).name,
        warehouse_id: part.warehouse_id,
        timestamp: daysAgo(randInt(0, 90)) + 059669">'T${String(randInt(8, 17)).padStart(2, '0')}:${String(randInt(0,
59)).padStart(2, '0')}:00Z',
        notes: 600">type === 059669">'adjusted' ? 059669">'Cycle count adjustment' : 600">undefined,
    });
}
});

059669">"color:#6b7280;font-style:italic">// — Purchase Requests & Orders

600">const requestStatuses: PurchaseRequestStatus[] = [059669">'draft', 059669">'pending_approval',
059669">'pending_approval', 059669">'approved', 059669">'approved', 059669">'rejected', 059669">'ordered',
059669">'received', 059669">'cancelled'];
600">const orderStatuses: PurchaseOrderStatus[] = [059669">'draft', 059669">'sent', 059669">'confirmed',
059669">'confirmed', 059669">'partial', 059669">'received', 059669">'received', 059669">'cancelled'];

600">export 600">const purchaseRequests: PurchaseRequest[] = [];
600">export 600">const purchaseOrders: PurchaseOrder[] = [];
600">for (600">let i = 0; i < 25; i++) {
    600">const items = pickN(spareParts, randInt(1, 4)).map((p) => ({
        part_id: p.id, part_name: p.name, part_number: p.part_number, quantity: randInt(5, 50), unit_cost: p.unit_cost,
total: 0,
    }));
    items.forEach((it) => (it.total = it.quantity * it.unit_cost));
    600">const totalValue = items.reduce((s, it) => s + it.total, 0);
    600">const status = pick(requestStatuses);
    purchaseRequests.push({
        id: 059669">'prq-${String(i + 1).padStart(3, '0')}\`,
        request_number: 059669">'PR-${String(i + 1001).padStart(5, '0')}\`,
        requested_by: pick(users.filter((u) => u.role === 059669">'procurement_officer' || u.role ===
059669">'inventory_manager')).name,
        department: pick(departments).name,
        status,
        priority: pick(priorities),
        items,
        total_value: parseFloat(totalValue.toFixed(2)),
        created_date: daysAgo(randInt(1, 60)),
        required_date: daysFromNow(randInt(5, 45)),
        approved_by: status === 059669">'approved' || status === 059669">'ordered' || status === 059669">'received' ?
059669">'Sarah Mitchell' : 600">undefined,
        justification: pick([059669">'Stock below reorder point', 059669">'Scheduled maintenance requirement',
059669">'Breakdown repair – urgent need', 059669">'New machine setup', 059669">'Routine restock']),
    });
}

600">for (600">let i = 0; i < 20; i++) {
    600">const supplier = pick(suppliers.filter((s) => s.status !== 059669">'blacklisted'));
    600">const items = pickN(spareParts, randInt(1, 5)).map((p) => ({
        part_id: p.id, part_name: p.name, part_number: p.part_number, quantity: randInt(10, 100), unit_cost: p.unit_cost,
total: 0,
    }));
    items.forEach((it) => (it.total = it.quantity * it.unit_cost));
    600">const totalValue = items.reduce((s, it) => s + it.total, 0);
    600">const status = pick(orderStatuses);
    600">const expectedDelivery = daysFromNow(randInt(-5, 30));
    purchaseOrders.push({
        id: 059669">'po-${String(i + 1).padStart(3, '0')}\`,
        order_number: 059669">'PO-${String(i + 2001).padStart(5, '0')}\`,
        supplier_id: supplier.id,
        supplier_name: supplier.name,
        status,
        items,
        total_value: parseFloat(totalValue.toFixed(2)),
        order_date: daysAgo(randInt(1, 45)),

```

```

    expected_delivery: expectedDelivery,
    actual_delivery: status === 059669">'received' ? daysAgo(randInt(0, 10)) : 600">undefined,
    created_by: pick(users.filter((u) => u.role === 059669">'procurement_officer')).name,
    delivery_progress: status === 059669">'received' ? 100 : status === 059669">'partial' ? randInt(30, 70) : status
    === 059669">'confirmed' ? 15 : status === 059669">'sent' ? 5 : 0,
  });
}

059669">"color:#6b7280;font-style:italic">// — Warranties

600">const warrantyProviders = [059669">'Siemens Service', 059669">'Bosch Rexroth Care', 059669">'Mitsubishi Electric
Warranty', 059669">'ABB Service Pro', 059669">'Fanuc Maintenance Plus', 059669">'Kuka Care Plus'];
600">const warrantyCoverages = [059669">'Full parts and labor', 059669">'Parts only', 059669">'Parts and labor with
on-site support', 059669">'Extended coverage with preventive maintenance'];
600">export 600">const warranties: Warranty[] = [];
machines.forEach((machine, idx) => {
  600">if (rand() > 0.3) {
    600">const startOffset = randInt(-1000, -100);
    600">const durationDays = randInt(365, 2555);
    600">const start = daysAgo(-startOffset);
    600">const end = daysFromNow(startOffset + durationDays);
    600">const daysRemaining = Math.ceil((600">new Date(end).getTime() - today.getTime()) / 86400000);
    600">const status: WarrantyStatus = daysRemaining < 0 ? 059669">'expired' : daysRemaining < 60 ?
059669">'expiring_soon' : 059669">'active';

    warranties.push({
      id: 059669">`war-${String(idx + 1).padStart(3, '0')}`,
      warranty_number: 059669">`WR-${String(idx + 3001).padStart(5, '0')}`,
      machine_id: machine.id,
      machine_name: machine.name,
      machine_id_code: machine.machine_id,
      600">type: 059669">'machine',
      start_date: start,
      end_date: end,
      status,
      provider: pick(warrantyProviders),
      coverage: pick(warrantyCoverages),
      terms: 059669">'Standard manufacturer warranty terms and conditions apply. Covers manufacturing defects and
component failures under normal use.',
      days_remaining: daysRemaining,
    });
  }
  059669">"color:#6b7280;font-style:italic">// Component warranties
  machine.components.filter((c) => rand() > 0.6).forEach((comp, ci) => {
    600">const startOffset = randInt(-800, -50);
    600">const durationDays = randInt(180, 1095);
    600">const start = daysAgo(-startOffset);
    600">const end = daysFromNow(startOffset + durationDays);
    600">const daysRemaining = Math.ceil((600">new Date(end).getTime() - today.getTime()) / 86400000);
    600">const status: WarrantyStatus = daysRemaining < 0 ? 059669">'expired' : daysRemaining < 60 ?
059669">'expiring_soon' : 059669">'active';

    warranties.push({
      id: 059669">`warc-${String(idx).padStart(3, '0')}-${ci}`,
      warranty_number: 059669">`WRC-${String(idx * 10 + ci + 4001).padStart(5, '0')}`,
      machine_id: machine.id,
      machine_name: machine.name,
      machine_id_code: machine.machine_id,
      component_id: comp.id,
      component_name: comp.name,
      600">type: 059669">'component',
      start_date: start,
      end_date: end,
      status,
      provider: pick(warrantyProviders),
      coverage: pick(warrantyCoverages),
      terms: 059669">'Component-level warranty covering defects and premature failure.',
      days_remaining: daysRemaining,
    });
  });
});
});
});

059669">"color:#6b7280;font-style:italic">// — Service Records

```

```

600">export 600">const serviceRecords: ServiceRecord[] = [];
machines.forEach((machine, idx) => {
  600">const recordCount = randInt(3, 12);
  600">for (600">let i = 0; i < recordCount; i++) {
    600">const partsReplaced: WorkOrderPart[] = rand() > 0.5 ? pickN(spareParts, randInt(1, 3)).map((p) => ({
      part_id: p.id, part_name: p.name, part_number: p.part_number, quantity: randInt(1, 3), unit_cost: p.unit_cost,
    })) : [];
    600">const laborHours = randFloat(1, 24, 1);
    600">const laborCost = laborHours * randFloat(45, 95, 2);
    600">const partsCost = partsReplaced.reduce((s, p) => s + p.quantity * p.unit_cost, 0);
    600">const serviceDate = daysAgo(randInt(5, 365));
    600">const 600">type = pick(maintenanceTypes);

    serviceRecords.push({
      id: 059669">'srv-${String(idx * 12 + i + 1).padStart(5, '0')}',
      service_id: 059669">'SR-${String(idx * 12 + i + 5001).padStart(5, '0')}',
      machine_id: machine.id,
      machine_name: machine.name,
      machine_id_code: machine.machine_id,
      date: serviceDate,
      technician: pick(technicians).name,
      600">type,
      problem_description: pick(symptomsList),
      diagnosis: pick(rootCauses),
      parts_replaced: partsReplaced,
      labor_hours: laborHours,
      labor_cost: parseFloat(laborCost.toFixed(2)),
      parts_cost: parseFloat(partsCost.toFixed(2)),
      total_cost: parseFloat((laborCost + partsCost).toFixed(2)),
      downtime_hours: randFloat(0.5, 48, 1),
      root_cause: pick(rootCauses),
      corrective_action: pick(correctiveActions),
      attachments: rand() > 0.7 ? [059669">'service_report.pdf', 059669">'photos.zip'] : [],
      factory_id: machine.factory_id,
    });
  }
});

```

059669">"color:#6b7280;font-style:italic">// — Notifications

```

600">export 600">const notifications: AppNotification[] = [
  { id: 059669">'ntf-001', 600">type: 059669">'maintenance_overdue', priority: 059669">'critical', title:
    059669">'Overdue Maintenance Alert', message: 059669">'M-0003 (DMC-630 V #3) has overdue preventive maintenance – 5
    days past due', module: 059669">'Maintenance', timestamp: daysAgo(0) + 059669">'T08:30:00Z', read: 600">false,
    action_url: 059669">'./maintenance', target_roles: [059669">'super_admin', 059669">'factory_manager',
    059669">'maintenance_manager'] },
  { id: 059669">'ntf-002', 600">type: 059669">'critical_stock', priority: 059669">'critical', title:
    059669">'Critical Stock Shortage', message: 059669">'Spindle Bearing Set(MEC-0003) is at 2 units – below minimum
    600">of 10', module: 059669">'Inventory', timestamp: daysAgo(0) + 059669">'T07:15:00Z', read: 600">false, action_url:
    059669">'./inventory', target_roles: [059669">'super_admin', 059669">'inventory_manager',
    059669">'procurement_officer'] },
  { id: 059669">'ntf-003', 600">type: 059669">'breakdown', priority: 059669">'critical', title: 059669">'Machine
    Breakdown Reported', message: 059669">'M-0007 (KR-210 R2700 #7) reported with severe vibration – repair 600">in
    progress', module: 059669">'Breakdowns', timestamp: daysAgo(0) + 059669">'T06:45:00Z', read: 600">false, action_url:
    059669">'./breakdowns', target_roles: [059669">'super_admin', 059669">'factory_manager',
    059669">'maintenance_manager', 059669">'maintenance_engineer'] },
  { id: 059669">'ntf-004', 600">type: 059669">'warranty_expiry', priority: 059669">'warning', title:
    059669">'Warranty Expiring Soon', message: 059669">'M-0012 (HPS-400 #12) warranty expires 600">in 28 days', module:
    059669">'Warranty', timestamp: daysAgo(1) + 059669">'T14:20:00Z', read: 600">false, action_url: 059669">'./warranty',
    target_roles: [059669">'super_admin', 059669">'factory_manager', 059669">'maintenance_manager'] },
  { id: 059669">'ntf-005', 600">type: 059669">'maintenance_due', priority: 059669">'warning', title:
    059669">'Upcoming Maintenance Reminder', message: 059669">'5 machines have preventive maintenance scheduled within 7
    days', module: 059669">'Maintenance', timestamp: daysAgo(1) + 059669">'T09:00:00Z', read: 600">true, action_url:
    059669">'./maintenance', target_roles: [059669">'super_admin', 059669">'factory_manager',
    059669">'maintenance_manager', 059669">'maintenance_engineer'] },
  { id: 059669">'ntf-006', 600">type: 059669">'low_stock', priority: 059669">'warning', title: 059669">'Low Stock
    Alert', message: 059669">'Hydraulic Pump A10VSO(HYD-0012) is at 15 units – below reorder point 600">of 25', module:
    059669">'Inventory', timestamp: daysAgo(1) + 059669">'T11:30:00Z', read: 600">false, action_url:
    059669">'./inventory', target_roles: [059669">'super_admin', 059669">'inventory_manager',
    059669">'procurement_officer'] },
  { id: 059669">'ntf-007', 600">type: 059669">'purchase_approval', priority: 059669">'info', title: 059669">'Purchase
    Request Awaiting Approval', message: 059669">'PR-1003 600">for $24,500 requires manager approval', module:

```

```

059669">'Procurement', timestamp: daysAgo(2) + 059669">'T10:15:00Z', read: 600">false, action_url:
059669">'./suppliers', target_roles: [059669">'super_admin', 059669">'factory_manager', 059669">'procurement_officer']
},
{ id: 059669">'ntf-008', 600">type: 059669">'supplier_delay', priority: 059669">'warning', title: 059669">'Delayed
Supplier Delivery', message: 059669">'PO-2007 600">from Omron Electronics is 3 days past expected delivery', module:
059669">'Procurement', timestamp: daysAgo(2) + 059669">'T15:45:00Z', read: 600">true, action_url:
059669">'./suppliers', target_roles: [059669">'super_admin', 059669">'procurement_officer'] },
{ id: 059669">'ntf-009', 600">type: 059669">'critical_stock', priority: 059669">'critical', title: 059669">'Stock-
Out Risk Predicted', message: 059669">'Forecast predicts Main Drive Motor 15kw(ELE-0008) will stock out 600">in 12
days', module: 059669">'Forecasting', timestamp: daysAgo(3) + 059669">'T08:00:00Z', read: 600">true, action_url:
059669">'./forecasting', target_roles: [059669">'super_admin', 059669">'inventory_manager',
059669">'procurement_officer'] },
{ id: 059669">'ntf-010', 600">type: 059669">'system', priority: 059669">'info', title: 059669">'System Update',
message: 059669">'Predictive maintenance engine recalculated health scores 600">for all machines', module:
059669">'System', timestamp: daysAgo(3) + 059669">'T03:00:00Z', read: 600">true, target_roles:
[059669">'super_admin', 059669">'factory_manager', 059669">'maintenance_manager'] },
];

```

059669">"color:#6b7280;font-style:italic">// — Audit Logs

```

600">const auditActions = [
  { action: 059669">'CREATE', module: 059669">'Machines' },
  { action: 059669">'UPDATE', module: 059669">'Machines' },
  { action: 059669">'CREATE', module: 059669">'Maintenance' },
  { action: 059669">'UPDATE', module: 059669">'Maintenance' },
  { action: 059669">'COMPLETE', module: 059669">'Maintenance' },
  { action: 059669">'CREATE', module: 059669">'Breakdowns' },
  { action: 059669">'UPDATE', module: 059669">'Breakdowns' },
  { action: 059669">'RESOLVE', module: 059669">'Breakdowns' },
  { action: 059669">'STOCK_OUT', module: 059669">'Inventory' },
  { action: 059669">'STOCK_IN', module: 059669">'Inventory' },
  { action: 059669">'ADJUST', module: 059669">'Inventory' },
  { action: 059669">'CREATE', module: 059669">'Procurement' },
  { action: 059669">'APPROVE', module: 059669">'Procurement' },
  { action: 059669">'REJECT', module: 059669">'Procurement' },
  { action: 059669">'CREATE', module: 059669">'Warranty' },
  { action: 059669">'CLAIM', module: 059669">'Warranty' },
  { action: 059669">'LOGIN', module: 059669">'Auth' },
  { action: 059669">'LOGOUT', module: 059669">'Auth' },
  { action: 059669">'UPDATE', module: 059669">'Settings' },
];
600">const ipAddresses = [059669">'192.168.1.10', 059669">'192.168.1.24', 059669">'10.0.0.45', 059669">'172.16.5.20',
059669">'192.168.2.33', 059669">'10.0.1.12'];
600">const devices = [059669">'Chrome / Windows 11', 059669">'Firefox / macOS 14', 059669">'Edge / Windows 11',
059669">'Safari / iOS 17', 059669">'Chrome / Android 14', 059669">'Chrome / macOS 14'];

600">export 600">const auditLogs: AuditLog[] = [];
600">for (600">let i = 0; i < 60; i++) {
  600">const user = pick(users);
  600">const auditAction = pick(auditActions);
  auditLogs.push({
    id: 059669">'`aud-${String(i + 1).padStart(4, '0')}`',
    user: user.name,
    user_role: user.role,
    action: auditAction.action,
    module: auditAction.module,
    entity_id: auditAction.module !== 059669">'Auth' && auditAction.module !== 059669">'Settings' ?
059669">'`ENT-${randInt(1000, 9999)}`' : 600">undefined,
    entity_type: auditAction.module !== 059669">'Auth' && auditAction.module !== 059669">'Settings' ?
auditAction.module.toLowerCase().replace(/s$/, 059669">'') : 600">undefined,
    previous_value: auditAction.action === 059669">'UPDATE' ? 059669">'Status: scheduled' : 600">undefined,
    updated_value: auditAction.action === 059669">'UPDATE' ? 059669">'Status: in_progress' : 600">undefined,
    timestamp: daysAgo(randInt(0, 14)) + 059669">'T${String(randInt(6, 20)).padStart(2, '0')}:${String(randInt(0,
59)).padStart(2, '0')}:${String(randInt(0, 59)).padStart(2, '0')}`',
    ip_address: pick(ipAddresses),
    device: pick(devices),
  });
}
auditLogs.sort((a, b) => b.timestamp.localeCompare(a.timestamp));

```

059669">"color:#6b7280;font-style:italic">// — KPI Computation

```

600">export 600">const kpiData: KPIData = {

```

```

total_machines: machines.length,
operational_machines: machines.filter((m) => m.status === 059669">'operational').length,
machines_under_maintenance: machines.filter((m) => m.status === 059669">'maintenance').length,
breakdown_count: machines.filter((m) => m.status === 059669">'breakdown').length,
total_downtime_hours: parseFloat(machines.reduce((s, m) => s + m.total_downtime_hours, 0).toFixed(0)),
upcoming_maintenance: maintenanceSchedules.filter((s) => s.status === 059669">'scheduled' && 600">new
Date(s.scheduled_date) >= today).length,
overdue_maintenance: maintenanceSchedules.filter((s) => s.status === 059669">'overdue').length,
low_stock_parts: spareParts.filter((p) => p.stock_status === 059669">'low_stock').length,
critical_stock_parts: spareParts.filter((p) => p.stock_status === 059669">'critical' || p.stock_status ===
059669">'out_of_stock').length,
maintenance_cost_ytd: parseFloat(serviceRecords.reduce((s, r) => s + r.total_cost, 0).toFixed(0)),
inventory_value: parseFloat(spareParts.reduce((s, p) => s + p.current_stock * p.unit_cost, 0).toFixed(0)),
avg_supplier_performance: parseFloat((suppliers.filter((s) => s.status !== 059669">'blacklisted').reduce((s, sup)
=> s + sup.on_time_delivery_rate, 0) / suppliers.filter((s) => s.status !==
059669">'blacklisted').length).toFixed(1)),
open_purchase_orders: purchaseOrders.filter((p) => p.status === 059669">'sent' || p.status === 059669">'confirmed'
|| p.status === 059669">'partial').length,
active_warranties: warranties.filter((w) => w.status === 059669">'active').length,
expiring_warranties: warranties.filter((w) => w.status === 059669">'expiring_soon').length,
};

```

059669">"color:#6b7280;font-style:italic">// — Forecast Data

```

600">export 600">const forecastData: ForecastData[] = spareParts.map((part) => {
  600">const avgMonthly = part.monthly_consumption.reduce((s, v) => s + v, 0) / 12;
  600">const forecasted = Array.600">from({ length: 3 }, () => Math.max(0, Math.round(avgMonthly * (0.8 + rand() *
0.5))));
  600">const safetyStock = Math.ceil(avgMonthly * (part.lead_time_days / 30) * 1.5);
  600">const recommendedReorder = Math.max(0, Math.ceil((avgMonthly * 3 + safetyStock) - part.current_stock));
  600">const stockOutRisk = part.current_stock <= part.min_stock ? randInt(60, 95) : part.current_stock <=
part.reorder_point ? randInt(25, 60) : randInt(0, 25);
  600">const excess = part.current_stock > part.max_stock * 1.2;

  600">return {
    part_id: part.id,
    part_name: part.name,
    part_number: part.part_number,
    historical_consumption: part.monthly_consumption,
    forecasted_consumption: forecasted,
    recommended_reorder_qty: recommendedReorder,
    safety_stock: safetyStock,
    stock_out_risk: stockOutRisk,
    excess_inventory: excess,
    movement_class: part.movement_class,
    current_stock: part.current_stock,
  };
});

```

059669">"color:#6b7280;font-style:italic">// — Chart Data

```

600">export 600">const downtimeTrendData = Array.600">from({ length: 12 }, (_, i) => {
  600">const month = 600">new Date(today);
  month.setMonth(month.getMonth() - (11 - i));
  600">return {
    month: month.toLocaleString(059669">'en', { month: 059669">'short' }),
    planned: randInt(20, 80),
    unplanned: randInt(10, 60),
  };
});

600">export 600">const maintenanceCostTrendData = Array.600">from({ length: 12 }, (_, i) => {
  600">const month = 600">new Date(today);
  month.setMonth(month.getMonth() - (11 - i));
  600">return {
    month: month.toLocaleString(059669">'en', { month: 059669">'short' }),
    preventive: randInt(5000, 25000),
    corrective: randInt(8000, 45000),
  };
});

600">export 600">const breakdownFrequencyData = Array.600">from({ length: 8 }, (_, i) => ({
  week: 059669">`W${i + 1}`,

```



```

breakdowns: randInt(1, 12),
});

600">export 600">const sparePartConsumptionData = spareParts
  .filter((p) => p.movement_class === 059669">'fast')
  .slice(0, 8)
  .map((p) => ({
    name: p.name.length > 20 ? p.name.substring(0, 20) + 059669">'...' : p.name,
    consumed: p.monthly_consumption.reduce((s, v) => s + v, 0),
  })))
  .sort((a, b) => b.consumed - a.consumed);

600">export 600">const inventoryTurnoverData = Array.600">from({ length: 6 }, (_, i) => ({
  quarter: 059669">`Q${i + 1}`,
  turnover: randFloat(2, 8, 1),
}));

600">export 600">const preventiveVsCorrectiveData = [
  { name: 059669">'Preventive', value: maintenanceSchedules.filter((s) => s.600">type ===
059669">'preventive').length, color: 059669">'#3479ff' },
  { name: 059669">'Corrective', value: maintenanceSchedules.filter((s) => s.600">type ===
059669">'corrective').length, color: 059669">'#f59e0b' },
  { name: 059669">'Predictive', value: maintenanceSchedules.filter((s) => s.600">type ===
059669">'predictive').length, color: 059669">'#10b981' },
  { name: 059669">'Inspection', value: maintenanceSchedules.filter((s) => s.600">type ===
059669">'inspection').length, color: 059669">'#8b5cf6' },
];

600">export 600">const machineReliabilityData = machines
  .slice(0, 10)
  .map((m) => ({
    name: m.machine_id,
    reliability: m.health_score,
    mtbf: m.mtbf_hours,
  })))
  .sort((a, b) => b.reliability - a.reliability);

```

```
import { useEffect, type ReactNode } from 'react';
import { useAppSelector } from '@store';

export function ThemeProvider({ children }: { children: ReactNode }) {
  const theme = useAppSelector((s) => s.theme.theme);

  useEffect(() => {
    const root = document.documentElement;
    if (theme === 'dark') {
      root.classList.add('dark');
    } else {
      root.classList.remove('dark');
    }
    localStorage.setItem('theme', theme);
  }, [theme]);

  return <>{children}</>;
}
```



```
600">import { useState } 600">from 059669">'react';
600">import { Outlet } 600">from 059669">'react-router-dom';
600">import { Sidebar } 600">from 059669">'./Sidebar';
600">import { TopBar } 600">from 059669">'./TopBar';

600">export 600">function AppLayout() {
  600">const [sidebarOpen, setSidebarOpen] = useState(600">>false);

  600">return (
    <div className=059669">"min-h-screen bg-slate-50 dark:bg-slate-950">
      <Sidebar open={sidebarOpen} onClose={() => setSidebarOpen(600">>false)} />
      <div className=059669">"lg:pl-64">
        <TopBar onMenuClick={() => setSidebarOpen(600">true)} />
        <main className=059669">"p-4 lg:p-6">
          <div className=059669">"mx-auto max-w-[1600px] animate-fade-600">in">
            <Outlet />
          </div>
        </main>
      </div>
    </div>
  );
}
```

```

600">import { useState } 600">from 059669">'react';
600">import { NavLink, useNavigate } 600">from 059669">'react-router-dom';
600">import { clsx } 600">from 059669">'clsx';
600">import { Cog, X, ChevronLeft, ChevronRight, Logout } 600">from 059669">'lucide-react';
600">import { getNavItemsForRole } 600">from 059669">'@/lib/navigation';
600">import { useAppSelector, useAppDispatch } 600">from 059669">'@/store';
600">import { logout } 600">from 059669">'@/store/authSlice';

600">export 600">function Sidebar({ open, onClose }: { open: boolean; onClose: () => 600">void }) {
  600">const role = useAppSelector((s) => s.auth.user?.role);
  600">const userName = useAppSelector((s) => s.auth.user?.name);
  600">const userRoleName = useAppSelector((s) => s.auth.user?.role_name);
  600">const avatarColor = useAppSelector((s) => s.auth.user?.avatar_color);
  600">const dispatch = useAppDispatch();
  600">const navigate = useNavigate();
  600">const [collapsed, setCollapsed] = useState(600">>false);

  600">if (!role) 600">return 600">>null;
  600">const sections = getNavItemsForRole(role);

  600">const handleLogout = () => {
    dispatch(logout());
    navigate(059669">' /login');
  };

  600">return (
    <>
      {open && <div className=059669">"fixed inset-0 z-30 bg-slate-900/50 backdrop-blur-sm lg:hidden" onClick=
{onClose} />}

      <aside
        className={clsx(
          059669">'fixed inset-y-0 left-0 z-40 flex flex-col border-r border-slate-200 bg-white transition-all
duration-300 dark:border-slate-800 dark:bg-slate-900',
          collapsed ? 059669">'w-16' : 059669">'w-64',
          open ? 059669">'translate-x-0' : 059669">' -translate-x-full lg:translate-x-0',
        )}
      >
        {059669">"color:#6b7280;font-style:italic"> /* Logo */}
        <div className=059669">"flex h-16 items-center justify-between border-b border-slate-200 px-4 dark:border-
slate-800">
          <div className=059669">"flex items-center gap-2.5">
            <div className=059669">"flex h-9 w-9 shrink-0 items-center justify-center rounded-xl bg-gradient-to-br
600">from-primary-500 to-primary-700 text-white shadow-lg shadow-primary-500/20">
              <Cog className=059669">"h-5 w-5 animate-[spin_8s_linear_infinite]" />
            </div>
            {!collapsed && (
              <div className=059669">"min-w-0">
                <p className=059669">"truncate text-sm font-bold text-slate-900 dark:text-white">SmartMaint</p>
                <p className=059669">"truncate text-[10px] text-slate-400 dark:text-slate-500">Industrial
Management</p>
              </div>
            )}
          </div>
          <button onClick={onClose} className=059669">"rounded-lg p-1 text-slate-400 hover:bg-slate-100
dark:hover:bg-slate-800 lg:hidden">
            <X className=059669">"h-5 w-5" />
          </button>
        </div>

        {059669">"color:#6b7280;font-style:italic"> /* Nav */}
        <nav className=059669">"flex-1 overflow-y-auto px-3 py-4">
          {sections.map((section) => (
            <div key={section.label} className=059669">"mb-5">
              {!collapsed && (
                <p className=059669">"mb-1.5 px-3 text-[10px] font-semibold uppercase tracking-wider text-slate-400
dark:text-slate-600">
                  {section.label}
                </p>
              )}
            )}
          )}
        </nav>
      </div>
    )}
  )}

```

```

    })
    <div className=059669>"space-y-0.5">
      {section.items.map((item) => (
        <NavLink
          key={item.key}
          to={item.path}
          onClick={onClose}
          title={collapsed ? item.label : 600}>undefined}
          className={({ isActive }) =>
            clsx(059669>'sidebar-link', isActive ? 059669>'sidebar-link-active' : 059669>'sidebar-link-
inactive', collapsed && 059669>'justify-center px-2')}
        >
          <item.icon className={clsx(059669>'h-[18px] w-[18px] shrink-0', collapsed && 059669>'mx-auto')}>
/>
          {!collapsed && <span className=059669>"truncate">{item.label}</span>}
        </NavLink>
      )})
    </div>
  </div>
  )})
</nav>

{059669>"color:#6b7280;font-style:italic"/* User section */}
<div className=059669>"border-t border-slate-200 p-3 dark:border-slate-800">
  <div className={clsx(059669>'flex items-center gap-2.5 rounded-lg px-2 py-2', !collapsed && 059669>'bg-
slate-50 dark:bg-slate-800/50')}>
    <div
      className=059669>"flex h-9 w-9 shrink-0 items-center justify-center rounded-full text-xs font-bold
text-white"
      style={{ backgroundColor: avatarColor }}
    >
      {userName?.split(059669>' ').map((n) => n[0]).join(059669>'').slice(0, 2)}
    </div>
    {!collapsed && (
      <div className=059669>"min-w-0 flex-1">
        <p className=059669>"truncate text-xs font-semibold text-slate-900 dark:text-white">{userName}</p>
        <p className=059669>"truncate text-[10px] text-slate-400 dark:text-slate-500">{userRoleName}</p>
      </div>
    )}
    {!collapsed && (
      <button onClick={handleLogout} className=059669>"rounded-lg p-1.5 text-slate-400 transition-colors
hover:bg-slate-200 hover:text-danger-600 dark:hover:bg-slate-700 dark:hover:text-danger-400" title=059669>"Sign
out">
        <LogOut className=059669>"h-4 w-4" />
      </button>
    )}
  </div>
  <button
    onClick={() => setCollapsed(!collapsed)}
    className=059669>"mt-2 hidden w-full items-center justify-center rounded-lg p-1.5 text-slate-400
transition-colors hover:bg-slate-100 dark:hover:bg-slate-800 lg:flex"
  >
    {collapsed ? <ChevronRight className=059669>"h-4 w-4" /> : <ChevronLeft className=059669>"h-4 w-4" />}
  </button>
</div>
</aside>
</>
);
}

```

```

600">import { useState, useEffect, useRef } 600">from 059669">'react';
600">import { useNavigate } 600">from 059669">'react-router-dom';
600">import { clsx } 600">from 059669">'clsx';
600">import { Menu, Bell, Sun, Moon, Search, ChevronDown } 600">from 059669">'lucide-react';
600">import { useAppSelector, useAppDispatch } 600">from 059669">'@/store';
600">import { toggleTheme } 600">from 059669">'@/store/themeSlice';
600">import { notifications } 600">from 059669">'@/data/seed';
600">import { ROLE_MAP } 600">from 059669">'@/lib/rbac';
600">import 600">type { RoleKey } 600">from 059669">'@/types';

600">export 600">function TopBar({ onMenuClick }: { onMenuClick: () => 600">void }) {
  600">const theme = useAppSelector((s) => s.theme.theme);
  600">const user = useAppSelector((s) => s.auth.user);
  600">const dispatch = useAppDispatch();
  600">const navigate = useNavigate();
  600">const [notifOpen, setNotifOpen] = useState(600">>false);
  600">const [profileOpen, setProfileOpen] = useState(600">>false);
  600">const notifRef = useRef<HTMLDivElement>(600">>null);
  600">const profileRef = useRef<HTMLDivElement>(600">>null);

  600">const unreadCount = notifications.filter((n) => !n.read && user && n.target_roles.includes(user.role)).length;

  useEffect(() => {
    600">function handler(e: MouseEvent) {
      600">if (notifRef.current && !notifRef.current.contains(e.target 600">as Node)) setNotifOpen(600">>false);
      600">if (profileRef.current && !profileRef.current.contains(e.target 600">as Node)) setProfileOpen(600">>false);
    }
    document.addEventListener(059669">'mousedown', handler);
    600">return () => document.removeEventListener(059669">'mousedown', handler);
  }, []);

  600">if (!user) 600">return 600">>null;
  600">const userNotifs = notifications.filter((n) => n.target_roles.includes(user.role)).slice(0, 5);

  600">return (
    <header className=059669">"sticky top-0 z-20 flex h-16 items-center justify-between border-b border-slate-200 bg-white/80 px-4 backdrop-blur-md dark:border-slate-800 dark:bg-slate-900/80 lg:px-6">
      <div className=059669">"flex items-center gap-3">
        <button onClick={onMenuClick} className=059669">"rounded-lg p-2 text-slate-500 hover:bg-slate-100 dark:text-slate-400 dark:hover:bg-slate-800 lg:hidden">
          <Menu className=059669">"h-5 w-5" />
        </button>
        <div className=059669">"relative hidden md:block">
          <Search className=059669">"pointer-events-none absolute left-3 top-1/2 h-4 w-4 -translate-y-1/2 text-slate-400" />
          <input
            600">type=059669">"text"
            placeholder=059669">"Search machines, parts, orders..."
            className=059669">"w-64 rounded-lg border border-slate-200 bg-slate-50 py-2 pl-9 pr-3 text-sm text-slate-700 placeholder:text-slate-400 focus:border-primary-500 focus:bg-white focus:outline-none focus:ring-1 focus:ring-primary-500 dark:border-slate-700 dark:bg-slate-800 dark:text-slate-300"
            />
          </div>
        </div>
        <div className=059669">"flex items-center gap-2">
          <button
            onClick={() => dispatch(toggleTheme())}
            className=059669">"rounded-lg p-2 text-slate-500 transition-colors hover:bg-slate-100 dark:text-slate-400 dark:hover:bg-slate-800"
            title=059669">"Toggle theme"
          >
            {theme === 059669">'light' ? <Moon className=059669">"h-5 w-5" /> : <Sun className=059669">"h-5 w-5" />}
          </button>

          {059669">"color:#6b7280;font-style:italic">/* Notifications */}
          <div className=059669">"relative" ref={notifRef}>
            <button
              onClick={() => setNotifOpen(!notifOpen)}

```

```

class=059669">"relative rounded-lg p-2 text-slate-500 transition-colors hover:bg-slate-100 dark:text-slate-400 dark:hover:bg-slate-800"
>
<Bell className=059669">"h-5 w-5" />
{unreadCount > 0 && (
  <span className=059669">"absolute right-1 top-1 flex h-4 min-w-4 items-center justify-center rounded-full bg-danger-500 px-1 text-[10px] font-bold text-white animate-pulse-ring">
    {unreadCount}
  </span>
)}
</button>
{notifOpen && (
  <div className=059669">"absolute right-0 mt-2 w-80 overflow-hidden rounded-xl border border-slate-200 bg-white shadow-xl dark:border-slate-800 dark:bg-slate-900 animate-slide-600">in">
    <div className=059669">"border-b border-slate-200 px-4 py-3 dark:border-slate-800">
      <p className=059669">"text-sm font-semibold text-slate-900 dark:text-white">Notifications</p>
      <p className=059669">"text-xs text-slate-400">{unreadCount} unread</p>
    </div>
    <div className=059669">"max-h-80 overflow-y-auto">
      {userNotifs.map((n) => (
        <div
          key={n.id}
          onClick={() => { navigate(059669">'/'notifications'); setNotifOpen(600">false); }}
          className={clsx(059669">'cursor-pointer border-b border-slate-100 px-4 py-3 transition-colors hover:bg-slate-50 dark:border-slate-800/70 dark:hover:bg-slate-800/50', !n.read && 059669">'bg-primary-50/50 dark:bg-primary-600/5')}>
            <div className=059669">"flex items-start gap-2">
              <span className={clsx(059669">'mt-1.5 h-2 w-2 shrink-0 rounded-full', n.priority === 059669">'critical' ? 059669">'bg-danger-500' : n.priority === 059669">'warning' ? 059669">'bg-warning-500' : 059669">'bg-sky-500')} />
              <div className=059669">"min-w-0">
                <p className=059669">"text-xs font-semibold text-slate-900 dark:text-white">{n.title}</p>
                <p className=059669">"mt-0.5 text-xs text-slate-500 dark:text-slate-400 line-clamp-2">
{n.message}</p>
                <p className=059669">"mt-1 text-[10px] text-slate-400">{n.timestamp.replace(059669">'T', 059669">' ').replace(059669">'Z', 059669">'')}</p>
              </div>
            </div>
          </div>
        </div>
      )})}
    </div>
    <button
      onClick={() => { navigate(059669">'/'notifications'); setNotifOpen(600">false); }}
      className=059669">"block w-full border-t border-slate-200 py-2.5 text-center text-xs font-medium text-primary-600 hover:bg-slate-50 dark:border-slate-800 dark:text-primary-400 dark:hover:bg-slate-800"
    >
      View all notifications
    </button>
  </div>
)}
</div>

{059669">"color:#6b7280;font-style:italic"/>* Profile */}
<div className=059669">"relative" ref={profileRef}>
  <button
    onClick={() => setProfileOpen(!profileOpen)}
    className=059669">"flex items-center gap-2 rounded-lg p-1 pr-2 transition-colors hover:bg-slate-100 dark:hover:bg-slate-800"
  >
    <div
      className=059669">"flex h-8 w-8 items-center justify-center rounded-full text-xs font-bold text-white"
      style={{ backgroundColor: user.avatar_color }}
    >
      {user.name.split(059669">' ').map((n) => n[0]).join(059669">'').slice(0, 2)}
    </div>
    <div className=059669">"hidden text-left sm:block">
      <p className=059669">"text-xs font-semibold text-slate-900 dark:text-white">{user.name}</p>
      <p className=059669">"text-[10px] text-slate-400 dark:text-slate-500">{user.role_name}</p>
    </div>
    <ChevronDown className=059669">"hidden h-4 w-4 text-slate-400 sm:block" />
  </button>
  {profileOpen && (

```

```

    <div className=059669">"absolute right-0 mt-2 w-56 overflow-hidden rounded-xl border border-slate-200 bg-
white shadow-xl dark:border-slate-800 dark:bg-slate-900 animate-slide-600">in">
      <div className=059669">"border-b border-slate-200 px-4 py-3 dark:border-slate-800">
        <p className=059669">"text-sm font-semibold text-slate-900 dark:text-white">{user.name}</p>
        <p className=059669">"text-xs text-slate-400">{user.email}</p>
        <span className=059669">"mt-2 inline-block rounded-full bg-primary-50 px-2 py-0.5 text-[10px] font-
medium text-primary-700 dark:bg-primary-600/15 dark:text-primary-300">
          {ROLE_MAP[user.role 600]>as RoleKey]?.name}
        </span>
      </div>
      <div className=059669">"py-1">
        <button onClick={() => { navigate(059669">' /settings'); setProfileOpen(600">false); }}
className=059669">"block w-full px-4 py-2 text-left text-sm text-slate-600 hover:bg-slate-50 dark:text-slate-300
dark: hover:bg-slate-800">
          Profile Settings
        </button>
        <button onClick={() => { navigate(059669">' /login'); setProfileOpen(600">false); }}
className=059669">"block w-full px-4 py-2 text-left text-sm text-danger-600 hover:bg-danger-50 dark:text-danger-400
dark: hover:bg-danger-600/10">
          Sign Out
        </button>
      </div>
    </div>
  )}
</div>
</div>
</header>
);
}

```

```

600">import 600">type { ReactNode } 600">from 059669">'react';
600">import { clsx } 600">from 059669">'clsx';

600">type Tone = 059669">'primary' | 059669">'success' | 059669">'warning' | 059669">'danger' | 059669">'neutral' |
059669">'info';

600">const toneClasses: Record<Tone, string> = {
  primary: 059669">'bg-primary-50 text-primary-700 dark:bg-primary-600/15 dark:text-primary-300',
  success: 059669">'bg-accent-50 text-accent-700 dark:bg-accent-600/15 dark:text-accent-300',
  warning: 059669">'bg-warning-50 text-warning-700 dark:bg-warning-600/15 dark:text-warning-300',
  danger: 059669">'bg-danger-50 text-danger-700 dark:bg-danger-600/15 dark:text-danger-300',
  neutral: 059669">'bg-slate-100 text-slate-600 dark:bg-slate-800 dark:text-slate-400',
  info: 059669">'bg-sky-50 text-sky-700 dark:bg-sky-600/15 dark:text-sky-300',
};

600">export 600">function Badge({
  children,
  tone = 059669">'neutral',
  className,
  dot = 600">false,
}: {
  children: ReactNode;
  tone?: Tone;
  className?: string;
  dot?: boolean;
}) {
  600">return (
    <span className={clsx(059669">'badge', toneClasses[tone], className)}>
      {dot && <span className=059669">"h-1.5 w-1.5 rounded-full bg-current" />}
      {children}
    </span>
  );
}

600">export 600">function StatusBadge({ status }: { status: string }) {
  600">const statusMap: Record<string, { tone: Tone; label: string }> = {
    operational: { tone: 059669">'success', label: 059669">'Operational' },
    maintenance: { tone: 059669">'warning', label: 059669">'Maintenance' },
    breakdown: { tone: 059669">'danger', label: 059669">'Breakdown' },
    idle: { tone: 059669">'neutral', label: 059669">'Idle' },
    decommissioned: { tone: 059669">'neutral', label: 059669">'Decommissioned' },
    scheduled: { tone: 059669">'info', label: 059669">'Scheduled' },
    in_progress: { tone: 059669">'warning', label: 059669">'In Progress' },
    completed: { tone: 059669">'success', label: 059669">'Completed' },
    overdue: { tone: 059669">'danger', label: 059669">'Overdue' },
    cancelled: { tone: 059669">'neutral', label: 059669">'Cancelled' },
    reported: { tone: 059669">'danger', label: 059669">'Reported' },
    diagnosing: { tone: 059669">'warning', label: 059669">'Diagnosing' },
    in_repair: { tone: 059669">'warning', label: 059669">'In Repair' },
    resolved: { tone: 059669">'success', label: 059669">'Resolved' },
    closed: { tone: 059669">'neutral', label: 059669">'Closed' },
    active: { tone: 059669">'success', label: 059669">'Active' },
    expiring_soon: { tone: 059669">'warning', label: 059669">'Expiring Soon' },
    expired: { tone: 059669">'danger', label: 059669">'Expired' },
    claimed: { tone: 059669">'info', label: 059669">'Claimed' },
    in_stock: { tone: 059669">'success', label: 059669">'In Stock' },
    low_stock: { tone: 059669">'warning', label: 059669">'Low Stock' },
    critical: { tone: 059669">'danger', label: 059669">'Critical' },
    out_of_stock: { tone: 059669">'danger', label: 059669">'Out 600">of Stock' },
    overstock: { tone: 059669">'info', label: 059669">'Overstock' },
    preferred: { tone: 059669">'primary', label: 059669">'Preferred' },
    inactive: { tone: 059669">'neutral', label: 059669">'Inactive' },
    blacklisted: { tone: 059669">'danger', label: 059669">'Blacklisted' },
    draft: { tone: 059669">'neutral', label: 059669">'Draft' },
    pending_approval: { tone: 059669">'warning', label: 059669">'Pending Approval' },
    approved: { tone: 059669">'success', label: 059669">'Approved' },
    rejected: { tone: 059669">'danger', label: 059669">'Rejected' },
    ordered: { tone: 059669">'info', label: 059669">'Ordered' },
    received: { tone: 059669">'success', label: 059669">'Received' },
  };

```

```

    sent: { tone: 059669">'info', label: 059669">'Sent' },
    confirmed: { tone: 059669">'primary', label: 059669">'Confirmed' },
    partial: { tone: 059669">'warning', label: 059669">'Partial' },
    ok: { tone: 059669">'success', label: 059669">'OK' },
    warning: { tone: 059669">'warning', label: 059669">'Warning' },
    replaced: { tone: 059669">'neutral', label: 059669">'Replaced' },
  };

  600">const config = statusMap[status] || { tone: 059669">'neutral' 600">as Tone, label: status };
  600">return <Badge tone={config.tone} dot={config.label}</Badge>;
}

600">export 600">function PriorityBadge({ priority }: { priority: string }) {
  600">const map: Record<string, Tone> = {
    low: 059669">'neutral',
    medium: 059669">'info',
    high: 059669">'warning',
    urgent: 059669">'danger',
    minor: 059669">'neutral',
    major: 059669">'warning',
    severe: 059669">'danger',
  };
  600">return <Badge tone={map[priority] || 059669">'neutral'}>{priority.charAt(0).toUpperCase() + priority.slice(1)}
</Badge>;
}

600">export 600">function RiskBadge({ level }: { level: string }) {
  600">const map: Record<string, { tone: Tone; label: string }> = {
    low: { tone: 059669">'success', label: 059669">'Low Risk' },
    medium: { tone: 059669">'warning', label: 059669">'Medium Risk' },
    high: { tone: 059669">'danger', label: 059669">'High Risk' },
    critical: { tone: 059669">'danger', label: 059669">'Critical Risk' },
  };
  600">const config = map[level] || { tone: 059669">'neutral' 600">as Tone, label: level };
  600">return <Badge tone={config.tone} dot={config.label}</Badge>;
}

```



```

600">import 600">type { ReactNode } 600">from 059669">'react';
600">import { clsx } 600">from 059669">'clsx';

600">export 600">interface Column<T> {
  key: string;
  header: string;
  render?: (row: T) => ReactNode;
  sortable?: boolean;
  width?: string;
  align?: 059669">'left' | 059669">'right' | 059669">'center';
}

600">export 600">function DataTable<T 600">extends { id: string }>({
  columns,
  rows,
  onClick,
  emptyMessage = 059669">'No records found',
  isLoading = 600">false,
}): {
  columns: Column<T>[];
  rows: T[];
  onClick?: (row: T) => 600">void;
  emptyMessage?: string;
  isLoading?: boolean;
}) {
  600">if (isLoading) {
    600">return (
      <div className=059669">"flex h-64 items-center justify-center">
        <div className=059669">"h-8 w-8 animate-spin rounded-full border-2 border-slate-200 border-t-primary-600
dark:border-slate-700 dark:border-t-primary-500" />
      </div>
    );
  }

  600">if (rows.length === 0) {
    600">return (
      <div className=059669">"flex h-40 items-center justify-center text-sm text-slate-400 dark:text-slate-500">
        {emptyMessage}
      </div>
    );
  }

  600">return (
    <div className=059669">"overflow-x-auto">
      <table className=059669">"w-full">
        <thead>
          <tr className=059669">"border-b border-slate-200 dark:border-slate-800">
            {columns.map((col) => (
              <th
                key={col.key}
                className={clsx(
                  059669">'table-th whitespace-nowrap',
                  col.align === 059669">'right' && 059669">'text-right',
                  col.align === 059669">'center' && 059669">'text-center',
                )}
                style={col.width ? { width: col.width } : 600">undefined}
              >
                {col.header}
              </th>
            )})}
          </tr>
        </thead>
        <tbody className=059669">"divide-y divide-slate-100 dark:divide-slate-800/70">
          {rows.map((row) => (
            <tr
              key={row.id}
              onClick={() => onClick?.(row)}
              className={clsx(
                059669">'transition-colors',

```

```

        onClick && 059669">'cursor-pointer hover:bg-slate-50 dark:hover:bg-slate-800/50',
      )}
    >
    {columns.map((col) => (
      <td
        key={col.key}
        className={clsx(
          059669">'table-td whitespace-nowrap',
          col.align === 059669">'right' && 059669">'text-right',
          col.align === 059669">'center' && 059669">'text-center',
        )}
      >
        {col.render ? col.render(row) : String((row 600">as Record<string, unknown>)[col.key] ??
059669">'')}
      </td>
    )}
  </tr>
  )}
</tbody>
</table>
</div>
);
}

```

```

600">import { clsx } 600">from 059669">'clsx';

600">export 600">function HealthScoreRing({ score, size = 64 }: { score: number; size?: number }) {
  600">const radius = (size - 8) / 2;
  600">const circumference = 2 * Math.PI * radius;
  600">const offset = circumference - (score / 100) * circumference;
  600">const color = score >= 80 ? 059669">'#10b981' : score >= 60 ? 059669">'#f59e0b' : score >= 40 ?
059669">'#f97316' : 059669">'#ef4444';

  600">return (
    <div className=059669">"relative inline-flex items-center justify-center" style={{ width: size, height: size }}>
      <svg width={size} height={size} className=059669">"-rotate-90">
        <circle
          cx={size / 2}
          cy={size / 2}
          r={radius}
          fill=059669">"none"
          strokeWidth={4}
          className=059669">"stroke-slate-200 dark:stroke-slate-800"
        />
        <circle
          cx={size / 2}
          cy={size / 2}
          r={radius}
          fill=059669">"none"
          strokeWidth={4}
          stroke={color}
          strokeDasharray={circumference}
          strokeDashoffset={offset}
          strokeLinecap=059669">"round"
          style={{ transition: 059669">'stroke-dashoffset 0.6s ease' }}
        />
      </svg>
      <span className=059669">"absolute text-sm font-bold" style={{ color }}>
        {score}
      </span>
    </div>
  );
}

600">export 600">function ProgressBar({
  value,
  max = 100,
  tone = 059669">'primary',
  className,
}: {
  value: number;
  max?: number;
  tone?: 059669">'primary' | 059669">'success' | 059669">'warning' | 059669">'danger';
  className?: string;
}) {
  600">const pct = Math.min(100, (value / max) * 100);
  600">const colors = {
    primary: 059669">'bg-primary-500',
    success: 059669">'bg-accent-500',
    warning: 059669">'bg-warning-500',
    danger: 059669">'bg-danger-500',
  };
  600">return (
    <div className={clsx(059669">'h-2 w-full overflow-hidden rounded-full bg-slate-200 dark:bg-slate-800',
className)}>
      <div
        className={clsx(059669">'h-full rounded-full transition-all duration-500', colors[tone])}
        style={{ width: 059669">'${pct}%' }}
      />
    </div>
  );
}

```

```

600">export 600">function StockLevelBar({
  current,
  min,
  reorder,
  max,
}): {
  current: number;
  min: number;
  max: number;
  reorder: number;
}) {
  600">const pct = Math.min(100, (current / max) * 100);
  600">const minPct = (min / max) * 100;
  600">const reorderPct = (reorder / max) * 100;
  600">const tone = current <= min ? 059669">'danger' : current <= reorder ? 059669">'warning' : 059669">'success';
  600">const colors = { success: 059669">'bg-accent-500', warning: 059669">'bg-warning-500', danger: 059669">'bg-
danger-500' };

  600">return (
    <div className=059669">"relative h-2 w-full overflow-hidden rounded-full bg-slate-200 dark:bg-slate-800">
      <div
        className={clsx(059669">'h-full rounded-full transition-all', colors[tone])}
        style={{ width: 059669">`${pct}%` }}
      />
      <div className=059669">"absolute top-0 h-full w-px bg-slate-400 dark:bg-slate-500" style={{ left:
059669">`${minPct}%` }} />
      <div className=059669">"absolute top-0 h-full w-px bg-warning-400 dark:bg-warning-600" style={{ left:
059669">`${reorderPct}%` }} />
    </div>
  );
}

```

```

600">import 600">type { ReactNode } 600">from 059669">'react';
600">import { clsx } 600">from 059669">'clsx';
600">import { TrendingUp, TrendingDown, Minus } 600">from 059669">'lucide-react';

600">type Tone = 059669">'primary' | 059669">'success' | 059669">'warning' | 059669">'danger' | 059669">'info' |
059669">'neutral';

600">const toneConfig: Record<Tone, { bg: string; text: string; ring: string }> = {
  primary: { bg: 059669">'bg-primary-50 dark:bg-primary-600/10', text: 059669">'text-primary-600 dark:text-primary-400', ring: 059669">'ring-primary-100 dark:ring-primary-600/20' },
  success: { bg: 059669">'bg-accent-50 dark:bg-accent-600/10', text: 059669">'text-accent-600 dark:text-accent-400', ring: 059669">'ring-accent-100 dark:ring-accent-600/20' },
  warning: { bg: 059669">'bg-warning-50 dark:bg-warning-600/10', text: 059669">'text-warning-600 dark:text-warning-400', ring: 059669">'ring-warning-100 dark:ring-warning-600/20' },
  danger: { bg: 059669">'bg-danger-50 dark:bg-danger-600/10', text: 059669">'text-danger-600 dark:text-danger-400', ring: 059669">'ring-danger-100 dark:ring-danger-600/20' },
  info: { bg: 059669">'bg-sky-50 dark:bg-sky-600/10', text: 059669">'text-sky-600 dark:text-sky-400', ring: 059669">'ring-sky-100 dark:ring-sky-600/20' },
  neutral: { bg: 059669">'bg-slate-100 dark:bg-slate-800', text: 059669">'text-slate-600 dark:text-slate-400', ring: 059669">'ring-slate-200 dark:ring-slate-700' },
};

600">export 600">function KPICard({
  label,
  value,
  icon,
  tone = 059669">'primary',
  trend,
  trendValue,
  subtitle,
}): {
  label: string;
  value: string | number;
  icon: ReactNode;
  tone?: Tone;
  trend?: 059669">'up' | 059669">'down' | 059669">'flat';
  trendValue?: string;
  subtitle?: string;
}) {
  600">const config = toneConfig[tone];
  600">const TrendIcon = trend === 059669">'up' ? TrendingUp : trend === 059669">'down' ? TrendingDown : Minus;
  600">const trendColor = trend === 059669">'up' ? 059669">'text-accent-600 dark:text-accent-400' : trend ===
059669">'down' ? 059669">'text-danger-600 dark:text-danger-400' : 059669">'text-slate-400';

  600">return (
    <div className=059669">"card card-hover p-5 animate-fade-600">in">
      <div className=059669">"flex items-start justify-between">
        <div className=059669">"min-w-0">
          <p className=059669">"text-xs font-medium uppercase tracking-wider text-slate-500 dark:text-slate-400">
{label}</p>
          <p className=059669">"mt-2 text-2xl font-bold tracking-tight text-slate-900 dark:text-white">{value}</p>
          {subtitle && <p className=059669">"mt-1 text-xs text-slate-400 dark:text-slate-500">{subtitle}</p>}
        </div>
        <div className={clsx(059669">'flex h-11 w-11 shrink-0 items-center justify-center rounded-xl ring-1',
config.bg, config.text, config.ring)}>
          {icon}
        </div>
      </div>
      {trend && trendValue && (
        <div className=059669">"mt-3 flex items-center gap-1.5">
          <TrendIcon className={clsx(059669">'h-3.5 w-3.5', trendColor)} />
          <span className={clsx(059669">'text-xs font-semibold', trendColor)}>{trendValue}</span>
          <span className=059669">"text-xs text-slate-400 dark:text-slate-500">vs last period</span>
        </div>
      )}
    </div>
  );
}

```

```

600">import 600">type { ReactNode } 600">from 059669">'react';
600">import { clsx } 600">from 059669">'clsx';
600">import { X } 600">from 059669">'lucide-react';
600">import { useEffect } 600">from 059669">'react';

600">export 600">function Modal({
  open,
  onClose,
  title,
  children,
  size = 059669">'md',
}): {
  open: boolean;
  onClose: () => 600">void;
  title: string;
  children: ReactNode;
  size?: 059669">'sm' | 059669">'md' | 059669">'lg' | 059669">'xl';
}) {
  useEffect(() => {
    600">if (open) {
      document.body.style.overflow = 059669">'hidden';
      600">return () => { document.body.style.overflow = 059669">''; };
    }
  }, [open]);

  600">if (!open) 600">return 600">>null;

  600">const sizeClass = {
    sm: 059669">'max-w-md',
    md: 059669">'max-w-lg',
    lg: 059669">'max-w-2xl',
    xl: 059669">'max-w-4xl',
  }[size];

  600">return (
    <div className=059669">"fixed inset-0 z-50 flex items-center justify-center p-4">
      <div className=059669">"absolute inset-0 bg-slate-900/50 backdrop-blur-sm animate-fade-600">in" onClick=
{onClose} />
      <div className={clsx(059669">'relative w-full max-h-[90vh] overflow-y-auto rounded-2xl border border-slate-200
bg-white shadow-2xl dark:border-slate-800 dark:bg-slate-900 animate-scale-600">in', sizeClass)}>
        <div className=059669">"sticky top-0 z-10 flex items-center justify-between border-b border-slate-200 bg-
white px-6 py-4 dark:border-slate-800 dark:bg-slate-900">
          <h2 className=059669">"text-lg font-semibold text-slate-900 dark:text-white">{title}</h2>
          <button onClick={onClose} className=059669">"rounded-lg p-1.5 text-slate-400 transition-colors hover:bg-
slate-100 hover:text-slate-600 dark:hover:bg-slate-800 dark:hover:text-slate-300">
            <X className=059669">"h-5 w-5" />
          </button>
        </div>
        <div className=059669">"px-6 py-5">{children}</div>
      </div>
    </div>
  );
}

```

```

600">import { clsx } 600">from 059669">'clsx';
600">import { ChevronLeft, ChevronRight } 600">from 059669">'lucide-react';

600">export 600">function Pagination({
  page,
  totalPages,
  total,
  pageSize,
  onPageChange,
}): {
  page: number;
  totalPages: number;
  total: number;
  pageSize: number;
  onPageChange: (page: number) => 600">void;
} {
  600">if (totalPages <= 1) 600">return 600">>null;

  600">const start = (page - 1) * pageSize + 1;
  600">const end = Math.min(page * pageSize, total);

  600">return (
    <div className=059669">"flex items-center justify-between border-t border-slate-200 px-4 py-3 dark:border-slate-
800">
      <p className=059669">"text-xs text-slate-500 dark:text-slate-400">
        Showing <span className=059669">"font-medium text-slate-700 dark:text-slate-300">{start}</span>-<span
className=059669">"font-medium text-slate-700 dark:text-slate-300">{end}</span> 600">of <span
className=059669">"font-medium text-slate-700 dark:text-slate-300">{total}</span>
      </p>
      <div className=059669">"flex items-center gap-1">
        <button
          onClick={() => onPageChange(page - 1)}
          disabled={page <= 1}
          className={clsx(059669">'rounded-lg p-1.5 transition-colors', page <= 1 ? 059669">'cursor-not-allowed text-
slate-300 dark:text-slate-700' : 059669">'text-slate-600 hover:bg-slate-100 dark:text-slate-400 dark:hover:bg-slate-
800')}
        >
          <ChevronLeft className=059669">"h-4 w-4" />
        </button>
        {Array.600">from({ length: totalPages }, (_, i) => i + 1)
          .filter((p) => p === 1 || p === totalPages || Math.abs(p - page) <= 1)
          .map((p, idx, arr) => (
            <span key={p} className=059669">"flex items-center">
              {idx > 0 && arr[idx - 1] !== p - 1 && <span className=059669">"px-1 text-slate-400">"...</span>}
              <button
                onClick={() => onPageChange(p)}
                className={clsx(
                  059669">'min-w-[32px] rounded-lg px-2 py-1.5 text-sm font-medium transition-colors',
                  p === page
                    ? 059669">'bg-primary-600 text-white'
                    : 059669">'text-slate-600 hover:bg-slate-100 dark:text-slate-400 dark:hover:bg-slate-800',
                )}
              >
                {p}
              </button>
            </span>
          )))
        <button
          onClick={() => onPageChange(page + 1)}
          disabled={page >= totalPages}
          className={clsx(059669">'rounded-lg p-1.5 transition-colors', page >= totalPages ? 059669">'cursor-not-
allowed text-slate-300 dark:text-slate-700' : 059669">'text-slate-600 hover:bg-slate-100 dark:text-slate-400
dark:hover:bg-slate-800')}
        >
          <ChevronRight className=059669">"h-4 w-4" />
        </button>
      </div>
    </div>
  )

```

```
);  
}
```



```

600">import 600">type { ReactNode } 600">from 059669">'react';
600">import { clsx } 600">from 059669">'clsx';
600">import { Search, X } 600">from 059669">'lucide-react';

600">export 600">function Toolbar({
  children,
  searchValue,
  onSearchChange,
  searchPlaceholder = 059669">'Search...',
  actions,
}: {
  children?: ReactNode;
  searchValue: string;
  onSearchChange: (v: string) => 600">void;
  searchPlaceholder?: string;
  actions?: ReactNode;
}) {
  600">return (
    <div className=059669">"flex flex-col gap-3 sm:flex-row sm:items-center sm:justify-between">
      <div className=059669">"flex flex-wrap items-center gap-2">
        {children}
        <div className=059669">"relative">
          <Search className=059669">"pointer-events-none absolute left-3 top-1/2 h-4 w-4 -translate-y-1/2 text-slate-
400" />
          <input
            600">type=059669">"text"
            value={searchValue}
            onChange={(e) => onSearchChange(e.target.value)}
            placeholder={searchPlaceholder}
            className=059669">"input pl-9 pr-9"
          />
          {searchValue && (
            <button
              onClick={() => onSearchChange(059669">'')}
              className=059669">"absolute right-2.5 top-1/2 -translate-y-1/2 text-slate-400 hover:text-slate-600
dark: hover:text-slate-300"
            >
              <X className=059669">"h-4 w-4" />
            </button>
          )}
        </div>
      </div>
      {actions && <div className=059669">"flex items-center gap-2">{actions}</div>}
    </div>
  );
}

600">export 600">function Select({
  value,
  onChange,
  options,
  placeholder,
  className,
}: {
  value: string;
  onChange: (v: string) => 600">void;
  options: { value: string; label: string }[];
  placeholder?: string;
  className?: string;
}) {
  600">return (
    <select
      value={value}
      onChange={(e) => onChange(e.target.value)}
      className={clsx(059669">'input cursor-pointer', className)}
    >
      {placeholder && <option value=059669">">{placeholder}</option>}
      {options.map((opt) => (
        <option key={opt.value} value={opt.value}>{opt.label}</option>

```

```

    )))
  </select>
);
}

600">export 600">function PageHeader({
  title,
  subtitle,
  actions,
}): {
  title: string;
  subtitle?: string;
  actions?: ReactNode;
}) {
  600">return (
    <div className=059669">"mb-6 flex flex-col gap-3 sm:flex-row sm:items-center sm:justify-between">
      <div>
        <h1 className=059669">"text-2xl font-bold tracking-tight text-slate-900 dark:text-white">{title}</h1>
        {subtitle} && <p className=059669">"mt-1 text-sm text-slate-500 dark:text-slate-400">{subtitle}</p>
      </div>
      {actions} && <div className=059669">"flex items-center gap-2">{actions}</div>
    </div>
  );
}

```

```

600">import { useState } 600">from 059669">'react';
600">import { useNavigate } 600">from 059669">'react-router-dom';
600">import { clsx } 600">from 059669">'clsx';
600">import { Cog, Lock, Mail, ChevronRight, ShieldCheck, Factory, Wrench, Package, Truck, User } 600">from
059669">'lucide-react';
600">import { useAppDispatch } 600">from 059669">'@/store';
600">import { login } 600">from 059669">'@/store/authSlice';
600">import { ROLE_CATALOG } 600">from 059669">'@/lib/rbac';
600">import 600">type { RoleKey } 600">from 059669">'@/types';

600">const roleIcons: Record<RoleKey, 600">typeof Cog> = {
  super_admin: ShieldCheck,
  factory_manager: Factory,
  maintenance_manager: Wrench,
  maintenance_engineer: User,
  inventory_manager: Package,
  procurement_officer: Truck,
  supplier: Truck,
};

600">export 600">function LoginPage() {
  600">const dispatch = useAppDispatch();
  600">const navigate = useNavigate();
  600">const [selectedRole, setSelectedRole] = useState<RoleKey>(059669">'super_admin');
  600">const [email, setEmail] = useState(059669">'sarah.mitchell@smartmaint.com');
  600">const [password, setPassword] = useState(059669">'demo1234');
  600">const [loading, setLoading] = useState(600">false);

  600">const handleRoleSelect = (role: RoleKey) => {
    setSelectedRole(role);
    600">const user = ROLE_CATALOG.find((r) => r.key === role);
    600">if (role === 059669">'super_admin') setEmail(059669">'sarah.mitchell@smartmaint.com');
    600">if (role === 059669">'factory_manager') setEmail(059669">'robert.chen@smartmaint.com');
    600">if (role === 059669">'maintenance_manager') setEmail(059669">'james.patterson@smartmaint.com');
    600">if (role === 059669">'maintenance_engineer') setEmail(059669">'david.kim@smartmaint.com');
    600">if (role === 059669">'inventory_manager') setEmail(059669">'michael.brown@smartmaint.com');
    600">if (role === 059669">'procurement_officer') setEmail(059669">'daniel.foster@smartmaint.com');
    600">if (role === 059669">'supplier') setEmail(059669">'sales@bosch-industrial.com');
  };

  600">const handleSubmit = (e: React.FormEvent) => {
    e.preventDefault();
    setLoading(600">true);
    setTimeout(() => {
      dispatch(login(selectedRole));
      navigate(059669">' /dashboard');
    }, 600);
  };

  600">return (
    <div className=059669">"flex min-h-screen">
      {059669">"color:#6b7280;font-style:italic"> /* Left panel */}
      <div className=059669">"relative hidden w-1/2 overflow-hidden bg-gradient-to-br 600">from-slate-900 via-
primary-950 to-slate-900 lg:flex lg:flex-col lg:justify-between lg:p-12">
        <div className=059669">"absolute inset-0 opacity-10" style={{ backgroundImage: 059669">'radial-
gradient(circle at 20% 50%, #3479ff 0%, transparent 50%), radial-gradient(circle at 80% 20%, #10b981 0%, transparent
50%), radial-gradient(circle at 40% 80%, #f59e0b 0%, transparent 50%)' }} />
        <div className=059669">"relative z-10 flex items-center gap-3">
          <div className=059669">"flex h-11 w-11 items-center justify-center rounded-xl bg-white/10 backdrop-blur-sm
ring-1 ring-white/20">
            <Cog className=059669">"h-6 w-6 animate-[spin_12s_linear_infinite] text-white" />
          </div>
          <div>
            <p className=059669">"text-lg font-bold text-white">SmartMaint</p>
            <p className=059669">"text-xs text-slate-400">Industrial Management System</p>
          </div>
        </div>

        <div className=059669">"relative z-10 max-w-md">

```

```

<h1 className=059669">"text-4xl font-bold leading-tight text-white">
  Minimize downtime.<br />
  Maximize production.
</h1>
<p className=059669">"mt-4 text-base text-slate-300">
  A centralized platform 600">for machine health monitoring, preventive maintenance,
  spare-parts optimization, and intelligent analytics across your manufacturing operations.
</p>
<div className=059669">"mt-8 grid grid-cols-3 gap-4">
  {[
    { value: 059669">'24/7', label: 059669">'Monitoring' },
    { value: 059669">'99.2%', label: 059669">'Uptime' },
    { value: 059669">'40%', label: 059669">'Less Downtime' },
  ]}.map((stat) => (
    <div key={stat.label}>
      <p className=059669">"text-2xl font-bold text-white">{stat.value}</p>
      <p className=059669">"text-xs text-slate-400">{stat.label}</p>
    </div>
  )))
</div>
</div>

<p className=059669">"relative z-10 text-xs text-slate-500">© 2026 SmartMaint Industrial. All rights
reserved.</p>
</div>

{059669">"color:#6b7280;font-style:italic">/* Right panel */}
<div className=059669">"flex w-full items-center justify-center bg-slate-50 p-6 dark:bg-slate-950 lg:w-1/2">
  <div className=059669">"w-full max-w-md">
    <div className=059669">"mb-8 text-center lg:hidden">
      <div className=059669">"mx-auto mb-3 flex h-12 w-12 items-center justify-center rounded-xl bg-gradient-
to-br 600">from-primary-500 to-primary-700 text-white shadow-lg">
        <Cog className=059669">"h-7 w-7" />
      </div>
    <h1 className=059669">"text-xl font-bold text-slate-900 dark:text-white">SmartMaint</h1>
  </div>

  <h2 className=059669">"text-2xl font-bold text-slate-900 dark:text-white">Sign 600">in to your account</h2>
  <p className=059669">"mt-1.5 text-sm text-slate-500 dark:text-slate-400">Select your role to access the
platform</p>

  {059669">"color:#6b7280;font-style:italic">/* Role selector */}
  <div className=059669">"mt-6 grid grid-cols-2 gap-2 sm:grid-cols-3">
    {ROLE_CATALOG.map((role) => {
      600">const Icon = roleIcons[role.key];
      600">return (
        <button
          key={role.key}
          onClick={() => handleRoleSelect(role.key)}
          className={clsx(
            059669">'flex flex-col items-center gap-1.5 rounded-xl border p-3 text-center transition-all',
            selectedRole === role.key
              ? 059669">'border-primary-500 bg-primary-50 ring-2 ring-primary-500/20 dark:border-primary-500
dark:bg-primary-600/10'
              : 059669">'border-slate-200 bg-white hover:border-slate-300 dark:border-slate-700 dark:bg-
slate-900 dark:hover:border-slate-600',
            )}
        >
          <Icon className={clsx(059669">'h-5 w-5', selectedRole === role.key ? 059669">'text-primary-600
dark:text-primary-400' : 059669">'text-slate-400')} />
          <span className={clsx(059669">'text-[10px] font-medium leading-tight', selectedRole === role.key ?
059669">'text-primary-700 dark:text-primary-300' : 059669">'text-slate-500 dark:text-slate-400')}>
            {role.name}
          </span>
        </button>
      )
    })}
  </div>

  <form onSubmit={handleSubmit} className=059669">"mt-6 space-y-4">
    <div>
      <label className=059669">"label">Email address</label>
      <div className=059669">"relative">

```

```

    <Mail className=059669">"pointer-events-none absolute left-3 top-1/2 h-4 w-4 -translate-y-1/2 text-
    slate-400" />
    <input
      600">type=059669">"email"
      value={email}
      onChange={(e) => setEmail(e.target.value)}
      className=059669">"input pl-9"
      placeholder=059669">"you@company.com"
      required
    />
  </div>
</div>
<div>
  <label className=059669">"label">Password</label>
  <div className=059669">"relative">
    <Lock className=059669">"pointer-events-none absolute left-3 top-1/2 h-4 w-4 -translate-y-1/2 text-
    slate-400" />
    <input
      600">type=059669">"password"
      value={password}
      onChange={(e) => setPassword(e.target.value)}
      className=059669">"input pl-9"
      placeholder=059669">"....."
      required
    />
  </div>
</div>
<div className=059669">"flex items-center justify-between text-xs">
  <label className=059669">"flex items-center gap-2 text-slate-500 dark:text-slate-400">
    <input 600">type=059669">"checkbox" defaultChecked className=059669">"rounded border-slate-300 text-
    primary-600 focus:ring-primary-500 dark:border-slate-600 dark:bg-slate-800" />
    Remember me
  </label>
  <button 600">type=059669">"button" className=059669">"font-medium text-primary-600 hover:underline
  dark:text-primary-400">Forgot password?</button>
</div>
<button 600">type=059669">"submit" disabled={loading} className=059669">"btn-primary w-full py-2.5">
  {loading ? (
    <span className=059669">"h-4 w-4 animate-spin rounded-full border-2 border-white/30 border-t-white"
  />
  ) : (
    <Sign In <ChevronRight className=059669">"h-4 w-4" /></>
  )}
</button>
</form>

  <p className=059669">"mt-6 text-center text-xs text-slate-400 dark:text-slate-500">
    Demo platform – credentials are pre-filled. Select any role to explore.
  </p>
</div>
</div>
);
}

```

```

600">import { useEffect, useState } 600">from 059669">'react';
600">import { Link } 600">from 059669">'react-router-dom';
600">import {
  Cpu, Wrench, AlertTriangle, Clock, Package, TrendingUp,
  DollarSign, Truck, ShieldCheck, ArrowRight, Activity,
} 600">from 059669">'lucide-react';
600">import {
  ResponsiveContainer, AreaChart, Area, BarChart, Bar, PieChart, Pie, Cell,
  XAxis, YAxis, CartesianGrid, Tooltip, Legend, LineChart, Line,
} 600">from 059669">'recharts';
600">import { KPICard } 600">from 059669">'@/components/ui/KPICard';
600">import { StatusBadge, RiskBadge } 600">from 059669">'@/components/ui/Badge';
600">import { PageHeader } 600">from 059669">'@/components/ui/Toolbar';
600">import { api } 600">from 059669">'@/lib/api';
600">import 600">type { KPIData } 600">from 059669">'@/types';
600">import { machines, maintenanceSchedules, spareParts, notifications } 600">from 059669">'@/data/seed';
600">import { useAppSelector } 600">from 059669">'@/store';

600">export 600">function DashboardPage() {
  600">const [kpis, setKpis] = useState<KPIData | 600">null>(600">null);
  600">const [downtimeTrend, setDowntimeTrend] = useState<{ month: string; planned: number; unplanned: number }[]>
  ([]);
  600">const [costTrend, setCostTrend] = useState<{ month: string; preventive: number; corrective: number }[]>([]);
  600">const [breakdownFreq, setBreakdownFreq] = useState<{ week: string; breakdowns: number }[]>([]);
  600">const [partConsumption, setPartConsumption] = useState<{ name: string; consumed: number }[]>([]);
  600">const [maintenancePie, setMaintenancePie] = useState<{ name: string; value: number; color: string }[]>([]);
  600">const [reliability, setReliability] = useState<{ name: string; reliability: number; mtbf: number }[]>([]);
  600">const user = useAppSelector((s) => s.auth.user);

  useEffect(() => {
    (600">async () => {
      600">const [k, dt, ct, bf, pc, mp, rl] = 600">await Promise.all([
        api.getKPIs(), api.getDowntimeTrend(), api.getMaintenanceCostTrend(),
        api.getBreakdownFrequency(), api.getSparePartConsumption(),
        api.getPreventiveVsCorrective(), api.getMachineReliability(),
      ]);
      setKpis(k); setDowntimeTrend(dt); setCostTrend(ct); setBreakdownFreq(bf);
      setPartConsumption(pc); setMaintenancePie(mp); setReliability(rl);
    })();
  }, []);

  600">if (!kpis) {
    600">return <div className=059669">"flex h-64 items-center justify-center"><div className=059669">"h-8 w-8
    animate-spin rounded-full border-2 border-slate-200 border-t-primary-600 dark:border-slate-700 dark:border-t-primary-
    500" /></div>
  }

  600">const upcomingMaintenance = maintenanceSchedules
    .filter((s) => s.status === 059669">'scheduled' && 600">new Date(s.scheduled_date) >= 600">new
    Date(059669">'2026-08-28'))
    .sort((a, b) => a.scheduled_date.localeCompare(b.scheduled_date))
    .slice(0, 5);

  600">const criticalParts = spareParts
    .filter((p) => p.stock_status === 059669">'critical' || p.stock_status === 059669">'out_of_stock')
    .slice(0, 5);

  600">const userNotifs = notifications.filter((n) => user && n.target_roles.includes(user.role)).slice(0, 4);

  600">return (
    <div>
      <PageHeader title=059669">"Operations Dashboard" subtitle=059669">"Real-time overview 600">of factory
      maintenance, inventory, and performance metrics" />

      {059669">"color:#6b7280;font-style:italic"/* KPI Row 1 */}
      <div className=059669">"grid grid-cols-1 gap-4 sm:grid-cols-2 lg:grid-cols-4">
        <KPICard label=059669">"Total Machines" value={kpis.total_machines} icon={<Cpu className=059669">"h-5 w-5"
        /> tone=059669">"primary" trend=059669">"up" trendValue=059669">"+3" subtitle={059669">`${kpis.operational_machines}
        operational`} />

```

```

        <KPICard label=059669>"Under Maintenance" value={kpis.machines_under_maintenance} icon={<Wrench
className=059669>"h-5 w-5" />} tone=059669>"warning" subtitle=059669>"In progress" />
        <KPICard label=059669>"Active Breakdowns" value={kpis.breakdown_count} icon={<AlertTriangle
className=059669>"h-5 w-5" />} tone=059669>"danger" trend=059669>"down" trendValue=059669>"-2"
subtitle=059669>"Requires attention" />
        <KPICard label=059669>"Total Downtime(hrs)" value={kpis.total_downtime_hours.toLocaleString()} icon={<Clock
className=059669>"h-5 w-5" />} tone=059669>"info" trend=059669>"down" trendValue=059669>"-12%"
subtitle=059669>"Year to date" />
    </div>

    {059669>"color:#6b7280;font-style:italic">/* KPI Row 2 */}
    <div className=059669>"mt-4 grid grid-cols-1 gap-4 sm:grid-cols-2 lg:grid-cols-4">
        <KPICard label=059669>"Maintenance Cost YTD" value={059669>`$${(kpis.maintenance_cost_ytd /
1000).toFixed(0)}K`} icon={<DollarSign className=059669>"h-5 w-5" />} tone=059669>"primary" trend=059669>"up"
trendValue=059669>"8%" />
        <KPICard label=059669>"Inventory Value" value={059669>`$${(kpis.inventory_value / 1000000).toFixed(2)}M`}
icon={<Package className=059669>"h-5 w-5" />} tone=059669>"success" />
        <KPICard label=059669>"Low-Stock Parts" value={kpis.low_stock_parts + kpis.critical_stock_parts} icon=
{<TrendingUp className=059669>"h-5 w-5" />} tone=059669>"warning" subtitle={059669>`$${kpis.critical_stock_parts}
critical`} />
        <KPICard label=059669>"Avg Supplier Performance" value={059669>`$${kpis.avg_supplier_performance}%`} icon=
{<Truck className=059669>"h-5 w-5" />} tone=059669>"success" trend=059669>"up" trendValue=059669>"2%" />
    </div>

    {059669>"color:#6b7280;font-style:italic">/* Charts Row 1 */}
    <div className=059669>"mt-6 grid grid-cols-1 gap-4 lg:grid-cols-2">
        <div className=059669>"card p-5">
            <div className=059669>"mb-4 flex items-center justify-between">
                <div>
                    <h3 className=059669>"text-sm font-semibold text-slate-900 dark:text-white">Machine Downtime
Trends</h3>
                    <p className=059669>"text-xs text-slate-400">Planned vs unplanned downtime(hours)</p>
                </div>
            </div>
            <ResponsiveContainer width=059669>"100%" height={260}>
                <AreaChart data={downtimeTrend}>
                    <defs>
                        <linearGradient id=059669>"plannedGrad" x1=059669>"0" y1=059669>"0" x2=059669>"0" y2=059669>"1">
                            <stop offset=059669>"0%" stopColor=059669>"#3479ff" stopOpacity={0.3} />
                            <stop offset=059669>"100%" stopColor=059669>"#3479ff" stopOpacity={0} />
                        </linearGradient>
                        <linearGradient id=059669>"unplannedGrad" x1=059669>"0" y1=059669>"0" x2=059669>"0"
y2=059669>"1">
                            <stop offset=059669>"0%" stopColor=059669>"#ef4444" stopOpacity={0.3} />
                            <stop offset=059669>"100%" stopColor=059669>"#ef4444" stopOpacity={0} />
                        </linearGradient>
                    </defs>
                    <CartesianGrid strokeDasharray=059669>"3 3" stroke=059669>"#e2e8f0" className=059669>"dark:!stroke-
slate-800" />
                    <XAxis dataKey=059669>"month" tick={{ fontSize: 12 }} stroke=059669>"#94a3b8" />
                    <YAxis tick={{ fontSize: 12 }} stroke=059669>"#94a3b8" />
                    <Tooltip contentStyle={{ borderRadius: 12, border: 059669>'1px solid #e2e8f0', fontSize: 12 }} />
                    <Legend wrapperStyle={{ fontSize: 12 }} />
                    <Area 600>"type=059669>"monotone" dataKey=059669>"planned" stroke=059669>"#3479ff"
fill=059669>"url(#plannedGrad)" strokeWidth={2} name=059669>"Planned" />
                    <Area 600>"type=059669>"monotone" dataKey=059669>"unplanned" stroke=059669>"#ef4444"
fill=059669>"url(#unplannedGrad)" strokeWidth={2} name=059669>"Unplanned" />
                </AreaChart>
            </ResponsiveContainer>
        </div>

        <div className=059669>"card p-5">
            <div className=059669>"mb-4 flex items-center justify-between">
                <div>
                    <h3 className=059669>"text-sm font-semibold text-slate-900 dark:text-white">Maintenance Cost
Trends</h3>
                    <p className=059669>"text-xs text-slate-400">Preventive vs corrective($)</p>
                </div>
            </div>
            <ResponsiveContainer width=059669>"100%" height={260}>
                <BarChart data={costTrend}>
                    <CartesianGrid strokeDasharray=059669>"3 3" stroke=059669>"#e2e8f0" className=059669>"dark:!stroke-
slate-800" />

```

```

<XAxis dataKey=059669>"month" tick={{ fontSize: 12 }} stroke=059669">"#94a3b8" />
<YAxis tick={{ fontSize: 12 }} stroke=059669">"#94a3b8" tickFormatter={(v) => 059669">`$$${v / 1000}K`}
/>

<Tooltip contentStyle={{ borderRadius: 12, border: 059669">'1px solid #e2e8f0', fontSize: 12 }}
formatter={(v) => 059669">`$${Number(v).toLocaleString()}`} />
<Legend wrapperStyle={{ fontSize: 12 }} />
<Bar dataKey=059669">"preventive" fill=059669">"#3479ff" name=059669">"Preventive" radius=[4, 4, 0,
0]] />

<Bar dataKey=059669">"corrective" fill=059669">"#f59e0b" name=059669">"Corrective" radius=[4, 4, 0,
0]] />

</BarChart>
</ResponsiveContainer>
</div>
</div>

{059669">"color:#6b7280;font-style:italic"/* Charts Row 2 */}
<div className=059669">"mt-4 grid grid-cols-1 gap-4 lg:grid-cols-3">
  <div className=059669">"card p-5">
    <div className=059669">"mb-4">
      <h3 className=059669">"text-sm font-semibold text-slate-900 dark:text-white">Maintenance
Distribution</h3>
      <p className=059669">"text-xs text-slate-400">By 600">type</p>
    </div>
    <ResponsiveContainer width=059669">"100%" height={220}>
      <PieChart>
        <Pie data={maintenancePie} dataKey=059669">"value" nameKey=059669">"name" cx=059669">"50%"
cy=059669">"50%" innerRadius={50} outerRadius={80} paddingAngle={3}>
          {maintenancePie.map((entry, i) => (
            <Cell key={i} fill={entry.color} />
          ))}
        </Pie>
        <Tooltip contentStyle={{ borderRadius: 12, border: 059669">'1px solid #e2e8f0', fontSize: 12 }} />
        <Legend wrapperStyle={{ fontSize: 11 }} />
      </PieChart>
    </ResponsiveContainer>
  </div>

  <div className=059669">"card p-5">
    <div className=059669">"mb-4">
      <h3 className=059669">"text-sm font-semibold text-slate-900 dark:text-white">Breakdown Frequency</h3>
      <p className=059669">"text-xs text-slate-400">Weekly count</p>
    </div>
    <ResponsiveContainer width=059669">"100%" height={220}>
      <BarChart data={breakdownFreq}>
        <CartesianGrid strokeDasharray=059669">"3 3" stroke=059669">"#e2e8f0" className=059669">"dark:!stroke-
slate-800" />
        <XAxis dataKey=059669">"week" tick={{ fontSize: 12 }} stroke=059669">"#94a3b8" />
        <YAxis tick={{ fontSize: 12 }} stroke=059669">"#94a3b8" />
        <Tooltip contentStyle={{ borderRadius: 12, border: 059669">'1px solid #e2e8f0', fontSize: 12 }} />
        <Bar dataKey=059669">"breakdowns" fill=059669">"#ef4444" radius=[4, 4, 0, 0]] />
      </BarChart>
    </ResponsiveContainer>
  </div>

  <div className=059669">"card p-5">
    <div className=059669">"mb-4">
      <h3 className=059669">"text-sm font-semibold text-slate-900 dark:text-white">Machine Reliability</h3>
      <p className=059669">"text-xs text-slate-400">Health score by machine</p>
    </div>
    <ResponsiveContainer width=059669">"100%" height={220}>
      <LineChart data={reliability}>
        <CartesianGrid strokeDasharray=059669">"3 3" stroke=059669">"#e2e8f0" className=059669">"dark:!stroke-
slate-800" />
        <XAxis dataKey=059669">"name" tick={{ fontSize: 10 }} stroke=059669">"#94a3b8" angle={-45}
textAnchor=059669">"end" height={50} />
        <YAxis tick={{ fontSize: 12 }} stroke=059669">"#94a3b8" domain={[0, 100]} />
        <Tooltip contentStyle={{ borderRadius: 12, border: 059669">'1px solid #e2e8f0', fontSize: 12 }} />
        <Line 600">type=059669">"monotone" dataKey=059669">"reliability" stroke=059669">"#10b981" strokeWidth=
{2} dot={{ r: 4 }} />
      </LineChart>
    </ResponsiveContainer>
  </div>
</div>

```



```

{059669}">"color:#6b7280;font-style:italic">/* Bottom row: Upcoming maintenance + Critical parts + Notifications
*/}

<div className=059669">"mt-4 grid grid-cols-1 gap-4 lg:grid-cols-3">
  {059669}">"color:#6b7280;font-style:italic">/* Upcoming maintenance */}
  <div className=059669">"card p-5">
    <div className=059669">"mb-4 flex items-center justify-between">
      <h3 className=059669">"text-sm font-semibold text-slate-900 dark:text-white">Upcoming Maintenance</h3>
      <Link to=059669">"/maintenance" className=059669">"flex items-center gap-1 text-xs font-medium text-
primary-600 hover:underline dark:text-primary-400">
        View all <ArrowRight className=059669">"h-3 w-3" />
      </Link>
    </div>
    <div className=059669">"space-y-3">
      {upcomingMaintenance.map((m) => (
        <div key={m.id} className=059669">"flex items-center gap-3 rounded-lg border border-slate-100 p-3
dark:border-slate-800">
          <div className=059669">"flex h-9 w-9 shrink-0 items-center justify-center rounded-lg bg-primary-50
dark:bg-primary-600/10">
            <Wrench className=059669">"h-4 w-4 text-primary-600 dark:text-primary-400" />
          </div>
          <div className=059669">"min-w-0 flex-1">
            <p className=059669">"truncate text-xs font-semibold text-slate-900 dark:text-white">
{m.machine_name}</p>
            <p className=059669">"text-[10px] text-slate-400">{m.scheduled_date} · {m.assigned_technician}</p>
          </div>
          <StatusBadge status={m.status} />
        </div>
      )))
    </div>
  </div>

  {059669}">"color:#6b7280;font-style:italic">/* Critical parts */}
  <div className=059669">"card p-5">
    <div className=059669">"mb-4 flex items-center justify-between">
      <h3 className=059669">"text-sm font-semibold text-slate-900 dark:text-white">Critical Stock Alerts</h3>
      <Link to=059669">"/inventory" className=059669">"flex items-center gap-1 text-xs font-medium text-
primary-600 hover:underline dark:text-primary-400">
        View all <ArrowRight className=059669">"h-3 w-3" />
      </Link>
    </div>
    <div className=059669">"space-y-3">
      {criticalParts.map((p) => (
        <div key={p.id} className=059669">"flex items-center gap-3 rounded-lg border border-slate-100 p-3
dark:border-slate-800">
          <div className=059669">"flex h-9 w-9 shrink-0 items-center justify-center rounded-lg bg-danger-50
dark:bg-danger-600/10">
            <Package className=059669">"h-4 w-4 text-danger-600 dark:text-danger-400" />
          </div>
          <div className=059669">"min-w-0 flex-1">
            <p className=059669">"truncate text-xs font-semibold text-slate-900 dark:text-white">{p.name}</p>
            <p className=059669">"text-[10px] text-slate-400">{p.part_number} · {p.current_stock} /
{p.min_stock} min</p>
          </div>
          <StatusBadge status={p.stock_status} />
        </div>
      )))
    </div>
  </div>

  {059669}">"color:#6b7280;font-style:italic">/* Notifications */}
  <div className=059669">"card p-5">
    <div className=059669">"mb-4 flex items-center justify-between">
      <h3 className=059669">"text-sm font-semibold text-slate-900 dark:text-white">Recent Alerts</h3>
      <Link to=059669">"/notifications" className=059669">"flex items-center gap-1 text-xs font-medium text-
primary-600 hover:underline dark:text-primary-400">
        View all <ArrowRight className=059669">"h-3 w-3" />
      </Link>
    </div>
    <div className=059669">"space-y-3">
      {userNotifs.map((n) => (
        <div key={n.id} className=059669">"flex items-start gap-3 rounded-lg border border-slate-100 p-3
dark:border-slate-800">

```

```

        <div className={059669}>`flex h-9 w-9 shrink-0 items-center justify-center rounded-lg ${
          n.priority === 'critical' ? 'bg-danger-50 dark:bg-danger-600/10' :
          n.priority === 'warning' ? 'bg-warning-50 dark:bg-warning-600/10' :
          'bg-sky-50 dark:bg-sky-600/10'
        }`>
          {n.priority === 059669}>'critical' ? <AlertTriangle className=059669>"h-4 w-4 text-danger-600
dark:text-danger-400" /> :
          n.priority === 059669}>'warning' ? <AlertTriangle className=059669>"h-4 w-4 text-warning-600
dark:text-warning-400" /> :
          <Activity className=059669>"h-4 w-4 text-sky-600 dark:text-sky-400" />
        </div>
        <div className=059669>"min-w-0 flex-1">
          <p className=059669>"text-xs font-semibold text-slate-900 dark:text-white">{n.title}</p>
          <p className=059669>"text-[10px] text-slate-400 line-clamp-2">{n.message}</p>
        </div>
      </div>
    )}}
  </div>
</div>
</div>

{059669}>"color:#6b7280;font-style:italic">/* Machine status overview */}
<div className=059669>"mt-4 card p-5">
  <div className=059669>"mb-4 flex items-center justify-between">
    <h3 className=059669>"text-sm font-semibold text-slate-900 dark:text-white">Machine Status Overview</h3>
    <Link to=059669>" /machines" className=059669>"flex items-center gap-1 text-xs font-medium text-primary-
600 hover:underline dark:text-primary-400">
      View all <ArrowRight className=059669>"h-3 w-3" />
    </Link>
  </div>
  <div className=059669>"grid grid-cols-1 gap-3 sm:grid-cols-2 lg:grid-cols-3 xl:grid-cols-4">
    {machines.slice(0, 8).map((m) => (
      <Link key={m.id} to={059669}>` /machines/${m.id}` className=059669>"flex items-center gap-3 rounded-xl
border border-slate-100 p-3 transition-colors hover:bg-slate-50 dark:border-slate-800 dark:hover:bg-slate-800/50">
        <div className=059669>"flex h-10 w-10 shrink-0 items-center justify-center rounded-lg bg-slate-100
dark:bg-slate-800">
          <CPU className=059669>"h-5 w-5 text-slate-500 dark:text-slate-400" />
        </div>
        <div className=059669>"min-w-0 flex-1">
          <p className=059669>"truncate text-xs font-semibold text-slate-900 dark:text-white">{m.machine_id}
</p>
          <p className=059669>"truncate text-[10px] text-slate-400">{m.name}</p>
        </div>
        <div className=059669>"flex flex-col items-end gap-1">
          <StatusBadge status={m.status} />
          <RiskBadge level={m.risk_level} />
        </div>
      </Link>
    )}}
  </div>
</div>
</div>
);
}

```

```

600">import { useState } 600">from 059669">'react';
600">import { useNavigate } 600">from 059669">'react-router-dom';
600">import { Cpu, Plus, Download, Filter } 600">from 059669">'lucide-react';
600">import { DataTable, 600">type Column } 600">from 059669">'@/components/ui/DataTable';
600">import { Pagination } 600">from 059669">'@/components/ui/Pagination';
600">import { Toolbar, Select, PageHeader } 600">from 059669">'@/components/ui/Toolbar';
600">import { StatusBadge, RiskBadge, PriorityBadge } 600">from 059669">'@/components/ui/Badge';
600">import { HealthScoreRing } 600">from 059669">'@/components/ui/Indicators';
600">import { usePaginatedData } 600">from 059669">'@/hooks/usePaginatedData';
600">import { api } 600">from 059669">'@/lib/api';
600">import { factories } 600">from 059669">'@/data/seed';
600">import 600">type { Machine } 600">from 059669">'@/types';

600">export 600">function MachinesPage() {
  600">const navigate = useNavigate();
  600">const {
    data, total, page, pageSize, totalPages, search, sortBy, sortDir, filters, isLoading,
    setPage, setSortBy, setFilters,
  } = usePaginatedData<Machine>(api.getMachines, [059669">'name', 059669">'machine_id']);

  600">const columns: Column<Machine>[] = [
    {
      key: 059669">'machine_id', header: 059669">'Machine', sortable: 600">true,
      render: (m) => (
        <div className=059669">"flex items-center gap-3">
          <div className=059669">"flex h-9 w-9 items-center justify-center rounded-lg bg-slate-100 dark:bg-slate-
800">
            <Cpu className=059669">"h-4 w-4 text-slate-500 dark:text-slate-400" />
          </div>
          <div>
            <p className=059669">"font-medium text-slate-900 dark:text-white">{m.machine_id}</p>
            <p className=059669">"text-xs text-slate-400">{m.name}</p>
          </div>
        </div>
      ),
    },
    { key: 059669">'600">type', header: 059669">'Type', sortable: 600">true, render: (m) => <span
className=059669">"text-slate-600 dark:text-slate-300">{m.600">type}</span> },
    { key: 059669">'manufacturer', header: 059669">'Manufacturer', sortable: 600">true, render: (m) => <span
className=059669">"text-slate-600 dark:text-slate-300">{m.manufacturer}</span> },
    { key: 059669">'factory_name', header: 059669">'Factory', sortable: 600">true, render: (m) => <span
className=059669">"text-slate-600 dark:text-slate-300">{m.factory_name}</span> },
    { key: 059669">'department_name', header: 059669">'Department', sortable: 600">true, render: (m) => <span
className=059669">"text-slate-600 dark:text-slate-300">{m.department_name}</span> },
    {
      key: 059669">'criticality', header: 059669">'Criticality', sortable: 600">true,
      render: (m) => <PriorityBadge priority={m.criticality} />,
    },
    { key: 059669">'status', header: 059669">'Status', sortable: 600">true, render: (m) => <StatusBadge status=
{m.status} /> },
    {
      key: 059669">'health_score', header: 059669">'Health', sortable: 600">true, align: 059669">'center',
      render: (m) => <HealthScoreRing score={m.health_score} size={48} />,
    },
    { key: 059669">'risk_level', header: 059669">'Risk', sortable: 600">true, render: (m) => <RiskBadge level=
{m.risk_level} /> },
  ];

  600">const factoryOptions = factories.map((f) => ({ value: f.id, label: f.name }));

  600">return (
    <div>
      <PageHeader
        title=059669">"Factory & Machine Management"
        subtitle=059669">"Register, monitor, and manage all machines across your factories"
        actions={
          <>
            <button className=059669">"btn-secondary"><Download className=059669">"h-4 w-4" /> Export</button>
            <button className=059669">"btn-primary"><Plus className=059669">"h-4 w-4" /> Register Machine</button>
          </>
        }
      </PageHeader>

```

```

    </>
  }
/>

<div className=059669">"card p-4">
  <Toolbar searchValue={search} onChange={(v) => { setPage(1); }} >
    <Select
      value={filters.factory_id || 059669">"'}
      onChange={(v) => setFilters(059669">"'factory_id', v)}
      options={factoryOptions}
      placeholder=059669">"All Factories"
    />
    <Select
      value={filters.status || 059669">"'}
      onChange={(v) => setFilters(059669">"'status', v)}
      options={[
        { value: 059669">"'operational', label: 059669">"'Operational' },
        { value: 059669">"'maintenance', label: 059669">"'Maintenance' },
        { value: 059669">"'breakdown', label: 059669">"'Breakdown' },
        { value: 059669">"'idle', label: 059669">"'Idle' },
      ]}
      placeholder=059669">"All Statuses"
    />
    <Select
      value={filters.criticality || 059669">"'}
      onChange={(v) => setFilters(059669">"'criticality', v)}
      options={[
        { value: 059669">"'critical', label: 059669">"'Critical' },
        { value: 059669">"'high', label: 059669">"'High' },
        { value: 059669">"'medium', label: 059669">"'Medium' },
        { value: 059669">"'low', label: 059669">"'Low' },
      ]}
      placeholder=059669">"All Criticality"
    />
  </Toolbar>
</div>

<div className=059669">"mt-4 card">
  <DataTable
    columns={columns}
    rows={data}
    onClick={m => navigate(059669">"`/machines/${m.id}`)}
    isLoading={isLoading}
    emptyMessage=059669">"No machines found matching your filters"
  />
  <Pagination page={page} totalPages={totalPages} total={total} pageSize={pageSize} onChange={setPage} />
</div>
</div>
);
}

```

```

600">import { useEffect, useState } 600">from 059669">'react';
600">import { useParams, Link, useNavigate } 600">from 059669">'react-router-dom';
600">import {
  ArrowLeft, Cpu, Wrench, AlertTriangle, Clock, Calendar, Factory,
  Settings, Activity, TrendingUp, Package, FileText, ChevronRight,
} 600">from 059669">'lucide-react';
600">import {
  ResponsiveContainer, AreaChart, Area, XAxis, YAxis, CartesianGrid, Tooltip,
} 600">from 059669">'recharts';
600">import { HealthScoreRing, ProgressBar } 600">from 059669">'@/components/ui/Indicators';
600">import { StatusBadge, RiskBadge, PriorityBadge } 600">from 059669">'@/components/ui/Badge';
600">import { PageHeader } 600">from 059669">'@/components/ui/Toolbar';
600">import { api } 600">from 059669">'@/lib/api';
600">import { maintenanceSchedules, breakdowns, serviceRecords } 600">from 059669">'@/data/seed';
600">import 600">type { Machine } 600">from 059669">'@/types';

600">export 600">function MachineDetailPage() {
  600">const { id } = useParams();
  600">const navigate = useNavigate();
  600">const [machine, setMachine] = useState<Machine | 600">null>(600">null);

  useEffect(() => {
    600">if (id) api.getMachine(id).then(setMachine);
  }, [id]);

  600">if (!machine) {
    600">return <div className=059669">"flex h-64 items-center justify-center"><div className=059669">"h-8 w-8
animate-spin rounded-full border-2 border-slate-200 border-t-primary-600 dark:border-slate-700 dark:border-t-primary-
500" /></div>;
  }

  600">const machineMaintenance = maintenanceSchedules.filter((s) => s.machine_id === machine.id).slice(0, 5);
  600">const machineBreakdowns = breakdowns.filter((b) => b.machine_id === machine.id).slice(0, 5);
  600">const machineServices = serviceRecords.filter((s) => s.machine_id === machine.id).slice(0, 5);

  600">const healthTrend = Array.600">from({ length: 12 }, (_, i) => ({
    month: 600">new Date(2026, i, 1).toLocaleString(059669">'en', { month: 059669">'short' }),
    score: Math.max(30, Math.min(100, machine.health_score + Math.round((Math.random() - 0.5) * 20))),
  }));

  600">const recommendedAction =
    machine.health_score >= 80 ? 059669">'No action needed – schedule routine inspection' :
    machine.health_score >= 60 ? 059669">'Schedule inspection within 30 days' :
    machine.health_score >= 40 ? 059669">'Schedule inspection within 7 days' :
    059669">'Immediate attention required – schedule emergency maintenance';

  600">return (
    <div>
      <button onClick={() => navigate(059669">' /machines')} className=059669">"mb-4 flex items-center gap-1.5 text-sm
text-slate-500 hover:text-slate-700 dark:text-slate-400 dark:hover:text-slate-200">
        <ArrowLeft className=059669">"h-4 w-4" /> Back to Machines
      </button>

      <div className=059669">"grid grid-cols-1 gap-6 lg:grid-cols-3">
        {059669">"color:#6b7280;font-style:italic"> /* Left: Machine info */}
        <div className=059669">"lg:col-span-2 space-y-6">
          {059669">"color:#6b7280;font-style:italic"> /* Header card */}
          <div className=059669">"card p-6">
            <div className=059669">"flex flex-col gap-4 sm:flex-row sm:items-start sm:justify-between">
              <div className=059669">"flex items-start gap-4">
                <div className=059669">"flex h-14 w-14 shrink-0 items-center justify-center rounded-xl bg-primary-50
dark:bg-primary-600/10">
                  <Cpu className=059669">"h-7 w-7 text-primary-600 dark:text-primary-400" />
                </div>
              </div>
              <h1 className=059669">"text-xl font-bold text-slate-900 dark:text-white">{machine.name}</h1>
              <p className=059669">"text-sm text-slate-400">{machine.machine_id} · {machine.600">type}</p>
              <div className=059669">"mt-2 flex flex-wrap gap-2">
                <StatusBadge status={machine.status} />
            </div>
          </div>
        </div>
      </div>
    </div>
  );
}

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        <RiskBadge level={machine.risk_level} />
        <PriorityBadge priority={machine.criticality} />
      </div>
    </div>
  </div>
  <div className=059669>"flex flex-col items-center">
    <HealthScoreRing score={machine.health_score} size={80} />
    <p className=059669>"mt-1 text-xs font-medium text-slate-500">Health Score</p>
  </div>
</div>

{059669}>"color:#6b7280;font-style:italic"/* Info grid */}
<div className=059669>"mt-6 grid grid-cols-2 gap-4 sm:grid-cols-3">
  [[
    { label: 059669>'Manufacturer', value: machine.manufacturer, icon: Factory },
    { label: 059669>'Model', value: machine.model, icon: Settings },
    { label: 059669>'Serial Number', value: machine.serial_number, icon: FileText },
    { label: 059669>'Factory', value: machine.factory_name, icon: Factory },
    { label: 059669>'Department', value: machine.department_name, icon: Settings },
    { label: 059669>'Production Line', value: machine.production_line, icon: TrendingUp },
    { label: 059669>'Installation Date', value: machine.installation_date, icon: Calendar },
    { label: 059669>'Operating Hours', value: 059669>`${machine.operating_hours.toLocaleString()} hrs`,
    icon: Clock },
    { label: 059669>'Total Downtime', value: 059669>`${machine.total_downtime_hours} hrs`, icon:
AlertTriangle },
  ].map((item) => (
    <div key={item.label} className=059669>"rounded-lg border border-slate-100 p-3 dark:border-slate-
800">
      <div className=059669>"flex items-center gap-1.5 text-xs text-slate-400">
        <item.icon className=059669>"h-3.5 w-3.5" /> {item.label}
      </div>
      <p className=059669>"mt-1 text-sm font-medium text-slate-900 dark:text-white">{item.value}</p>
    </div>
  )))
</div>
</div>

{059669}>"color:#6b7280;font-style:italic"/* Health trend */}
<div className=059669>"card p-5">
  <h3 className=059669>"mb-4 text-sm font-semibold text-slate-900 dark:text-white">Health Score Trend(12
months)</h3>
  <ResponsiveContainer width=059669>"100%" height={240}>
    <AreaChart data={healthTrend}>
      <defs>
        <linearGradient id=059669>"healthGrad" x1=059669>"0" y1=059669>"0" x2=059669>"0"
y2=059669>"1">
          <stop offset=059669>"0%" stopColor=059669>"#10b981" stopOpacity={0.3} />
          <stop offset=059669>"100%" stopColor=059669>"#10b981" stopOpacity={0} />
        </linearGradient>
      </defs>
      <CartesianGrid strokeDasharray=059669>"3 3" stroke=059669>"#e2e8f0"
className=059669>"dark:!stroke-slate-800" />
      <XAxis dataKey=059669>"month" tick={{ fontSize: 12 }} stroke=059669>"#94a3b8" />
      <YAxis domain={{ [0, 100] }} tick={{ fontSize: 12 }} stroke=059669>"#94a3b8" />
      <Tooltip contentStyle={{ borderRadius: 12, border: 059669>'1px solid #e2e8f0', fontSize: 12 }} />
      <Area 600>"type=059669>"monotone" dataKey=059669>"score" stroke=059669>"#10b981"
fill=059669>"url(#healthGrad)" strokeWidth={2} />
    </AreaChart>
  </ResponsiveContainer>
</div>

{059669}>"color:#6b7280;font-style:italic"/* Components */}
<div className=059669>"card p-5">
  <h3 className=059669>"mb-4 text-sm font-semibold text-slate-900 dark:text-white">Machine Components</h3>
  <div className=059669>"space-y-2">
    {machine.components.map((c) => (
      <div key={c.id} className=059669>"flex items-center justify-between rounded-lg border border-slate-
100 p-3 dark:border-slate-800">
        <div className=059669>"flex items-center gap-3">
          <Package className=059669>"h-4 w-4 text-slate-400" />
        </div>
        <p className=059669>"text-sm font-medium text-slate-900 dark:text-white">{c.name}</p>
        <p className=059669>"text-xs text-slate-400">{c.part_number} · {c.category}</p>
      </div>
    ))
  </div>

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        </div>
      </div>
      <div className=059669>"flex items-center gap-3">
        {c.last_replaced && <span className=059669>"text-xs text-slate-400">Last replaced:
{c.last_replaced}</span>}
        <StatusBadge status={c.current_status} />
      </div>
    </div>
  )))
</div>
</div>
</div>

{059669>"color:#6b7280;font-style:italic">/* Right: Predictive + maintenance */}
<div className=059669>"space-y-6">
  {059669>"color:#6b7280;font-style:italic">/* Predictive card */}
  <div className=059669>"card p-5">
    <div className=059669>"mb-4 flex items-center gap-2">
      <Activity className=059669>"h-4 w-4 text-primary-600 dark:text-primary-400" />
      <h3 className=059669>"text-sm font-semibold text-slate-900 dark:text-white">Predictive Analytics</h3>
    </div>
    <div className=059669>"space-y-3">
      <div>
        <div className=059669>"flex items-center justify-between text-xs">
          <span className=059669>"text-slate-500">MTBF</span>
          <span className=059669>"font-semibold text-slate-900 dark:text-white">
{machine.mtbf_hours.toLocaleString()} hrs</span>
        </div>
        <p className=059669>"mt-0.5 text-[10px] text-slate-400">Mean Time Between Failures</p>
      </div>
      <div>
        <div className=059669>"flex items-center justify-between text-xs">
          <span className=059669>"text-slate-500">MTTR</span>
          <span className=059669>"font-semibold text-slate-900 dark:text-white">{machine.mttr_hours}
hrs</span>
        </div>
        <p className=059669>"mt-0.5 text-[10px] text-slate-400">Mean Time To Repair</p>
      </div>
      <div>
        <div className=059669>"flex items-center justify-between text-xs">
          <span className=059669>"text-slate-500">Failures(30d)</span>
          <span className=059669>"font-semibold text-slate-900 dark:text-white">{machine.failure_count_30d}
</span>
        </div>
      </div>
    </div>
    <div className=059669>"rounded-lg bg-slate-50 p-3 dark:bg-slate-800/50">
      <p className=059669>"text-xs font-medium text-slate-700 dark:text-slate-300">Recommended Action</p>
      <p className=059669>"mt-1 text-xs text-slate-500 dark:text-slate-400">{recommendedAction}</p>
    </div>
  </div>

  {059669>"color:#6b7280;font-style:italic">/* Recent maintenance */}
  <div className=059669>"card p-5">
    <div className=059669>"mb-4 flex items-center justify-between">
      <h3 className=059669>"text-sm font-semibold text-slate-900 dark:text-white">Recent Maintenance</h3>
      <Link to="/maintenance" className=059669>"text-xs text-primary-600 hover:underline dark:text-
primary-400">View all</Link>
    </div>
    <div className=059669>"space-y-2">
      {machineMaintenance.map((m) => (
        <div key={m.id} className=059669>"flex items-center justify-between rounded-lg border border-slate-
100 p-2.5 dark:border-slate-800">
          <div className=059669>"flex items-center gap-2">
            <Wrench className=059669>"h-3.5 w-3.5 text-slate-400" />
            <div>
              <p className=059669>"text-xs font-medium text-slate-900 dark:text-white">{m.600}>type</p>
              <p className=059669>"text-[10px] text-slate-400">{m.scheduled_date}</p>
            </div>
            <StatusBadge status={m.status} />
          </div>
        </div>
      ))}

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    </div>
  </div>

  {059669}>"color:#6b7280;font-style:italic">/* Recent breakdowns */</div>
  <div className=059669>"card p-5">
    <div className=059669>"mb-4 flex items-center justify-between">
      <h3 className=059669>"text-sm font-semibold text-slate-900 dark:text-white">Breakdown History</h3>
      <Link to=059669>"/breakdowns" className=059669>"text-xs text-primary-600 hover:underline dark:text-
primary-400">View all</Link>
    </div>
    <div className=059669>"space-y-2">
      {machineBreakdowns.length === 0 && <p className=059669>"text-xs text-slate-400">No breakdowns
recorded</p>}
      {machineBreakdowns.map((b) => (
        <div key={b.id} className=059669>"flex items-center justify-between rounded-lg border border-slate-
100 p-2.5 dark:border-slate-800">
          <div className=059669>"flex items-center gap-2">
            <AlertTriangle className=059669>"h-3.5 w-3.5 text-danger-500" />
            <div>
              <p className=059669>"text-xs font-medium text-slate-900 dark:text-white">{b.breakdown_id}</p>
              <p className=059669>"text-[10px] text-slate-400">{b.reported_date} · {b.downtime_hours}h
downtime</p>
            </div>
          </div>
          <StatusBadge status={b.status} />
        </div>
      )})
    </div>
  </div>

  {059669}>"color:#6b7280;font-style:italic">/* Service history */</div>
  <div className=059669>"card p-5">
    <div className=059669>"mb-4 flex items-center justify-between">
      <h3 className=059669>"text-sm font-semibold text-slate-900 dark:text-white">Service History</h3>
      <Link to=059669>"/service-history" className=059669>"text-xs text-primary-600 hover:underline
dark:text-primary-400">View all</Link>
    </div>
    <div className=059669>"space-y-2">
      {machineServices.map((s) => (
        <div key={s.id} className=059669>"flex items-center justify-between rounded-lg border border-slate-
100 p-2.5 dark:border-slate-800">
          <div className=059669>"flex items-center gap-2">
            <FileText className=059669>"h-3.5 w-3.5 text-slate-400" />
            <div>
              <p className=059669>"text-xs font-medium text-slate-900 dark:text-white">{s.service_id}</p>
              <p className=059669>"text-[10px] text-slate-400">{s.date} · ${s.total_cost}</p>
            </div>
          </div>
          <span className=059669>"text-xs text-slate-400 capitalize">{s.600}>type</span>
        </div>
      )})
    </div>
  </div>
</div>
</div>
);
}

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```

600">import { useState } 600">from 059669">'react';
600">import { Wrench, Plus, Calendar, List, Clock, CheckCircle2 } 600">from 059669">'lucide-react';
600">import { clsx } 600">from 059669">'clsx';
600">import { DataTable, 600">type Column } 600">from 059669">'@/components/ui/DataTable';
600">import { Pagination } 600">from 059669">'@/components/ui/Pagination';
600">import { Toolbar, Select, PageHeader } 600">from 059669">'@/components/ui/Toolbar';
600">import { StatusBadge, PriorityBadge } 600">from 059669">'@/components/ui/Badge';
600">import { Modal } 600">from 059669">'@/components/ui/Modal';
600">import { usePaginatedData } 600">from 059669">'@/hooks/usePaginatedData';
600">import { api } 600">from 059669">'@/lib/api';
600">import { maintenanceSchedules, machines } 600">from 059669">'@/data/seed';
600">import 600">type { MaintenanceSchedule } 600">from 059669">'@/types';

600">export 600">function MaintenancePage() {
  600">const [view, setView] = useState<059669">'list' | 059669">'calendar'>(059669">'list');
  600">const [selected, setSelected] = useState<MaintenanceSchedule | 600">null>(600">null);
  600">const {
    data, total, page, pageSize, totalPages, search, filters, isLoading,
    setPage, setFilters, setSearch,
  } = usePaginatedData<MaintenanceSchedule>(api.getMaintenanceSchedules, [059669">'machine_name']);

  600">const machineOptions = machines.slice(0, 20).map((m) => ({ value: m.id, label: m.machine_id }));

  600">const columns: Column<MaintenanceSchedule>[] = [
    {
      key: 059669">'machine_name', header: 059669">'Machine', sortable: 600">true,
      render: (s) => (
        <div>
          <p className=059669">"font-medium text-slate-900 dark:text-white">{s.machine_name}</p>
          <p className=059669">"text-xs text-slate-400">{s.machine_id_code}</p>
        </div>
      ),
    },
    { key: 059669">'600">type', header: 059669">'Type', sortable: 600">true, render: (s) => <span
      className=059669">"capitalize text-slate-600 dark:text-slate-300">{s.600">type}</span> },
    { key: 059669">'priority', header: 059669">'Priority', sortable: 600">true, render: (s) => <PriorityBadge
      priority={s.priority} /> },
    { key: 059669">'scheduled_date', header: 059669">'Scheduled', sortable: 600">true, render: (s) => <span
      className=059669">"text-slate-600 dark:text-slate-300">{s.scheduled_date}</span> },
    { key: 059669">'assigned_technician', header: 059669">'Technician', sortable: 600">true, render: (s) => <span
      className=059669">"text-slate-600 dark:text-slate-300">{s.assigned_technician}</span> },
    { key: 059669">'estimated_hours', header: 059669">'Est. Hours', sortable: 600">true, align: 059669">'center',
      render: (s) => <span className=059669">"text-slate-600 dark:text-slate-300">{s.estimated_hours}h</span> },
    { key: 059669">'status', header: 059669">'Status', sortable: 600">true, render: (s) => <StatusBadge status=
      {s.status} /> },
  ];

  059669">"color:#6b7280;font-style:italic">// Calendar view
  600">const today = 600">new Date(059669">'2026-08-28');
  600">const monthStart = 600">new Date(today.getFullYear(), today.getMonth(), 1);
  600">const monthEnd = 600">new Date(today.getFullYear(), today.getMonth() + 1, 0);
  600">const daysInMonth = monthEnd.getDate();
  600">const startWeekday = monthStart.getDay();
  600">const calendarDays: (number | 600">null)[] = [
    ...Array(startWeekday).fill(600">null),
    ...Array.600">from({ length: daysInMonth }, (_, i) => i + 1),
  ];

  600">const getSchedulesForDay = (day: number) => {
    600">const dateStr = 059669">`${today.getFullYear()}-${String(today.getMonth() + 1).padStart(2,
    '0')}-${String(day).padStart(2, '0')}`;
    600">return maintenanceSchedules.filter((s) => s.scheduled_date === dateStr);
  };

  600">return (
    <div>
      <PageHeader
        title=059669">"Preventive Maintenance"
        subtitle=059669">"Schedule, track, and manage maintenance work orders"

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actions={
  <>
    <div className=059669">"flex rounded-lg border border-slate-200 p-0.5 dark:border-slate-700">
      <button onClick={() => setView(059669">'list')} className={clsx(059669">'rounded-md px-3 py-1.5 text-xs
font-medium', view === 059669">'list' ? 059669">'bg-primary-600 text-white' : 059669">'text-slate-500 dark:text-
slate-400')}>
        <List className=059669">"h-4 w-4" />
      </button>
      <button onClick={() => setView(059669">'calendar')} className={clsx(059669">'rounded-md px-3 py-1.5
text-xs font-medium', view === 059669">'calendar' ? 059669">'bg-primary-600 text-white' : 059669">'text-slate-500
dark:text-slate-400')}>
        <Calendar className=059669">"h-4 w-4" />
      </button>
    </div>
    <button className=059669">"btn-primary"><Plus className=059669">"h-4 w-4" /> Schedule
Maintenance</button>
  </>
}
/>

{view === 059669">'list' ? (
  <>
    <div className=059669">"card p-4">
      <Toolbar searchValue={search} onSearchChange={setSearch}>
        <Select value={filters.status || 059669">''} onChange={(v) => setFilters(059669">'status', v)} options=
[[{ value: 059669">'scheduled', label: 059669">'Scheduled' }, { value: 059669">'in_progress', label: 059669">'In
Progress' }, { value: 059669">'completed', label: 059669">'Completed' }, { value: 059669">'overdue', label:
059669">'Overdue' }]] placeholder=059669">"All Statuses" />
        <Select value={filters.600">'type' || 059669">''} onChange={(v) => setFilters(059669">'600">'type', v)}
options=[[{ value: 059669">'preventive', label: 059669">'Preventive' }, { value: 059669">'corrective', label:
059669">'Corrective' }, { value: 059669">'predictive', label: 059669">'Predictive' }, { value: 059669">'inspection',
label: 059669">'Inspection' }]] placeholder=059669">"All Types" />
        <Select value={filters.priority || 059669">''} onChange={(v) => setFilters(059669">'priority', v)}
options=[[{ value: 059669">'low', label: 059669">'Low' }, { value: 059669">'medium', label: 059669">'Medium' }, {
value: 059669">'high', label: 059669">'High' }, { value: 059669">'urgent', label: 059669">'Urgent' }]]
placeholder=059669">"All Priorities" />
      </Toolbar>
    </div>
    <div className=059669">"mt-4 card">
      <DataTable columns={columns} rows={data} onClick={setSelected} isLoading={isLoading} />
      <Pagination page={page} totalPages={totalPages} total={total} pageSize={pageSize} onChange={setPage}
/>
    </div>
  </>
) : (
  <div className=059669">"card p-5">
    <div className=059669">"mb-4 flex items-center justify-between">
      <h3 className=059669">"text-lg font-semibold text-slate-900 dark:text-white">
        {today.toLocaleString(059669">'en', { month: 059669">'long', year: 059669">'numeric' })}
      </h3>
    </div>
    <div className=059669">"flex gap-2">
      <button className=059669">"btn-secondary px-2 py-1 text-xs">Prev</button>
      <button className=059669">"btn-secondary px-2 py-1 text-xs">Today</button>
      <button className=059669">"btn-secondary px-2 py-1 text-xs">Next</button>
    </div>
  </div>
  <div className=059669">"grid grid-cols-7 gap-1">
    {[059669">'Sun', 059669">'Mon', 059669">'Tue', 059669">'Wed', 059669">'Thu', 059669">'Fri',
059669">'Sat'].map((d) => (
      <div key={d} className=059669">"pb-2 text-center text-xs font-semibold text-slate-400">{d}</div>
    ))}
    {calendarDays.map((day, i) => {
      600">const daySchedules = day ? getSchedulesForDay(day) : [];
      600">const isToday = day === today.getDate();
      600">return (
        <div
          key={i}
          className={clsx(
            059669">'min-h-[100px] rounded-lg border p-1.5',
            day === 600">null ? 059669">'border-transparent' : 059669">'border-slate-100 dark:border-slate-
800',
            isToday && 059669">'border-primary-300 bg-primary-50/50 dark:border-primary-700 dark:bg-primary-
600/5',

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    ))
  >
  {day && (
    <>
    <p className={clsx(059669">'mb-1 text-xs font-medium', isToday ? 059669">'text-primary-700
dark:text-primary-300' : 059669">'text-slate-500 dark:text-slate-400')}>>{day}</p>
    <div className=059669">"space-y-1">
      {daySchedules.slice(0, 3).map((s) => (
        <button
          key={s.id}
          onClick={() => setSelected(s)}
          className={clsx(
            059669">'block w-full truncate rounded px-1.5 py-0.5 text-left text-[10px] font-medium
transition-colors',
            s.status === 059669">'overdue' ? 059669">'bg-danger-100 text-danger-700 dark:bg-danger-
600/20 dark:text-danger-300' :
            s.status === 059669">'completed' ? 059669">'bg-accent-100 text-accent-700 dark:bg-
accent-600/20 dark:text-accent-300' :
            s.status === 059669">'in_progress' ? 059669">'bg-warning-100 text-warning-700 dark:bg-
warning-600/20 dark:text-warning-300' :
            059669">'bg-primary-100 text-primary-700 dark:bg-primary-600/20 dark:text-primary-300',
          )}
        >
        {s.machine_id_code}
      </button>
    )}}
    {daySchedules.length > 3 && <p className=059669">"text-[10px] text-slate-400">+
{daySchedules.length - 3} more</p>}
  </div>
</>
  )}
</div>
);
)}}
</div>
</div>
))
{059669">"color:#6b7280;font-style:italic">/* Detail modal */}
<Modal open={!selected} onClose={() => setSelected(600">null)} title=059669">"Maintenance Work Order"
size=059669">"lg">
  {selected && (
    <div className=059669">"space-y-4">
      <div className=059669">"flex items-center justify-between">
        <div>
          <h3 className=059669">"text-lg font-semibold text-slate-900 dark:text-white">{selected.machine_name}
</h3>
          <p className=059669">"text-sm text-slate-400">{selected.machine_id_code} · {selected.factory_name}
</p>
        </div>
        <StatusBadge status={selected.status} />
      </div>
      <div className=059669">"grid grid-cols-2 gap-4">
        <div><p className=059669">"label">Type</p><p className=059669">"text-sm capitalize text-slate-700
dark:text-slate-300">{selected.600">type}</p></div>
        <div><p className=059669">"label">Priority</p><PriorityBadge priority={selected.priority} /></div>
        <div><p className=059669">"label">Scheduled Date</p><p className=059669">"text-sm text-slate-700
dark:text-slate-300">{selected.scheduled_date}</p></div>
        <div><p className=059669">"label">Assigned Technician</p><p className=059669">"text-sm text-slate-700
dark:text-slate-300">{selected.assigned_technician}</p></div>
        <div><p className=059669">"label">Estimated Hours</p><p className=059669">"text-sm text-slate-700
dark:text-slate-300">{selected.estimated_hours}h</p></div>
        <div><p className=059669">"label">Cost</p><p className=059669">"text-sm text-slate-700 dark:text-slate-
300">${selected.cost}</p></div>
      </div>
      <div><p className=059669">"label">Description</p><p className=059669">"text-sm text-slate-700 dark:text-
slate-300">{selected.description}</p></div>
      <div>
        <p className=059669">"label">Checklist</p>
        <div className=059669">"space-y-2">
          {selected.checklist.map((c) => (
            <div key={c.id} className=059669">"flex items-center gap-2 rounded-lg border border-slate-100 p-2
dark:border-slate-800">

```

```

        {c.completed ? <CheckCircle2 className=059669>"h-4 w-4 text-accent-500" /> : <Clock
className=059669>"h-4 w-4 text-slate-300 dark:text-slate-600" />}
        <span className={clsx(059669>'text-sm', c.completed ? 059669>'text-slate-500 line-through
dark:text-slate-500' : 059669>'text-slate-700 dark:text-slate-300')}>{c.task}</span>
      </div>
    )}}
  </div>
</div>
{selected.recurring && (
  <div className=059669>"rounded-lg bg-primary-50 p-3 text-xs text-primary-700 dark:bg-primary-600/10
dark:text-primary-300">
    Recurring every {selected.interval_days} days
  </div>
)}
</div>
)}
</Modal>
</div>
);
}

```

```

600">import { useState } 600">from 059669">'react';
600">import { AlertTriangle, Plus, Download, Clock, Wrench } 600">from 059669">'lucide-react';
600">import { DataTable, 600">type Column } 600">from 059669">'@/components/ui/DataTable';
600">import { Pagination } 600">from 059669">'@/components/ui/Pagination';
600">import { Toolbar, Select, PageHeader } 600">from 059669">'@/components/ui/Toolbar';
600">import { StatusBadge, PriorityBadge } 600">from 059669">'@/components/ui/Badge';
600">import { Modal } 600">from 059669">'@/components/ui/Modal';
600">import { usePaginatedData } 600">from 059669">'@/hooks/usePaginatedData';
600">import { api } 600">from 059669">'@/lib/api';
600">import 600">type { Breakdown } 600">from 059669">'@/types';

600">export 600">function BreakdownsPage() {
  600">const [selected, setSelected] = useState<Breakdown | 600">null>(600">null);
  600">const {
    data, total, page, pageSize, totalPages, search, filters, isLoading,
    setPage, setFilters, setSearch,
  } = usePaginatedData<Breakdown>(api.getBreakdowns, [059669">'machine_name']);

  600">const columns: Column<Breakdown>[] = [
    {
      key: 059669">'breakdown_id', header: 059669">'ID', sortable: 600">true,
      render: (b) => <span className=059669">"font-mono text-xs font-medium text-slate-900 dark:text-white">
{b.breakdown_id}</span>,
    },
    {
      key: 059669">'machine_name', header: 059669">'Machine', sortable: 600">true,
      render: (b) => (
        <div>
          <p className=059669">"font-medium text-slate-900 dark:text-white">{b.machine_name}</p>
          <p className=059669">"text-xs text-slate-400">{b.machine_id_code}</p>
        </div>
      ),
    },
    { key: 059669">'factory_name', header: 059669">'Factory', sortable: 600">true, render: (b) => <span
className=059669">"text-slate-600 dark:text-slate-300">{b.factory_name}</span> },
    { key: 059669">'severity', header: 059669">'Severity', sortable: 600">true, render: (b) => <PriorityBadge
priority={b.severity} /> },
    { key: 059669">'reported_date', header: 059669">'Reported', sortable: 600">true, render: (b) => <span
className=059669">"text-slate-600 dark:text-slate-300">{b.reported_date}</span> },
    { key: 059669">'downtime_hours', header: 059669">'Downtime', sortable: 600">true, align: 059669">'center',
render: (b) => <span className=059669">"text-slate-600 dark:text-slate-300">{b.downtime_hours}h</span> },
    { key: 059669">'assigned_technician', header: 059669">'Technician', sortable: 600">true, render: (b) => <span
className=059669">"text-slate-600 dark:text-slate-300">{b.assigned_technician}</span> },
    { key: 059669">'status', header: 059669">'Status', sortable: 600">true, render: (b) => <StatusBadge status=
{b.status} /> },
  ];

  600">return (
    <div>
      <PageHeader
        title=059669">"Breakdown & Corrective Maintenance"
        subtitle=059669">"Report, track, and analyze machine breakdowns and repair activities"
        actions={
          <>
            <button className=059669">"btn-secondary"><Download className=059669">"h-4 w-4" /> Export</button>
            <button className=059669">"btn-primary"><Plus className=059669">"h-4 w-4" /> Report Breakdown</button>
          </>
        }
      />

      <div className=059669">"card p-4">
        <Toolbar searchValue={search} onSearchChange={setSearch}>
          <Select value={filters.status || 059669">''} onChange={(v) => setFilters(059669">'status', v)} options={[{
value: 059669">'reported', label: 059669">'Reported' }, { value: 059669">'diagnosing', label: 059669">'Diagnosing' },
{ value: 059669">'in_repair', label: 059669">'In Repair' }, { value: 059669">'resolved', label: 059669">'Resolved' },
{ value: 059669">'closed', label: 059669">'Closed' }]} placeholder=059669">"All Statuses" />
          <Select value={filters.severity || 059669">''} onChange={(v) => setFilters(059669">'severity', v)} options=
[[{ value: 059669">'minor', label: 059669">'Minor' }, { value: 059669">'major', label: 059669">'Major' }, { value:
059669">'severe', label: 059669">'Severe' }, { value: 059669">'critical', label: 059669">'Critical' }]]

```

```

placeholder=059669">"All Severities" />
    </Toolbar>
</div>

<div className=059669">"mt-4 card">
    <DataTable columns={columns} rows={data} onClick={setSelected} isLoading={isLoading} />
    <Pagination page={page} totalPages={totalPages} total={total} pageSize={pageSize} onPageChange={setPage} />
</div>

<Modal open={!selected} onClose={() => setSelected(600">null)} title=059669">"Breakdown Report"
size=059669">"lg">
    {selected && (
        <div className=059669">"space-y-4">
            <div className=059669">"flex items-center justify-between">
                <div>
                    <h3 className=059669">"text-lg font-semibold text-slate-900 dark:text-white">{selected.machine_name}
</h3>

                    <p className=059669">"text-sm text-slate-400">{selected.breakdown_id} · {selected.factory_name}</p>
                </div>
                <div className=059669">"flex gap-2">
                    <PriorityBadge priority={selected.severity} />
                    <StatusBadge status={selected.status} />
                </div>
            </div>
            <div className=059669">"grid grid-cols-2 gap-4">
                <div><p className=059669">"label">Reported Date</p><p className=059669">"text-sm text-slate-700
dark:text-slate-300">{selected.reported_date}</p></div>
                <div><p className=059669">"label">Reported By</p><p className=059669">"text-sm text-slate-700
dark:text-slate-300">{selected.reported_by}</p></div>
                <div><p className=059669">"label">Assigned Technician</p><p className=059669">"text-sm text-slate-700
dark:text-slate-300">{selected.assigned_technician}</p></div>
                <div><p className=059669">"label">Downtime</p><p className=059669">"text-sm text-slate-700 dark:text-
slate-300">{selected.downtime_hours} hours</p></div>
                <div><p className=059669">"label">Repair Cost</p><p className=059669">"text-sm text-slate-700
dark:text-slate-300">${selected.repair_cost.toLocaleString()}</p></div>
                <div><p className=059669">"label">Resolved Date</p><p className=059669">"text-sm text-slate-700
dark:text-slate-300">{selected.resolved_date || 059669">"-"}</p></div>
            </div>
            <div><p className=059669">"label">Symptoms</p><p className=059669">"text-sm text-slate-700 dark:text-
slate-300">{selected.symptoms}</p></div>
            {selected.root_cause && <div><p className=059669">"label">Root Cause Analysis</p><p
className=059669">"text-sm text-slate-700 dark:text-slate-300">{selected.root_cause}</p></div>}
            {selected.corrective_action && <div><p className=059669">"label">Corrective Action</p><p
className=059669">"text-sm text-slate-700 dark:text-slate-300">{selected.corrective_action}</p></div>}
            {selected.parts_replaced.length > 0 && (
                <div>
                    <p className=059669">"label">Parts Replaced</p>
                    <div className=059669">"space-y-1.5">
                        {selected.parts_replaced.map((p, i) => (
                            <div key={i} className=059669">"flex items-center justify-between rounded-lg border border-slate-
100 p-2 dark:border-slate-800">
                                <div className=059669">"flex items-center gap-2">
                                    <Wrench className=059669">"h-3.5 w-3.5 text-slate-400" />
                                    <span className=059669">"text-sm text-slate-700 dark:text-slate-300">{p.part_name}</span>
                                </div>
                                    <span className=059669">"text-xs text-slate-400">{p.quantity} × ${p.unit_cost}</span>
                                </div>
                            </div>
                        )
                    )
                </div>
            )
        </div>
    )
</Modal>
</div>
);
}

```

```

600">import { useState } 600">from 059669">'react';
600">import { Package, Plus, Download, ArrowDownToLine, ArrowUpFromLine } 600">from 059669">'lucide-react';
600">import { DataTable, 600">type Column } 600">from 059669">'@/components/ui/DataTable';
600">import { Pagination } 600">from 059669">'@/components/ui/Pagination';
600">import { Toolbar, Select, PageHeader } 600">from 059669">'@/components/ui/Toolbar';
600">import { StatusBadge } 600">from 059669">'@/components/ui/Badge';
600">import { StockLevelBar } 600">from 059669">'@/components/ui/Indicators';
600">import { Modal } 600">from 059669">'@/components/ui/Modal';
600">import { usePaginatedData } 600">from 059669">'@/hooks/usePaginatedData';
600">import { api } 600">from 059669">'@/lib/api';
600">import { inventoryTransactions } 600">from 059669">'@/data/seed';
600">import 600">type { SparePart } 600">from 059669">'@/types';

600">export 600">function InventoryPage() {
  600">const [selected, setSelected] = useState<SparePart | 600">null>(600">null);
  600">const [txView, setTxView] = useState(600">>false);
  600">const {
    data, total, page, pageSize, totalPages, search, filters, isLoading,
    setPage, setFilters, setSearch,
  } = usePaginatedData<SparePart>(api.getSpareParts, [059669">'name', 059669">'part_number']);

  600">const columns: Column<SparePart>[] = [
    {
      key: 059669">'part_number', header: 059669">'Part', sortable: 600">true,
      render: (p) => (
        <div>
          <p className=059669">"font-mono text-xs font-medium text-slate-900 dark:text-white">{p.part_number}</p>
          <p className=059669">"text-xs text-slate-400">{p.name}</p>
        </div>
      ),
    },
    { key: 059669">'category', header: 059669">'Category', sortable: 600">true, render: (p) => <span
className=059669">"capitalize text-slate-600 dark:text-slate-300">{p.category}</span> },
    {
      key: 059669">'current_stock', header: 059669">'Stock Level', sortable: 600">true,
      render: (p) => (
        <div className=059669">"w-32">
          <div className=059669">"mb-1 flex justify-between text-xs">
            <span className=059669">"font-medium text-slate-700 dark:text-slate-300">{p.current_stock}</span>
            <span className=059669">"text-slate-400">{p.max_stock}</span>
          </div>
          <StockLevelBar current={p.current_stock} min={p.min_stock} reorder={p.reorder_point} max={p.max_stock} />
        </div>
      ),
    },
    { key: 059669">'unit_cost', header: 059669">'Unit Cost', sortable: 600">true, align: 059669">'right', render: (p)
=> <span className=059669">"text-slate-600 dark:text-slate-300">${p.unit_cost}</span> },
    { key: 059669">'supplier_name', header: 059669">'Supplier', sortable: 600">true, render: (p) => <span
className=059669">"text-slate-600 dark:text-slate-300">{p.supplier_name}</span> },
    { key: 059669">'warehouse_name', header: 059669">'Warehouse', sortable: 600">true, render: (p) => <span
className=059669">"text-slate-600 dark:text-slate-300">{p.warehouse_name}</span> },
    { key: 059669">'movement_class', header: 059669">'Movement', sortable: 600">true, render: (p) => <span
className=059669">"capitalize text-slate-600 dark:text-slate-300">{p.movement_class}</span> },
    { key: 059669">'stock_status', header: 059669">'Status', sortable: 600">true, render: (p) => <StatusBadge status=
{p.stock_status} /> },
  ];

  600">const partTransactions = selected ? inventoryTransactions.filter((t) => t.part_id === selected.id).slice(0,
10) : [];

  600">return (
    <div>
      <PageHeader
        title=059669">"Spare Parts Inventory"
        subtitle=059669">"Manage spare-part catalog, stock levels, and warehouse locations"
        actions={
          <>
            <button className=059669">"btn-secondary" onClick={() => setTxView(600">true)}><ArrowDownToLine
className=059669">"h-4 w-4" /> Transactions</button>

```



```

        <button className=059669">"btn-secondary"><Download className=059669">"h-4 w-4" /> Export</button>
        <button className=059669">"btn-primary"><Plus className=059669">"h-4 w-4" /> Add Part</button>
    </>
}
/>

<div className=059669">"card p-4">
    <Toolbar searchValue={search} onChange={setSearch}>
        <Select value={filters.category || 059669">"'} onChange={(v) => setFilters(059669">"category', v)} options=
        [{ value: 059669">"mechanical', label: 059669">"Mechanical' }, { value: 059669">"electrical', label:
        059669">"Electrical' }, { value: 059669">"hydraulic', label: 059669">"Hydraulic' }, { value: 059669">"pneumatic',
        label: 059669">"Pneumatic' }, { value: 059669">"electronic', label: 059669">"Electronic' }, { value:
        059669">"consumable', label: 059669">"Consumable' }, { value: 059669">"lubricant', label: 059669">"Lubricant' }, {
        value: 059669">"safety', label: 059669">"Safety' }] placeholder=059669">"All Categories" />
        <Select value={filters.stock_status || 059669">"'} onChange={(v) => setFilters(059669">"stock_status', v)}
        options=[[{ value: 059669">"in_stock', label: 059669">"In Stock' }, { value: 059669">"low_stock', label: 059669">"Low
        Stock' }, { value: 059669">"critical', label: 059669">"Critical' }, { value: 059669">"out_of_stock', label:
        059669">"Out 600">of Stock' }, { value: 059669">"overstock', label: 059669">"Overstock' }] placeholder=059669">"All
        Stock Status" />
        <Select value={filters.movement_class || 059669">"'} onChange={(v) => setFilters(059669">"movement_class',
        v)} options=[[{ value: 059669">"fast', label: 059669">"Fast-Moving' }, { value: 059669">"medium', label:
        059669">"Medium' }, { value: 059669">"slow', label: 059669">"Slow-Moving' }] placeholder=059669">"All Movement" />
    </Toolbar>
</div>

<div className=059669">"mt-4 card">
    <DataTable columns={columns} rows={data} onClick={setSelected} isLoading={isLoading} />
    <Pagination page={page} totalPages={totalPages} total={total} pageSize={pageSize} onChange={setPage} />
</div>

{059669">"color:#6b7280;font-style:italic"/> /* Part detail modal */
<Modal open={!selected && !txView} onClose={() => setSelected(600">null)} title=059669">"Spare Part Details"
size=059669">"lg">
    {selected && (
        <div className=059669">"space-y-4">
            <div className=059669">"flex items-center justify-between">
                <div>
                    <h3 className=059669">"text-lg font-semibold text-slate-900 dark:text-white">{selected.name}</h3>
                    <p className=059669">"font-mono text-sm text-slate-400">{selected.part_number}</p>
                </div>
                <StatusBadge status={selected.stock_status} />
            </div>
            <div className=059669">"grid grid-cols-2 gap-4 sm:grid-cols-3">
                <div><p className=059669">"label">Category</p><p className=059669">"text-sm capitalize text-slate-700
                dark:text-slate-300">{selected.category}</p></div>
                <div><p className=059669">"label">Unit Cost</p><p className=059669">"text-sm text-slate-700 dark:text-
                slate-300">${selected.unit_cost}</p></div>
                <div><p className=059669">"label">Warehouse</p><p className=059669">"text-sm text-slate-700 dark:text-
                slate-300">{selected.warehouse_name}</p></div>
                <div><p className=059669">"label">Supplier</p><p className=059669">"text-sm text-slate-700 dark:text-
                slate-300">{selected.supplier_name}</p></div>
                <div><p className=059669">"label">Lead Time</p><p className=059669">"text-sm text-slate-700 dark:text-
                slate-300">{selected.lead_time_days} days</p></div>
                <div><p className=059669">"label">Batch Number</p><p className=059669">"text-sm text-slate-700
                dark:text-slate-300">{selected.batch_number}</p></div>
                <div><p className=059669">"label">Current Stock</p><p className=059669">"text-sm font-semibold text-
                slate-700 dark:text-slate-300">{selected.current_stock} units</p></div>
                <div><p className=059669">"label">Reserved</p><p className=059669">"text-sm text-slate-700 dark:text-
                slate-300">{selected.reserved_stock} units</p></div>
                <div><p className=059669">"label">Available</p><p className=059669">"text-sm font-semibold text-accent-
                600 dark:text-accent-400">{selected.current_stock - selected.reserved_stock} units</p></div>
                <div><p className=059669">"label">Min Stock</p><p className=059669">"text-sm text-slate-700 dark:text-
                slate-300">{selected.min_stock}</p></div>
                <div><p className=059669">"label">Reorder Point</p><p className=059669">"text-sm text-slate-700
                dark:text-slate-300">{selected.reorder_point}</p></div>
                <div><p className=059669">"label">Max Stock</p><p className=059669">"text-sm text-slate-700 dark:text-
                slate-300">{selected.max_stock}</p></div>
            </div>
            <div>
                <p className=059669">"label">Stock Level</p>
                <div className=059669">"w-full">
                    <StockLevelBar current={selected.current_stock} min={selected.min_stock} reorder=
                    {selected.reorder_point} max={selected.max_stock} />
                </div>
            </div>
        </div>
    )}

```



```

        <div className=059669>"mt-1 flex justify-between text-[10px] text-slate-400">
          <span>Min: {selected.min_stock}</span>
          <span>Reorder: {selected.reorder_point}</span>
          <span>Max: {selected.max_stock}</span>
        </div>
      </div>
    </div>
    <div>
      <p className=059669>"label">Inventory Valuation</p>
      <p className=059669>"text-2xl font-bold text-slate-900 dark:text-white">${(selected.current_stock *
selected.unit_cost).toLocaleString()}</p>
    </div>
    <div>
      <p className=059669>"label">Compatible Machines</p>
      <div className=059669>"flex flex-wrap gap-1.5">
        {selected.compatible_machines.map((m) => (
          <span key={m} className=059669>"rounded-md bg-slate-100 px-2 py-1 text-xs text-slate-600 dark:bg-
slate-800 dark:text-slate-400">{m}</span>
        ))}
      </div>
    </div>
    <div>
      <p className=059669>"label">Recent Transactions</p>
      <div className=059669>"space-y-1.5">
        {partTransactions.map((t) => (
          <div key={t.id} className=059669>"flex items-center justify-between rounded-lg border border-
slate-100 p-2 dark:border-slate-800">
            <div className=059669>"flex items-center gap-2">
              {t.600}>type === 059669>'stock_in' ? <ArrowDownToLine className=059669>"h-3.5 w-3.5 text-
accent-500" /> : <ArrowUpFromLine className=059669>"h-3.5 w-3.5 text-danger-500" />}
              <span className=059669>"text-xs capitalize text-slate-700 dark:text-slate-300">
{t.600}>type.replace(059669>'_', 059669>' ')</span>
            </div>
            <span className=059669>"text-xs text-slate-400">{t.quantity} units ·
{t.timestamp.split(059669>'T')[0]}</span>
          </div>
        ))}
      </div>
    </div>
  </div>
</Modal>

{059669>"color:#6b7280;font-style:italic">/* Transactions modal */}
<Modal open={txView} onClose={() => setTxView(600>false)} title=059669>"Inventory Transactions"
size=059669>"xl">
  <div className=059669>"overflow-x-auto">
    <table className=059669>"w-full">
      <thead>
        <tr className=059669>"border-b border-slate-200 dark:border-slate-800">
          <th className=059669>"table-th">Part</th>
          <th className=059669>"table-th">Type</th>
          <th className=059669>"table-th">Qty</th>
          <th className=059669>"table-th">Reference</th>
          <th className=059669>"table-th">By</th>
          <th className=059669>"table-th">Date</th>
        </tr>
      </thead>
      <tbody className=059669>"divide-y divide-slate-100 dark:divide-slate-800/70">
        {inventoryTransactions.slice(0, 30).map((t) => (
          <tr key={t.id} className=059669>"hover:bg-slate-50 dark:hover:bg-slate-800/50">
            <td className=059669>"table-td"><p className=059669>"font-medium text-slate-900 dark:text-white">
{t.part_name}</p><p className=059669>"text-xs text-slate-400">{t.part_number}</p></td>
            <td className=059669>"table-td"><span className=059669>"capitalize">
{t.600}>type.replace(059669>'_', 059669>' ')</span></td>
            <td className=059669>"table-td">{t.quantity}</td>
            <td className=059669>"table-td font-mono text-xs">{t.reference}</td>
            <td className=059669>"table-td">{t.performed_by}</td>
            <td className=059669>"table-td">{t.timestamp.split(059669>'T')[0]}</td>
          </tr>
        ))}
      </tbody>
    </table>

```

```
        </div>
      </Modal>
    </div>
  );
}
```

```

600">import { useEffect, useState } 600">from 059669">'react';
600">import { TrendingUp, AlertTriangle, Package, ShoppingCart, BarChart3 } 600">from 059669">'lucide-react';
600">import {
  ResponsiveContainer, LineChart, Line, BarChart, Bar, XAxis, YAxis, CartesianGrid, Tooltip, Legend,
} 600">from 059669">'recharts';
600">import { PageHeader } 600">from 059669">'@/components/ui/Toolbar';
600">import { KPICard } 600">from 059669">'@/components/ui/KPICard';
600">import { Badge } 600">from 059669">'@/components/ui/Badge';
600">import { api } 600">from 059669">'@/lib/api';
600">import 600">type { ForecastData } 600">from 059669">'@/types';

600">export 600">function ForecastingPage() {
  600">const [forecasts, setForecasts] = useState<ForecastData[]>([]);
  600">const [selected, setSelected] = useState<ForecastData | 600">null>(600">null);

  useEffect(() => { api.getForecastData().then(setForecasts); }, []);

  600">if (forecasts.length === 0) {
    600">return <div className=059669">"flex h-64 items-center justify-center"><div className=059669">"h-8 w-8
animate-spin rounded-full border-2 border-slate-200 border-t-primary-600 dark:border-slate-700 dark:border-t-primary-
500" /></div>;
  }

  600">const fastMoving = forecasts.filter((f) => f.movement_class === 059669">'fast').length;
  600">const highRisk = forecasts.filter((f) => f.stock_out_risk > 50).length;
  600">const excessItems = forecasts.filter((f) => f.excess_inventory).length;
  600">const totalReorder = forecasts.reduce((s, f) => s + f.recommended_reorder_qty, 0);

  600">const selectedChart = selected ? {
    labels: [...Array(12).keys()].map((i) => 059669">`M${i + 1}`),
    historical: selected.historical_consumption,
    forecast: [...selected.historical_consumption.slice(-3), ...selected.forecasted_consumption],
  } : 600">null;

  600">const chartData = selectedChart
  ? selectedChart.labels.map((label, i) => ({
    month: label,
    historical: selectedChart.historical[i] || 0,
    forecast: i >= 9 ? selectedChart.forecast[i - 9] || 0 : 600">null,
  }))
  : [];

  600">return (
    <div>
      <PageHeader title=059669">"Intelligent Spare-Part Forecasting" subtitle=059669">"Demand prediction, reorder
recommendations, and stock-out risk analysis" />

      <div className=059669">"grid grid-cols-1 gap-4 sm:grid-cols-2 lg:grid-cols-4">
        <KPICard label=059669">"Fast-Moving Parts" value={fastMoving} icon={<TrendingUp className=059669">"h-5 w-5"
/>} tone=059669">"primary" />
        <KPICard label=059669">"High Stock-Out Risk" value={highRisk} icon={<AlertTriangle className=059669">"h-5 w-
5" />} tone=059669">"danger" />
        <KPICard label=059669">"Excess Inventory" value={excessItems} icon={<Package className=059669">"h-5 w-5" />}
tone=059669">"warning" />
        <KPICard label=059669">"Reorder Recommendations" value={totalReorder} icon={<ShoppingCart
className=059669">"h-5 w-5" />} tone=059669">"info" subtitle=059669">"units to order" />
      </div>

      <div className=059669">"mt-6 grid grid-cols-1 gap-6 lg:grid-cols-3">
        {059669">"color:#6b7280;font-style:italic"/* Forecast list */}
        <div className=059669">"lg:col-span-1 card p-4">
          <h3 className=059669">"mb-3 text-sm font-semibold text-slate-900 dark:text-white">Parts Forecast</h3>
          <div className=059669">"max-h-[600px] space-y-2 overflow-y-auto">
            {forecasts.map((f) => (
              <button
                key={f.part_id}
                onClick={() => setSelected(f)}
                className={059669">`block w-full rounded-lg border p-3 text-left transition-colors
${selected?.part_id === f.part_id ? 'border-primary-300 bg-primary-50 dark:border-primary-700 dark:bg-primary-600/10'

```

```

: 'border-slate-100 hover:bg-slate-50 dark:border-slate-800 dark:hover:bg-slate-800/50'}```
    >
    <div className=059669>"flex items-center justify-between">
      <p className=059669>"text-xs font-medium text-slate-900 dark:text-white">{f.part_name}</p>
      <Badge tone={f.movement_class === 059669}>'fast' ? 059669>'danger' : f.movement_class ===
059669>'medium' ? 059669>'warning' : 059669>'neutral'}>{f.movement_class}</Badge>
    </div>
    <p className=059669>"mt-0.5 font-mono text-[10px] text-slate-400">{f.part_number}</p>
    <div className=059669>"mt-2 flex items-center justify-between">
      <span className=059669>"text-[10px] text-slate-400">Stock: {f.current_stock}</span>
      <span className={059669}>`text-[10px] font-semibold ${f.stock_out_risk > 50 ? 'text-danger-600
dark:text-danger-400' : f.stock_out_risk > 25 ? 'text-warning-600 dark:text-warning-400' : 'text-accent-600
dark:text-accent-400'}`>
        {f.stock_out_risk}% risk
      </span>
    </div>
  </button>
))}
</div>
</div>

{059669>"color:#6b7280;font-style:italic">/* Chart + details */}
<div className=059669>"lg:col-span-2 space-y-6">
  {selected && (
    <>
      <div className=059669>"card p-5">
        <div className=059669>"mb-4 flex items-center justify-between">
          <div>
            <h3 className=059669>"text-sm font-semibold text-slate-900 dark:text-white">{selected.part_name}
</h3>
            <p className=059669>"font-mono text-xs text-slate-400">{selected.part_number}</p>
          </div>
          <BarChart3 className=059669>"h-5 w-5 text-slate-400" />
        </div>
        <ResponsiveContainer width=059669>"100%" height={280}>
          <LineChart data={chartData}>
            <CartesianGrid strokeDasharray=059669>"3 3" stroke=059669>"#e2e8f0"
className=059669>"dark:!stroke-slate-800" />
            <XAxis dataKey=059669>"month" tick={{ fontSize: 11 }} stroke=059669>"#94a3b8" />
            <YAxis tick={{ fontSize: 11 }} stroke=059669>"#94a3b8" />
            <Tooltip contentStyle={{ borderRadius: 12, border: 059669>"1px solid #e2e8f0", fontSize: 12 }}
/>
            <Legend wrapperStyle={{ fontSize: 12 }} />
            <Line 600>"type=059669>"monotone" dataKey=059669>"historical" stroke=059669>"#3479ff"
strokeWidth={2} dot={{ r: 3 }} name=059669>"Historical" />
            <Line 600>"type=059669>"monotone" dataKey=059669>"forecast" stroke=059669>"#f59e0b"
strokeWidth={2} strokeDasharray=059669>"5 5" dot={{ r: 3 }} name=059669>"Forecast" />
          </LineChart>
        </ResponsiveContainer>
      </div>

      <div className=059669>"grid grid-cols-2 gap-4 sm:grid-cols-4">
        <div className=059669>"card p-4">
          <p className=059669>"text-xs text-slate-400">Current Stock</p>
          <p className=059669>"mt-1 text-xl font-bold text-slate-900 dark:text-white">
{selected.current_stock}</p>
        </div>
        <div className=059669>"card p-4">
          <p className=059669>"text-xs text-slate-400">Safety Stock</p>
          <p className=059669>"mt-1 text-xl font-bold text-slate-900 dark:text-white">
{selected.safety_stock}</p>
        </div>
        <div className=059669>"card p-4">
          <p className=059669>"text-xs text-slate-400">Reorder Qty</p>
          <p className=059669>"mt-1 text-xl font-bold text-primary-600 dark:text-primary-400">
{selected.recommended_reorder_qty}</p>
        </div>
        <div className=059669>"card p-4">
          <p className=059669>"text-xs text-slate-400">Stock-Out Risk</p>
          <p className={059669}>`mt-1 text-xl font-bold ${selected.stock_out_risk > 50 ? 'text-danger-600
dark:text-danger-400' : selected.stock_out_risk > 25 ? 'text-warning-600 dark:text-warning-400' : 'text-accent-600
dark:text-accent-400'}`>{selected.stock_out_risk}%</p>
        </div>
      </div>
    </>
  )}
</div>

```

```

    </div>

    {selected.excess_inventory && (
      <div className=059669>"card p-4">
        <div className=059669>"flex items-center gap-2">
          <AlertTriangle className=059669>"h-5 w-5 text-warning-500" />
          <div>
            <p className=059669>"text-sm font-semibold text-slate-900 dark:text-white">Excess Inventory
Detected</p>
            <p className=059669>"text-xs text-slate-400">This part has more stock than needed. Consider
reducing future orders.</p>
          </div>
        </div>
      </div>
    )}
  </>
)}
</div>
</div>
</div>
);
}

```

```

600">import { useState } 600">from 059669">'react';
600">import { Truck, Plus, Star, Clock, CheckCircle, XCircle, FileText } 600">from 059669">'lucide-react';
600">import { clsx } 600">from 059669">'clsx';
600">import { DataTable, 600">type Column } 600">from 059669">'@/components/ui/DataTable';
600">import { Pagination } 600">from 059669">'@/components/ui/Pagination';
600">import { Toolbar, Select, PageHeader } 600">from 059669">'@/components/ui/Toolbar';
600">import { StatusBadge, Badge } 600">from 059669">'@/components/ui/Badge';
600">import { Modal } 600">from 059669">'@/components/ui/Modal';
600">import { ProgressBar } 600">from 059669">'@/components/ui/Indicators';
600">import { usePaginatedData } 600">from 059669">'@/hooks/usePaginatedData';
600">import { api } 600">from 059669">'@/lib/api';
600">import { purchaseRequests, purchaseOrders } 600">from 059669">'@/data/seed';
600">import 600">type { Supplier, PurchaseRequest, PurchaseOrder } 600">from 059669">'@/types';

600">export 600">function SuppliersPage() {
  600">const [tab, setTab] = useState<059669">'suppliers' | 059669">'requests' | 059669">'orders'>
(059669">'suppliers');
  600">const [selectedSupplier, setSelectedSupplier] = useState<Supplier | 600">null>(600">null);
  600">const [selectedRequest, setSelectedRequest] = useState<PurchaseRequest | 600">null>(600">null);
  600">const [selectedOrder, setSelectedOrder] = useState<PurchaseOrder | 600">null>(600">null);

  600">const {
    data: suppliers, total: sTotal, page: sPage, pageSize: sPageSize, totalPages: sTotalPages, search: sSearch,
    filters: sFilters, isLoading: sLoading,
    setPage: sSetPage, setFilters: sSetFilters, setSearch: sSetSearch,
  } = usePaginatedData<Supplier>(api.getSuppliers, [059669">'name', 059669">'contact_person']);
  600">const {
    data: requests, total: rTotal, page: rPage, pageSize: rPageSize, totalPages: rTotalPages, search: rSearch,
    filters: rFilters, isLoading: rLoading,
    setPage: rSetPage, setFilters: rSetFilters, setSearch: rSetSearch,
  } = usePaginatedData<PurchaseRequest>(api.getPurchaseRequests, [059669">'request_number', 059669">'requested_by']);
  600">const {
    data: orders, total: oTotal, page: oPage, pageSize: oPageSize, totalPages: oTotalPages, search: oSearch, filters:
oFilters, isLoading: oLoading,
    setPage: oSetPage, setFilters: oSetFilters, setSearch: oSetSearch,
  } = usePaginatedData<PurchaseOrder>(api.getPurchaseOrders, [059669">'order_number', 059669">'supplier_name']);

  600">const supplierColumns: Column<Supplier>[] = [
    { key: 059669">'name', header: 059669">'Supplier', sortable: 600">true, render: (s) => <div><p
className=059669">"font-medium text-slate-900 dark:text-white">{s.name}</p><p className=059669">"text-xs text-slate-
400">{s.contact_person}</p></div> },
    { key: 059669">'rating', header: 059669">'Rating', sortable: 600">true, align: 059669">'center', render: (s) =>
<div className=059669">"flex items-center justify-center gap-1"><Star className=059669">"h-3.5 w-3.5 text-warning-400
fill-warning-400" /><span className=059669">"text-sm font-medium text-slate-700 dark:text-slate-300">{s.rating}
</span></div> },
    { key: 059669">'lead_time_days', header: 059669">'Lead Time', sortable: 600">true, align: 059669">'center',
render: (s) => <span className=059669">"text-slate-600 dark:text-slate-300">{s.lead_time_days}</span> },
    { key: 059669">'on_time_delivery_rate', header: 059669">'On-Time %', sortable: 600">true, align:
059669">'center', render: (s) => <div className=059669">"w-24"><ProgressBar value={s.on_time_delivery_rate} tone=
{s.on_time_delivery_rate >= 90 ? 059669">'success' : s.on_time_delivery_rate >= 80 ? 059669">'warning' :
059669">'danger'} /><span className=059669">"mt-1 block text-xs text-slate-500">{s.on_time_delivery_rate}%</span>
</div> },
    { key: 059669">'quality_rate', header: 059669">'Quality %', sortable: 600">true, align: 059669">'center', render:
(s) => <span className=059669">"text-slate-600 dark:text-slate-300">{s.quality_rate}%</span> },
    { key: 059669">'total_orders', header: 059669">'Orders', sortable: 600">true, align: 059669">'center', render:
(s) => <span className=059669">"text-slate-600 dark:text-slate-300">{s.total_orders}</span> },
    { key: 059669">'total_value', header: 059669">'Total Value', sortable: 600">true, align: 059669">'right', render:
(s) => <span className=059669">"text-slate-600 dark:text-slate-300">${(s.total_value / 1000000).toFixed(2)}M</span>
},
    { key: 059669">'status', header: 059669">'Status', sortable: 600">true, render: (s) => <StatusBadge status=
{s.status} /> },
  ];

  600">const requestColumns: Column<PurchaseRequest>[] = [
    { key: 059669">'request_number', header: 059669">'Request #', sortable: 600">true, render: (r) => <span
className=059669">"font-mono text-xs font-medium text-slate-900 dark:text-white">{r.request_number}</span> },
    { key: 059669">'requested_by', header: 059669">'Requested By', sortable: 600">true, render: (r) => <span
className=059669">"text-slate-600 dark:text-slate-300">{r.requested_by}</span> },
    { key: 059669">'department', header: 059669">'Department', sortable: 600">true, render: (r) => <span

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className=059669">"text-slate-600 dark:text-slate-300">{r.department}</span> },
  { key: 059669">'priority', header: 059669">'Priority', sortable: 600">true, render: (r) => <Badge tone=
{r.priority === 059669">'urgent' ? 059669">'danger' : r.priority === 059669">'high' ? 059669">'warning' :
059669">'info'}>{r.priority}</Badge> },
  { key: 059669">'total_value', header: 059669">'Value', sortable: 600">true, align: 059669">'right', render: (r)
=> <span className=059669">"text-slate-600 dark:text-slate-300">${r.total_value.toLocaleString()}</span> },
  { key: 059669">'required_date', header: 059669">'Required By', sortable: 600">true, render: (r) => <span
className=059669">"text-slate-600 dark:text-slate-300">{r.required_date}</span> },
  { key: 059669">'status', header: 059669">'Status', sortable: 600">true, render: (r) => <StatusBadge status=
{r.status} /> },
];

600">const orderColumns: Column<PurchaseOrder>[] = [
  { key: 059669">'order_number', header: 059669">'Order #', sortable: 600">true, render: (o) => <span
className=059669">"font-mono text-xs font-medium text-slate-900 dark:text-white">{o.order_number}</span> },
  { key: 059669">'supplier_name', header: 059669">'Supplier', sortable: 600">true, render: (o) => <span
className=059669">"text-slate-600 dark:text-slate-300">{o.supplier_name}</span> },
  { key: 059669">'total_value', header: 059669">'Value', sortable: 600">true, align: 059669">'right', render: (o)
=> <span className=059669">"text-slate-600 dark:text-slate-300">{o.total_value.toLocaleString()}</span> },
  { key: 059669">'order_date', header: 059669">'Order Date', sortable: 600">true, render: (o) => <span
className=059669">"text-slate-600 dark:text-slate-300">{o.order_date}</span> },
  { key: 059669">'expected_delivery', header: 059669">'Expected Delivery', sortable: 600">true, render: (o) =>
<span className=059669">"text-slate-600 dark:text-slate-300">{o.expected_delivery}</span> },
  { key: 059669">'delivery_progress', header: 059669">'Progress', sortable: 600">true, align: 059669">'center',
render: (o) => <div className=059669">"w-24"><ProgressBar value={o.delivery_progress} tone={o.delivery_progress ===
100 ? 059669">'success' : 059669">'primary'} /><span className=059669">"mt-1 block text-xs text-slate-500">
{o.delivery_progress}%</span></div> },
  { key: 059669">'status', header: 059669">'Status', sortable: 600">true, render: (o) => <StatusBadge status=
{o.status} /> },
];

600">return (
  <div>
    <PageHeader
      title=059669">"Supplier & Procurement Management"
      subtitle=059669">"Manage suppliers, purchase requests, and purchase orders"
      actions={tab === 059669">'suppliers' ? <button className=059669">"btn-primary"><Plus className=059669">"h-4
w-4" /> Add Supplier</button> : tab === 059669">'requests' ? <button className=059669">"btn-primary"><Plus
className=059669">"h-4 w-4" /> New Request</button> : <button className=059669">"btn-primary"><Plus
className=059669">"h-4 w-4" /> Create Order</button>}
    />

    {059669">"color:#6b7280;font-style:italic">/* Tabs */}
    <div className=059669">"mb-4 flex gap-1 rounded-lg border border-slate-200 p-1 dark:border-slate-800">
      {([059669">'suppliers', 059669">'Suppliers'], [059669">'requests', 059669">'Purchase Requests'],
[059669">'orders', 059669">'Purchase Orders']) 600">as 600">const).map(([key, label]) => (
        <button key={key} onClick={() => setTab(key)} className={clsx(059669">'flex-1 rounded-md px-4 py-2 text-sm
font-medium transition-colors', tab === key ? 059669">'bg-primary-600 text-white' : 059669">'text-slate-500 hover:bg-
slate-100 dark:text-slate-400 dark:hover:bg-slate-800')}>
          {label}
        </button>
      ))}
    </div>

    {tab === 059669">'suppliers' && (
      <div className=059669">"card p-4">
        <Toolbar searchValue={sSearch} onSearchChange={sSetSearch}>
          <Select value={sFilters.status || 059669">''} onChange={(v) => sSetFilters(059669">'status', v)}
options={([{ value: 059669">'active', label: 059669">'Active' }, { value: 059669">'preferred', label:
059669">'Preferred' }, { value: 059669">'inactive', label: 059669">'Inactive' }, { value: 059669">'blacklisted',
label: 059669">'Blacklisted' }]} placeholder=059669">'All Statuses' />
        </Toolbar>
      </div>
      <div className=059669">"mt-4 card">
        <DataTable columns={supplierColumns} rows={suppliers} onRowClick={setSelectedSupplier} isLoading=
{sLoading} />
        <Pagination page={sPage} totalPages={sTotalPages} total={sTotal} pageSize={sPageSize} onPageChange=
{sSetPage} />
      </div>
    </>
  )}
)

```



```

{tab === 059669}>'requests' && (
  <>
    <div className=059669>"card p-4">
      <Toolbar searchValue={rSearch} onSearchChange={rSetSearch}>
        <Select value={Filters.status || 059669}>' ' onChange={(v) => rSetFilters(059669}>'status', v)}
options=[[ { value: 059669}>'draft', label: 059669}>'Draft' }, { value: 059669}>'pending_approval', label:
059669}>'Pending Approval' }, { value: 059669}>'approved', label: 059669}>'Approved' }, { value: 059669}>'rejected',
label: 059669}>'Rejected' }, { value: 059669}>'ordered', label: 059669}>'Ordered' }, { value: 059669}>'received',
label: 059669}>'Received' }]] placeholder=059669}>"All Statuses" />
      </Toolbar>
    </div>
    <div className=059669>"mt-4 card">
      <DataTable columns={requestColumns} rows={requests} onRowClick={setSelectedRequest} isLoading={rLoading}
/>
      <Pagination page={rPage} totalPages={rTotalPages} total={rTotal} pageSize={rPageSize} onPageChange=
{rSetPage} />
    </div>
  </>
)}

{tab === 059669}>'orders' && (
  <>
    <div className=059669>"card p-4">
      <Toolbar searchValue={oSearch} onSearchChange={oSetSearch}>
        <Select value={oFilters.status || 059669}>' ' onChange={(v) => oSetFilters(059669}>'status', v)}
options=[[ { value: 059669}>'draft', label: 059669}>'Draft' }, { value: 059669}>'sent', label: 059669}>'Sent' }, {
value: 059669}>'confirmed', label: 059669}>'Confirmed' }, { value: 059669}>'partial', label: 059669}>'Partial' }, {
value: 059669}>'received', label: 059669}>'Received' }]] placeholder=059669}>"All Statuses" />
      </Toolbar>
    </div>
    <div className=059669>"mt-4 card">
      <DataTable columns={orderColumns} rows={orders} onRowClick={setSelectedOrder} isLoading={oLoading} />
      <Pagination page={oPage} totalPages={oTotalPages} total={oTotal} pageSize={oPageSize} onPageChange=
{oSetPage} />
    </div>
  </>
)}

{059669}>"color:#6b7280;font-style:italic">/* Supplier detail */}
<Modal open={!selectedSupplier} onClose={() => setSelectedSupplier(600>null)} title=059669>"Supplier
Profile" size=059669>"lg">
  {selectedSupplier && (
    <div className=059669>"space-y-4">
      <div className=059669>"flex items-center justify-between">
        <div>
          <h3 className=059669>"text-lg font-semibold text-slate-900 dark:text-white">{selectedSupplier.name}
</h3>
          <p className=059669>"text-sm text-slate-400">{selectedSupplier.contact_person} .
{selectedSupplier.email}</p>
        </div>
        <StatusBadge status={selectedSupplier.status} />
      </div>
      <div className=059669>"grid grid-cols-2 gap-4 sm:grid-cols-3">
        <div><p className=059669>"label">Rating</p><div className=059669>"flex items-center gap-1"><Star
className=059669>"h-4 w-4 text-warning-400 fill-warning-400" /><span className=059669>"text-sm font-medium text-
slate-700 dark:text-slate-300">{selectedSupplier.rating}</span></div></div>
        <div><p className=059669>"label">Lead Time</p><p className=059669>"text-sm text-slate-700 dark:text-
slate-300">{selectedSupplier.lead_time_days} days</p></div>
        <div><p className=059669>"label">Phone</p><p className=059669>"text-sm text-slate-700 dark:text-
slate-300">{selectedSupplier.phone}</p></div>
        <div><p className=059669>"label">Address</p><p className=059669>"text-sm text-slate-700 dark:text-
slate-300">{selectedSupplier.address}</p></div>
        <div><p className=059669>"label">Established</p><p className=059669>"text-sm text-slate-700
dark:text-slate-300">{selectedSupplier.established}</p></div>
        <div><p className=059669>"label">Parts Supplied</p><p className=059669>"text-sm text-slate-700
dark:text-slate-300">{selectedSupplier.parts_supplied}</p></div>
        <div><p className=059669>"label">Total Orders</p><p className=059669>"text-sm text-slate-700
dark:text-slate-300">{selectedSupplier.total_orders}</p></div>
        <div><p className=059669>"label">Total Value</p><p className=059669>"text-sm font-semibold text-
slate-700 dark:text-slate-300">${(selectedSupplier.total_value / 1000000).toFixed(2)}M</p></div>
      </div>
      <div className=059669>"grid grid-cols-2 gap-4">
        <div className=059669>"rounded-lg border border-slate-100 p-3 dark:border-slate-800">

```



```


<p className=059669">"text-xs text-slate-400">On-Time Delivery Rate</p>  

  <div className=059669">"mt-2"><ProgressBar value={selectedSupplier.on_time_delivery_rate}  

tone=059669">"success" /></div>  

  <p className=059669">"mt-1 text-sm font-semibold text-slate-700 dark:text-slate-300">  

{selectedSupplier.on_time_delivery_rate}%</p>  

</div>  

  <div className=059669">"rounded-lg border border-slate-100 p-3 dark:border-slate-800">  

    <p className=059669">"text-xs text-slate-400">Quality Rate</p>  

    <div className=059669">"mt-2"><ProgressBar value={selectedSupplier.quality_rate}  

tone=059669">"primary" /></div>  

    <p className=059669">"mt-1 text-sm font-semibold text-slate-700 dark:text-slate-300">  

{selectedSupplier.quality_rate}%</p>  

  </div>  

</div>  

})  

</Modal>


```



```

{059669">"color:#6b7280;font-style:italic"}/* Request detail */  

<Modal open={!selectedRequest} onClose={() => setSelectedRequest(600">>null)} title=059669">"Purchase Request"  

size=059669">"lg">  

  {selectedRequest && (  

    <div className=059669">"space-y-4">  

      <div className=059669">"flex items-center justify-between">  

        <h3 className=059669">"text-lg font-semibold text-slate-900 dark:text-white">  

{selectedRequest.request_number}</h3>  

        <StatusBadge status={selectedRequest.status} />  

      </div>  

      <div className=059669">"grid grid-cols-2 gap-4">  

        <div><p className=059669">"label">Requested By</p><p className=059669">"text-sm text-slate-700  

dark:text-slate-300">{selectedRequest.requested_by}</p></div>  

        <div><p className=059669">"label">Department</p><p className=059669">"text-sm text-slate-700 dark:text-  

slate-300">{selectedRequest.department}</p></div>  

        <div><p className=059669">"label">Created</p><p className=059669">"text-sm text-slate-700 dark:text-  

slate-300">{selectedRequest.created_date}</p></div>  

        <div><p className=059669">"label">Required By</p><p className=059669">"text-sm text-slate-700  

dark:text-slate-300">{selectedRequest.required_date}</p></div>  

        <div><p className=059669">"label">Priority</p><Badge tone={selectedRequest.priority ===  

059669">'urgent' ? 059669">'danger' : 059669">'warning'}>{selectedRequest.priority}</Badge></div>  

        <div><p className=059669">"label">Approved By</p><p className=059669">"text-sm text-slate-700  

dark:text-slate-300">{selectedRequest.approved_by || 059669">'-'}</p></div>  

      </div>  

      <div><p className=059669">"label">Justification</p><p className=059669">"text-sm text-slate-700  

dark:text-slate-300">{selectedRequest.justification}</p></div>  

      <div>  

        <p className=059669">"label">Items</p>  

        <div className=059669">"overflow-x-auto">  

          <table className=059669">"w-full">  

            <thead><tr className=059669">"border-b border-slate-200 dark:border-slate-800"><th  

className=059669">"table-th">Part</th><th className=059669">"table-th">Qty</th><th className=059669">"table-th">Unit  

Cost</th><th className=059669">"table-th">Total</th></tr></thead>  

            <tbody className=059669">"divide-y divide-slate-100 dark:divide-slate-800/70">  

              {selectedRequest.items.map((it, i) => (  

                <tr key={i}><td className=059669">"table-tid">{it.part_name}</td><td className=059669">"table-  

tid">{it.quantity}</td><td className=059669">"table-tid">${it.unit_cost}</td><td className=059669">"table-tid font-  

medium">${it.total.toLocaleString()}</td></tr>  

              ))}  

            </tbody>  

          </table>  

        </div>  

      </div>  

      <div className=059669">"flex justify-end"><p className=059669">"text-lg font-bold text-slate-900  

dark:text-white">Total: ${selectedRequest.total_value.toLocaleString()}</p></div>  

    </div>  

  )}  

</Modal>
```



```

{059669">"color:#6b7280;font-style:italic"}/* Order detail */  

<Modal open={!selectedOrder} onClose={() => setSelectedOrder(600">>null)} title=059669">"Purchase Order"  

size=059669">"lg">  

  {selectedOrder && (  

    <div className=059669">"space-y-4">  

      <div className=059669">"flex items-center justifv-between">
```

```

        <h3 className=059669>"text-lg font-semibold text-slate-900 dark:text-white">
{selectedOrder.order_number}</h3>
        <StatusBadge status={selectedOrder.status} />
    </div>
    <div className=059669>"grid grid-cols-2 gap-4">
        <div><p className=059669>"label">Supplier</p><p className=059669>"text-sm text-slate-700 dark:text-slate-300">{selectedOrder.supplier_name}</p></div>
        <div><p className=059669>"label">Created By</p><p className=059669>"text-sm text-slate-700 dark:text-slate-300">{selectedOrder.created_by}</p></div>
        <div><p className=059669>"label">Order Date</p><p className=059669>"text-sm text-slate-700 dark:text-slate-300">{selectedOrder.order_date}</p></div>
        <div><p className=059669>"label">Expected Delivery</p><p className=059669>"text-sm text-slate-700 dark:text-slate-300">{selectedOrder.expected_delivery}</p></div>
        <div><p className=059669>"label">Actual Delivery</p><p className=059669>"text-sm text-slate-700 dark:text-slate-300">{selectedOrder.actual_delivery || 059669}'-'}</p></div>
        <div><p className=059669>"label">Delivery Progress</p><div className=059669>"mt-1"><ProgressBar value={selectedOrder.delivery_progress} tone=059669>"primary" /></div></div>
    </div>
    <div>
        <p className=059669>"label">Items</p>
        <div className=059669>"overflow-x-auto">
            <table className=059669>"w-full">
                <thead><tr className=059669>"border-b border-slate-200 dark:border-slate-800"><th className=059669>"table-th">Part</th><th className=059669>"table-th">Qty</th><th className=059669>"table-th">Unit Cost</th><th className=059669>"table-th">Total</th></tr></thead>
                <tbody className=059669>"divide-y divide-slate-100 dark:divide-slate-800/70">
                    {selectedOrder.items.map((it, i) => (
                        <tr key={i}><td className=059669>"table-td">{it.part_name}</td><td className=059669>"table-td">{it.quantity}</td><td className=059669>"table-td">${it.unit_cost}</td><td className=059669>"table-td font-medium">${it.total.toLocaleString()}</td></tr>
                    ))}
                </tbody>
            </table>
        </div>
    </div>
    <div className=059669>"flex justify-end"><p className=059669>"text-lg font-bold text-slate-900 dark:text-white">Total: ${selectedOrder.total_value.toLocaleString()}</p></div>
</div>
    )}
</Modal>
</div>
);
}

```

```

600">import { useState } 600">from 059669">'react';
600">import { ShieldCheck, Plus, AlertTriangle, CalendarClock } 600">from 059669">'lucide-react';
600">import { DataTable, 600">type Column } 600">from 059669">'@/components/ui/DataTable';
600">import { Pagination } 600">from 059669">'@/components/ui/Pagination';
600">import { Toolbar, Select, PageHeader } 600">from 059669">'@/components/ui/Toolbar';
600">import { StatusBadge } 600">from 059669">'@/components/ui/Badge';
600">import { Modal } 600">from 059669">'@/components/ui/Modal';
600">import { usePaginatedData } 600">from 059669">'@/hooks/usePaginatedData';
600">import { api } 600">from 059669">'@/lib/api';
600">import 600">type { Warranty } 600">from 059669">'@/types';

600">export 600">function WarrantyPage() {
  600">const [selected, setSelected] = useState<Warranty | 600">null>(600">null);
  600">const {
    data, total, page, pageSize, totalPages, search, filters, isLoading,
    setPage, setFilters, setSearch,
  } = usePaginatedData<Warranty>(api.getWarranties, [059669">'machine_name', 059669">'warranty_number',
059669">'provider']));

  600">const columns: Column<Warranty>[] = [
    { key: 059669">'warranty_number', header: 059669">'Warranty #', sortable: 600">true, render: (w) => <span
className=059669">"font-mono text-xs font-medium text-slate-900 dark:text-white">{w.warranty_number}</span> },
    {
      key: 059669">'machine_name', header: 059669">'Machine / Component', sortable: 600">true,
      render: (w) => (
        <div>
          <p className=059669">"font-medium text-slate-900 dark:text-white">{w.machine_name}</p>
          <p className=059669">"text-xs text-slate-400">{w.component_name || w.machine_id_code}</p>
        </div>
      ),
    },
    { key: 059669">'600">type', header: 059669">'Type', sortable: 600">true, render: (w) => <span
className=059669">"capitalize text-slate-600 dark:text-slate-300">{w.600">type}</span> },
    { key: 059669">'provider', header: 059669">'Provider', sortable: 600">true, render: (w) => <span
className=059669">"text-slate-600 dark:text-slate-300">{w.provider}</span> },
    { key: 059669">'start_date', header: 059669">'Start', sortable: 600">true, render: (w) => <span
className=059669">"text-slate-600 dark:text-slate-300">{w.start_date}</span> },
    { key: 059669">'end_date', header: 059669">'End', sortable: 600">true, render: (w) => <span
className=059669">"text-slate-600 dark:text-slate-300">{w.end_date}</span> },
    {
      key: 059669">'days_remaining', header: 059669">'Days Left', sortable: 600">true, align: 059669">'center',
      render: (w) => (
        <span className={059669">`text-sm font-medium ${w.days_remaining < 0 ? 'text-danger-600 dark:text-danger-400'
: w.days_remaining < 60 ? 'text-warning-600 dark:text-warning-400' : 'text-slate-600 dark:text-slate-300'}`}>
          {w.days_remaining < 0 ? 059669">'Expired' : w.days_remaining}
        </span>
      ),
    },
    { key: 059669">'status', header: 059669">'Status', sortable: 600">true, render: (w) => <StatusBadge status=
{w.status} /> },
  ];

  600">return (
    <div>
      <PageHeader
        title=059669">"Warranty Management"
        subtitle=059669">"Track machine and component warranties, expiry alerts, and claims"
        actions={<button className=059669">"btn-primary"><Plus className=059669">"h-4 w-4" /> Add Warranty</button>}
      />

      <div className=059669">"card p-4">
        <Toolbar searchValue={search} onSearchChange={setSearch}>
          <Select value={filters.status || 059669">''} onChange={(v) => setFilters(059669">'status', v)} options={[{
value: 059669">'active', label: 059669">'Active' }, { value: 059669">'expiring_soon', label: 059669">'Expiring Soon'
}, { value: 059669">'expired', label: 059669">'Expired' }, { value: 059669">'claimed', label: 059669">'Claimed' }]}
placeholder=059669">"All Statuses" />
          <Select value={filters.600">type || 059669">''} onChange={(v) => setFilters(059669">'600">type', v)}
options={[{ value: 059669">'machine', label: 059669">'Machine' }, { value: 059669">'component', label:
059669">'Component' }]} placeholder=059669">"All Types" />

```

```

    </Toolbar>
  </div>

  <div className=059669">"mt-4 card">
    <DataTable columns={columns} rows={data} onClick={setSelected} isLoading={isLoading} />
    <Pagination page={page} totalPages={totalPages} total={total} pageSize={pageSize} onPageChange={setPage} />
  </div>

  <Modal open={!!selected} onClose={() => setSelected(600">null)} title=059669">"Warranty Details"
size=059669">"lg">
    {selected && (
      <div className=059669">"space-y-4">
        <div className=059669">"flex items-center justify-between">
          <div>
            <h3 className=059669">"text-lg font-semibold text-slate-900 dark:text-white">{selected.machine_name}
          </h3>
          <p className=059669">"font-mono text-sm text-slate-400">{selected.warranty_number}</p>
        </div>
        <StatusBadge status={selected.status} />
      </div>
      <div className=059669">"grid grid-cols-2 gap-4">
        <div><p className=059669">"label">Type</p><p className=059669">"text-sm capitalize text-slate-700
dark:text-slate-300">{selected.600">type}</p></div>
        <div><p className=059669">"label">Provider</p><p className=059669">"text-sm text-slate-700 dark:text-
slate-300">{selected.provider}</p></div>
        <div><p className=059669">"label">Start Date</p><p className=059669">"text-sm text-slate-700 dark:text-
slate-300">{selected.start_date}</p></div>
        <div><p className=059669">"label">End Date</p><p className=059669">"text-sm text-slate-700 dark:text-
slate-300">{selected.end_date}</p></div>
        <div><p className=059669">"label">Days Remaining</p><p className=059669">"text-sm font-semibold
${selected.days_remaining < 0 ? 'text-danger-600 dark:text-danger-400' : selected.days_remaining < 60 ? 'text-
warning-600 dark:text-warning-400' : 'text-accent-600 dark:text-accent-400'}`}>{selected.days_remaining < 0 ?
059669">'Expired' : 059669">`${selected.days_remaining} days`}</p></div>
        {selected.component_name && <div><p className=059669">"label">Component</p><p className=059669">"text-
sm text-slate-700 dark:text-slate-300">{selected.component_name}</p></div>}
      </div>
      <div><p className=059669">"label">Coverage</p><p className=059669">"text-sm text-slate-700 dark:text-
slate-300">{selected.coverage}</p></div>
      <div><p className=059669">"label">Terms & Conditions</p><p className=059669">"text-sm text-slate-700
dark:text-slate-300">{selected.terms}</p></div>
      {selected.status === 059669">'expiring_soon' && (
        <div className=059669">"flex items-center gap-2 rounded-lg bg-warning-50 p-3 dark:bg-warning-600/10">
          <AlertTriangle className=059669">"h-5 w-5 text-warning-600 dark:text-warning-400" />
          <p className=059669">"text-sm text-warning-700 dark:text-warning-300">This warranty expires 600">in
{selected.days_remaining} days. Consider renewal or claim filing.</p>
        </div>
      )}
    </div>
  )}
</Modal>
</div>
);
}

```

```

600">import { useState } 600">from 059669">'react';
600">import { ClipboardList, Plus, Download, Paperclip } 600">from 059669">'lucide-react';
600">import { DataTable, 600">type Column } 600">from 059669">'@/components/ui/DataTable';
600">import { Pagination } 600">from 059669">'@/components/ui/Pagination';
600">import { Toolbar, Select, PageHeader } 600">from 059669">'@/components/ui/Toolbar';
600">import { Badge } 600">from 059669">'@/components/ui/Badge';
600">import { Modal } 600">from 059669">'@/components/ui/Modal';
600">import { usePaginatedData } 600">from 059669">'@/hooks/usePaginatedData';
600">import { api } 600">from 059669">'@/lib/api';
600">import 600">type { ServiceRecord } 600">from 059669">'@/types';

600">export 600">function ServiceHistoryPage() {
  600">const [selected, setSelected] = useState<ServiceRecord | 600">null>(600">null);
  600">const {
    data, total, page, pageSize, totalPages, search, filters, isLoading,
    setPage, setFilters, setSearch,
  } = usePaginatedData<ServiceRecord>(api.getServiceRecords, [059669">'machine_name', 059669">'service_id',
059669">'technician']);

  600">const columns: Column<ServiceRecord>[] = [
    { key: 059669">'service_id', header: 059669">'Service #', sortable: 600">true, render: (s) => <span
className=059669">"font-mono text-xs font-medium text-slate-900 dark:text-white">{s.service_id}</span> },
    {
      key: 059669">'machine_name', header: 059669">'Machine', sortable: 600">true,
      render: (s) => <div><p className=059669">"font-medium text-slate-900 dark:text-white">{s.machine_name}</p><p
className=059669">"text-xs text-slate-400">{s.machine_id_code}</p></div>,
    },
    { key: 059669">'date', header: 059669">'Date', sortable: 600">true, render: (s) => <span className=059669">"text-
slate-600 dark:text-slate-300">{s.date}</span> },
    { key: 059669">'technician', header: 059669">'Technician', sortable: 600">true, render: (s) => <span
className=059669">"text-slate-600 dark:text-slate-300">{s.technician}</span> },
    { key: 059669">'600">type', header: 059669">'Type', sortable: 600">true, render: (s) => <Badge tone={s.600">type
=== 059669">'preventive' ? 059669">'primary' : s.600">type === 059669">'corrective' ? 059669">'warning' : s.600">type
=== 059669">'predictive' ? 059669">'success' : 059669">'neutral'}>{s.600">type}</Badge> },
    { key: 059669">'labor_hours', header: 059669">'Labor Hours', sortable: 600">true, align: 059669">'center',
render: (s) => <span className=059669">"text-slate-600 dark:text-slate-300">{s.labor_hours}h</span> },
    { key: 059669">'downtime_hours', header: 059669">'Downtime', sortable: 600">true, align: 059669">'center',
render: (s) => <span className=059669">"text-slate-600 dark:text-slate-300">{s.downtime_hours}h</span> },
    { key: 059669">'total_cost', header: 059669">'Cost', sortable: 600">true, align: 059669">'right', render: (s) =>
<span className=059669">"font-medium text-slate-700 dark:text-slate-300">${s.total_cost.toLocaleString()}</span> },
  ];

  600">return (
    <div>
      <PageHeader
        title=059669">"Service History"
        subtitle=059669">"Complete digital service record 600">for every machine"
        actions={<button className=059669">"btn-secondary"><Download className=059669">"h-4 w-4" /> Export</button>}
      />

      <div className=059669">"card p-4">
        <Toolbar searchValue={search} onSearchChange={setSearch}>
          <Select value={filters.600">type || 059669">''} onChange={(v) => setFilters(059669">'600">type', v)}
options={[[{ value: 059669">'preventive', label: 059669">'Preventive' }, { value: 059669">'corrective', label:
059669">'Corrective' }, { value: 059669">'predictive', label: 059669">'Predictive' }, { value: 059669">'inspection',
label: 059669">'Inspection' }]} placeholder=059669">"All Types" />
        </Toolbar>
      </div>

      <div className=059669">"mt-4 card">
        <DataTable columns={columns} rows={data} onClick={setSelected} isLoading={isLoading} />
        <Pagination page={page} totalPages={totalPages} total={total} pageSize={pageSize} onPageChange={setPage} />
      </div>

      <Modal open={!selected} onClose={() => setSelected(600">null)} title=059669">"Service Record"
size=059669">"lg">
        {selected && (
          <div className=059669">"space-y-4">
            <div className=059669">"flex items-center justify-between">

```

```


###


```



```

600">import { useEffect, useState } 600">from 059669">'react';
600">import { Activity, AlertTriangle, TrendingUp, Gauge, ShieldAlert } 600">from 059669">'lucide-react';
600">import {
  ResponsiveContainer, BarChart, Bar, XAxis, YAxis, CartesianGrid, Tooltip, Legend, RadarChart, PolarGrid,
  PolarAngleAxis, PolarRadiusAxis, Radar,
} 600">from 059669">'recharts';
600">import { PageHeader } 600">from 059669">'@/components/ui/Toolbar';
600">import { KPICard } 600">from 059669">'@/components/ui/KPICard';
600">import { RiskBadge } 600">from 059669">'@/components/ui/Badge';
600">import { HealthScoreRing, ProgressBar } 600">from 059669">'@/components/ui/Indicators';
600">import { machines } 600">from 059669">'@/data/seed';

600">export 600">function PredictivePage() {
  600">const [sortBy, setSortBy] = useState<059669">'risk' | 059669">'health' | 059669">'mtbf'>(059669">'risk');

  600">const sortedMachines = [...machines].sort((a, b) => {
    600">if (sortBy === 059669">'risk') {
      600">const order = { critical: 0, high: 1, medium: 2, low: 3 };
      600">return order[a.risk_level] - order[b.risk_level];
    }
    600">if (sortBy === 059669">'health') 600">return a.health_score - b.health_score;
    600">return a.mtbf_hours - b.mtbf_hours;
  });

  600">const topRisk = sortedMachines.slice(0, 12);

  600">const avgMTBF = Math.round(machines.reduce((s, m) => s + m.mtbf_hours, 0) / machines.length);
  600">const avgMTTR = Math.round(machines.reduce((s, m) => s + m.mttr_hours, 0) / machines.length);
  600">const avgHealth = Math.round(machines.reduce((s, m) => s + m.health_score, 0) / machines.length);
  600">const criticalCount = machines.filter((m) => m.risk_level === 059669">'critical' || m.risk_level ===
059669">'high').length;

  600">const radarData = machines.slice(0, 5).map((m) => ({
    machine: m.machine_id,
    health: m.health_score,
    reliability: Math.min(100, Math.round((m.mtbf_hours / 20000) * 100)),
    maintainability: Math.max(0, 100 - m.mttr_hours * 2),
  }));

  600">const recommendedActions = sortedMachines.filter((m) => m.health_score < 70).slice(0, 6);

  600">return (
    <div>
      <PageHeader title=059669">"Predictive Maintenance Analytics" subtitle=059669">"MTBF, MTTR, machine health
scoring, and risk assessment" />

      <div className=059669">"grid grid-cols-1 gap-4 sm:grid-cols-2 lg:grid-cols-4">
        <KPICard label=059669">"Avg MTBF" value={059669">`${avgMTBF.toLocaleString()}h`} icon={<TrendingUp
className=059669">"h-5 w-5" /> tone=059669">"primary" subtitle=059669">"Mean Time Between Failures" />
        <KPICard label=059669">"Avg MTTR" value={059669">`${avgMTTR}h`} icon={<Activity className=059669">"h-5 w-5"
/>} tone=059669">"warning" subtitle=059669">"Mean Time To Repair" />
        <KPICard label=059669">"Avg Health Score" value={059669">`${avgHealth}%`} icon={<Gauge className=059669">"h-5
w-5" />} tone=059669">"success" />
        <KPICard label=059669">"High-Risk Machines" value={criticalCount} icon={<ShieldAlert className=059669">"h-5
w-5" />} tone=059669">"danger" subtitle=059669">"Critical or high risk" />
      </div>

      {059669">"color:#6b7280;font-style:italic">/* Sort controls */}
      <div className=059669">"mt-6 flex items-center gap-2">
        <span className=059669">"text-sm text-slate-500 dark:text-slate-400">Sort by:</span>
        {([059669">'risk', 059669">'Risk Level', [059669">'health', 059669">'Health Score'], [059669">'mtbf',
059669">'MTBF']] 600">as 600">const).map(([key, label]) => (
          <button
            key={key}
            onClick={() => setSortBy(key)}
            className={059669">`rounded-lg px-3 py-1.5 text-xs font-medium transition-colors ${sortBy === key ? 'bg-
primary-600 text-white' : 'border border-slate-200 text-slate-500 dark:border-slate-700 dark:text-slate-400'}`}
            >
              {label}

```

```

    </button>
  )}
</div>

{059669}>"color:#6b7280;font-style:italic"/>
```



```

        <Radar name=059669>"Reliability" dataKey=059669>"reliability" stroke=059669>"#10b981"
fill=059669>"#10b981" fillOpacity={0.2} />
        <Radar name=059669>"Maintainability" dataKey=059669>"maintainability" stroke=059669>"#f59e0b"
fill=059669>"#f59e0b" fillOpacity={0.15} />
        <Legend wrapperStyle={{ fontSize: 11 }} />
        <Tooltip contentStyle={{ borderRadius: 12, border: 059669>'1px solid #e2e8f0', fontSize: 12 }} />
    </RadarChart>
</ResponsiveContainer>
</div>
</div>

{059669>"color:#6b7280;font-style:italic">/* Recommended actions table */}
<div className=059669>"mt-6 card p-5">
    <h3 className=059669>"mb-4 flex items-center gap-2 text-sm font-semibold text-slate-900 dark:text-white">
        <AlertTriangle className=059669>"h-4 w-4 text-warning-500" /> Maintenance Priority Recommendations
    </h3>
    <div className=059669>"space-y-2">
        {recommendedActions.map((m) => (
            <div key={m.id} className=059669>"flex items-center gap-4 rounded-lg border border-slate-100 p-3
dark:border-slate-800">
                <HealthScoreRing score={m.health_score} size={44} />
                <div className=059669>"min-w-0 flex-1">
                    <p className=059669>"text-sm font-medium text-slate-900 dark:text-white">{m.machine_id} · {m.name}
</p>

                    <p className=059669>"text-xs text-slate-400">{m.factory_name} · {m.department_name}</p>
                </div>
                <div className=059669>"hidden sm:block w-32">
                    <p className=059669>"mb-1 text-[10px] text-slate-400">Reliability</p>
                    <ProgressBar value={m.health_score} tone={m.health_score >= 60 ? 059669>'success' : m.health_score
>= 40 ? 059669>'warning' : 059669>'danger'} />
                </div>
                <RiskBadge level={m.risk_level} />
            </div>
        ))}
    </div>
</div>
</div>
);
}

```

```

600">import { useState } 600">from 059669">'react';
600">import { Bell, CheckCheck, AlertTriangle, Info, AlertCircle, Filter } 600">from 059669">'lucide-react';
600">import { clsx } 600">from 059669">'clsx';
600">import { PageHeader, Select } 600">from 059669">'@/components/ui/Toolbar';
600">import { Badge } 600">from 059669">'@/components/ui/Badge';
600">import { notifications 600">as allNotifications } 600">from 059669">'@/data/seed';
600">import { useAppSelector } 600">from 059669">'@/store';

600">export 600">function NotificationsPage() {
  600">const user = useAppSelector((s) => s.auth.user);
  600">const [filter, setFilter] = useState<string>(059669">'all');
  600">const [readFilter, setReadFilter] = useState<string>(059669">'all');

  600">if (!user) 600">return 600">>null;
  600">let notifs = allNotifications.filter((n) => n.target_roles.includes(user.role));

  600">if (filter !== 059669">'all') notifs = notifs.filter((n) => n.priority === filter);
  600">if (readFilter === 059669">'unread') notifs = notifs.filter((n) => !n.read);
  600">if (readFilter === 059669">'read') notifs = notifs.filter((n) => n.read);

  600">const unreadCount = allNotifications.filter((n) => n.target_roles.includes(user.role) && !n.read).length;

  600">const iconFor = (priority: string) => {
    600">if (priority === 059669">'critical') 600">return <AlertCircle className=059669">"h-5 w-5 text-danger-600
dark:text-danger-400" />;
    600">if (priority === 059669">'warning') 600">return <AlertTriangle className=059669">"h-5 w-5 text-warning-600
dark:text-warning-400" />;
    600">return <Info className=059669">"h-5 w-5 text-sky-600 dark:text-sky-400" />;
  };

  600">const bgFor = (priority: string) => {
    600">if (priority === 059669">'critical') 600">return 059669">'bg-danger-50 dark:bg-danger-600/10';
    600">if (priority === 059669">'warning') 600">return 059669">'bg-warning-50 dark:bg-warning-600/10';
    600">return 059669">'bg-sky-50 dark:bg-sky-600/10';
  };

  600">return (
    <div>
      <PageHeader
        title=059669">"Notification Center"
        subtitle={059669">`${unreadCount} unread notifications`}
        actions={<button className=059669">"btn-secondary"><CheckCheck className=059669">"h-4 w-4" /> Mark all
read</button>}
      />

      <div className=059669">"mb-4 flex flex-wrap items-center gap-2">
        <Filter className=059669">"h-4 w-4 text-slate-400" />
        <Select value={filter} onChange={setFilter} options={[{ value: 059669">'critical', label: 059669">'Critical'
}, { value: 059669">'warning', label: 059669">'Warning' }, { value: 059669">'info', label: 059669">'Info' }]}
placeholder=059669">"All Priorities" />
        <Select value={readFilter} onChange={setReadFilter} options={[{ value: 059669">'unread', label:
059669">'Unread' }, { value: 059669">'read', label: 059669">'Read' }]} placeholder=059669">"All" />
      </div>

      <div className=059669">"space-y-2">
        {notifs.map((n) => (
          <div
            key={n.id}
            className={clsx(
              059669">'card card-hover flex items-start gap-4 p-4',
              !n.read && 059669">'ring-1 ring-primary-200 dark:ring-primary-800',
            )}
          >
            <div className={clsx(059669">'flex h-10 w-10 shrink-0 items-center justify-center rounded-lg',
bgFor(n.priority))}>
              {iconFor(n.priority)}
            </div>
            <div className=059669">"min-w-0 flex-1">
              <div className=059669">"flex items-start justify-between gap-2">

```

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        <div>
          <p className=059669>"text-sm font-semibold text-slate-900 dark:text-white">{n.title}</p>
          <p className=059669>"mt-0.5 text-sm text-slate-500 dark:text-slate-400">{n.message}</p>
        </div>
        {!n.read && <span className=059669>"h-2 w-2 shrink-0 rounded-full bg-primary-500" />}
      </div>
      <div className=059669>"mt-2 flex items-center gap-2">
        <Badge tone={n.priority === 059669}>'critical' ? 059669>'danger' : n.priority === 059669>'warning'
? 059669>'warning' : 059669>'info'}>{n.module}</Badge>
        <span className=059669>"text-xs text-slate-400">{n.timestamp.replace(059669>'T', 059669>' '
').replace(059669>'Z', 059669>'')}</span>
      </div>
    </div>
    {n.action_url && (
      <a href={n.action_url} className=059669>"shrink-0 text-xs font-medium text-primary-600 hover:underline
dark:text-primary-400">
        View
      </a>
    )}
  </div>
)}}
```

```

{notifs.length === 0 && (
  <div className=059669>"card flex h-40 items-center justify-center text-sm text-slate-400">
    <Bell className=059669>"mr-2 h-5 w-5" /> No notifications
  </div>
)}
</div>
</div>
);
}

```

```

600">import { useState } 600">from 059669">'react';
600">import { FileClock, Download, Monitor, Globe } 600">from 059669">'lucide-react';
600">import { DataTable, 600">type Column } 600">from 059669">'@/components/ui/DataTable';
600">import { Pagination } 600">from 059669">'@/components/ui/Pagination';
600">import { Toolbar, Select, PageHeader } 600">from 059669">'@/components/ui/Toolbar';
600">import { Badge } 600">from 059669">'@/components/ui/Badge';
600">import { usePaginatedData } 600">from 059669">'@/hooks/usePaginatedData';
600">import { api } 600">from 059669">'@/lib/api';
600">import { ROLE_MAP } 600">from 059669">'@/lib/rbac';
600">import 600">type { AuditLog } 600">from 059669">'@/types';

600">export 600">function AuditPage() {
  600">const {
    data, total, page, pageSize, totalPages, search, filters, isLoading,
    setPage, setFilters, setSearch,
  } = usePaginatedData<AuditLog>(api.getAuditLogs, [059669">'user', 059669">'action', 059669">'module',
059669">'entity_id']);

  600">const columns: Column<AuditLog>[] = [
    {
      key: 059669">'timestamp', header: 059669">'Timestamp', sortable: 600">true,
      render: (l) => <span className=059669">"font-mono text-xs text-slate-600 dark:text-slate-300">
{l.timestamp.replace(059669">'T', 059669">' ').replace(059669">'Z', 059669">'')}</span>,
    },
    { key: 059669">'user', header: 059669">'User', sortable: 600">true, render: (l) => <div><p
className=059669">"font-medium text-slate-900 dark:text-white">{l.user}</p><p className=059669">"text-xs text-slate-
400">{ROLE_MAP[l.user_role]?.name || l.user_role}</p></div> },
    { key: 059669">'action', header: 059669">'Action', sortable: 600">true, render: (l) => <Badge tone={l.action ===
059669">'CREATE' ? 059669">'success' : l.action === 059669">'DELETE' || l.action === 059669">'REJECT' ?
059669">'danger' : l.action === 059669">'APPROVE' || l.action === 059669">'RESOLVE' || l.action ===
059669">'COMPLETE' ? 059669">'primary' : 059669">'neutral'}>{l.action}</Badge> },
    { key: 059669">'module', header: 059669">'Module', sortable: 600">true, render: (l) => <span
className=059669">"text-slate-600 dark:text-slate-300">{l.module}</span> },
    { key: 059669">'entity_id', header: 059669">'Entity', render: (l) => l.entity_id ? <span className=059669">"font-
mono text-xs text-slate-500">{l.entity_id}</span> : <span className=059669">"text-slate-300"></span> },
    {
      key: 059669">'changes', header: 059669">'Changes',
      render: (l) => l.previous_value || l.updated_value ? (
        <div className=059669">"text-xs">
          {l.previous_value && <span className=059669">"text-danger-600 dark:text-danger-400 line-through">
{l.previous_value}</span>}
          {l.previous_value && l.updated_value && <span className=059669">"mx-1 text-slate-400">.</span>}
          {l.updated_value && <span className=059669">"text-accent-600 dark:text-accent-400">{l.updated_value}
</span>}
        </div>
      ) : <span className=059669">"text-slate-300"></span>,
    },
    { key: 059669">'ip_address', header: 059669">'IP Address', render: (l) => <span className=059669">"font-mono
text-xs text-slate-500">{l.ip_address}</span> },
    { key: 059669">'device', header: 059669">'Device', render: (l) => <div className=059669">"flex items-center gap-
1.5"><Monitor className=059669">"h-3 w-3 text-slate-400" /><span className=059669">"text-xs text-slate-500">
{l.device}</span></div> },
  ];

  600">return (
    <div>
      <PageHeader
        title=059669">"Audit & Activity Tracking"
        subtitle=059669">"Complete audit trail 600">of all user actions and system changes"
        actions={<button className=059669">"btn-secondary"><Download className=059669">"h-4 w-4" /> Export
Logs</button>}
      />

      <div className=059669">"card p-4">
        <Toolbar searchValue={search} onSearchChange={setSearch}>
          <Select value={filters.module || 059669">''} onChange={(v) => setFilters(059669">'module', v)} options={[{
value: 059669">'Machines', label: 059669">'Machines' }, { value: 059669">'Maintenance', label: 059669">'Maintenance'
}, { value: 059669">'Breakdowns', label: 059669">'Breakdowns' }, { value: 059669">'Inventory', label:
059669">'Inventory' }, { value: 059669">'Procurement', label: 059669">'Procurement' }, { value: 059669">'Warranty',

```

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label: 059669">'Warranty' }, { value: 059669">'Auth', label: 059669">'Auth' }}} placeholder=059669">"All Modules" />
    <Select value={filters.action || 059669">''} onChange={(v) => setFilters(059669">'action', v)} options={[{
value: 059669">'CREATE', label: 059669">'Create' }, { value: 059669">'UPDATE', label: 059669">'Update' }, { value:
059669">'DELETE', label: 059669">'Delete' }, { value: 059669">'APPROVE', label: 059669">'Approve' }, { value:
059669">'LOGIN', label: 059669">'Login' }}} placeholder=059669">"All Actions" />
    </Toolbar>
</div>

<div className=059669">"mt-4 card">
    <DataTable columns={columns} rows={data} isLoading={isLoading} />
    <Pagination page={page} totalPages={totalPages} total={total} pageSize={pageSize} onPageChange={setPage} />
</div>
</div>
);
}

```

```
600">import { Users 600">as UsersIcon, ShieldCheck, Mail, Phone, Plus } 600">from 059669">'lucide-react';
600">import { PageHeader } 600">from 059669">'@/components/ui/Toolbar';
600">import { Badge } 600">from 059669">'@/components/ui/Badge';
600">import { users } 600">from 059669">'@/data/seed';
600">import { ROLE_MAP } 600">from 059669">'@/lib/rbac';

600">export 600">function UsersPage() {
  600">return (
    <div>
      <PageHeader
        title=059669">"User Management"
        subtitle=059669">"Manage system users, roles, and permissions"
        actions={<button className=059669">"btn-primary"><Plus className=059669">"h-4 w-4" /> Add User</button>}
      />

      <div className=059669">"grid grid-cols-1 gap-4 sm:grid-cols-2 lg:grid-cols-3">
        {users.map((u) => (
          <div key={u.id} className=059669">"card card-hover p-5">
            <div className=059669">"flex items-start gap-3">
              <div className=059669">"flex h-11 w-11 shrink-0 items-center justify-center rounded-full text-sm font-
bold text-white" style={{ backgroundColor: u.avatar_color }}>
                {u.name.split(059669">' ' ).map((n) => n[0]).join(059669">'').slice(0, 2)}
              </div>
              <div className=059669">"min-w-0 flex-1">
                <p className=059669">"text-sm font-semibold text-slate-900 dark:text-white">{u.name}</p>
                <Badge tone=059669">"primary" className=059669">"mt-1">{ROLE_MAP[u.role]?.name || u.role}</Badge>
              </div>
            </div>
            <div className=059669">"mt-4 space-y-1.5">
              <div className=059669">"flex items-center gap-2 text-xs text-slate-500 dark:text-slate-400"><Mail
className=059669">"h-3.5 w-3.5" /> {u.email}</div>
              {u.phone && <div className=059669">"flex items-center gap-2 text-xs text-slate-500 dark:text-slate-
400"><Phone className=059669">"h-3.5 w-3.5" /> {u.phone}</div>}
              <div className=059669">"flex items-center gap-2 text-xs text-slate-500 dark:text-slate-400">
<ShieldCheck className=059669">"h-3.5 w-3.5" /> Last active: {u.last_active}</div>
            </div>
          </div>
        )))
      </div>
    </div>
  );
}
```

```

600">import { Settings 600">as SettingsIcon, Moon, Sun, Bell, Shield, Database } 600">from 059669">'lucide-react';
600">import { PageHeader } 600">from 059669">'@/components/ui/Toolbar';
600">import { useAppSelector, useAppDispatch } 600">from 059669">'@/store';
600">import { toggleTheme } 600">from 059669">'@/store/themeSlice';

600">export 600">function SettingsPage() {
  600">const theme = useAppSelector((s) => s.theme.theme);
  600">const user = useAppSelector((s) => s.auth.user);
  600">const dispatch = useAppDispatch();

  600">return (
    <div>
      <PageHeader title=059669">"Settings" subtitle=059669">"Configure your account and system preferences" />

      <div className=059669">"grid grid-cols-1 gap-6 lg:grid-cols-2">
        {059669">"color:#6b7280;font-style:italic">/* Profile settings */}
        <div className=059669">"card p-6">
          <div className=059669">"mb-4 flex items-center gap-2">
            <SettingsIcon className=059669">"h-5 w-5 text-slate-400" />
            <h3 className=059669">"text-sm font-semibold text-slate-900 dark:text-white">Profile Settings</h3>
          </div>
          <div className=059669">"space-y-4">
            <div>
              <label className=059669">"label">Full Name</label>
              <input 600">type=059669">"text" className=059669">"input" defaultValue={user?.name || 059669">"'} />
            </div>
            <div>
              <label className=059669">"label">Email Address</label>
              <input 600">type=059669">"email" className=059669">"input" defaultValue={user?.email || 059669">"'} />
            </div>
            <div>
              <label className=059669">"label">Phone Number</label>
              <input 600">type=059669">"tel" className=059669">"input" placeholder=059669">" +1 555 000 0000" />
            </div>
            <button className=059669">"btn-primary">Save Changes</button>
          </div>
        </div>

        {059669">"color:#6b7280;font-style:italic">/* Appearance */}
        <div className=059669">"card p-6">
          <div className=059669">"mb-4 flex items-center gap-2">
            {theme === 059669">'light' ? <Sun className=059669">"h-5 w-5 text-slate-400" /> : <Moon
className=059669">"h-5 w-5 text-slate-400" />}
            <h3 className=059669">"text-sm font-semibold text-slate-900 dark:text-white">Appearance</h3>
          </div>
          <div className=059669">"space-y-3">
            <div className=059669">"flex items-center justify-between rounded-lg border border-slate-100 p-3
dark:border-slate-800">
              <div>
                <p className=059669">"text-sm font-medium text-slate-700 dark:text-slate-300">Dark Mode</p>
                <p className=059669">"text-xs text-slate-400">Toggle dark theme across the app</p>
              </div>
              <button
                onClick={() => dispatch(toggleTheme())}
                className={059669">`relative h-6 w-11 rounded-full transition-colors ${theme === 'dark' ? 'bg-
primary-600' : 'bg-slate-300'}`}
              >
                <span className={059669">`absolute top-0.5 h-5 w-5 rounded-full bg-white shadow transition-transform
${theme === 'dark' ? 'translate-x-5' : 'translate-x-0.5'}`} />
              </button>
            </div>
          </div>
        </div>

        {059669">"color:#6b7280;font-style:italic">/* Notification preferences */}
        <div className=059669">"card p-6">
          <div className=059669">"mb-4 flex items-center gap-2">
            <Bell className=059669">"h-5 w-5 text-slate-400" />
            <h3 className=059669">"text-sm font-semibold text-slate-900 dark:text-white">Notification

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Preferences</h3>
</div>
<div className=059669>"space-y-3">
  {[059669]>'Maintenance reminders', 059669>'Low-stock alerts', 059669>'Breakdown notifications',
059669>'Warranty expiry alerts', 059669>'Purchase approval requests'].map((item) => (
    <div key={item} className=059669>"flex items-center justify-between rounded-lg border border-slate-100
p-3 dark:border-slate-800">
      <p className=059669>"text-sm text-slate-700 dark:text-slate-300">{item}</p>
      <input 600>"type=059669>"checkbox" defaultChecked className=059669>"rounded border-slate-300 text-
primary-600 focus:ring-primary-500 dark:border-slate-600 dark:bg-slate-800" />
    </div>
  )]}
</div>
</div>

{059669>"color:#6b7280;font-style:italic">/* System info */}
<div className=059669>"card p-6">
  <div className=059669>"mb-4 flex items-center gap-2">
    <Database className=059669>"h-5 w-5 text-slate-400" />
    <h3 className=059669>"text-sm font-semibold text-slate-900 dark:text-white">System Information</h3>
  </div>
  <div className=059669>"space-y-2">
    {[
      { label: 059669>'System Version', value: 059669>'v2.4.1' },
      { label: 059669>'Database', value: 059669>'MongoDB 7.0' },
      { label: 059669>'API Version', value: 059669>'v1.0' },
      { label: 059669>'Last Backup', value: 059669>'2026-08-28 03:00 UTC' },
      { label: 059669>'Uptime', value: 059669>'99.97%' },
    ].map((item) => (
      <div key={item.label} className=059669>"flex items-center justify-between border-b border-slate-100
py-2 dark:border-slate-800">
        <span className=059669>"text-sm text-slate-500 dark:text-slate-400">{item.label}</span>
        <span className=059669>"text-sm font-medium text-slate-700 dark:text-slate-300">{item.value}</span>
      </div>
    )]}
  </div>
</div>
</div>
</div>
);
}

```



```

@600">import url(059669">'https://fonts.googleapis.com/css2?
family=Inter:wght@400;500;600;700&family=JetBrains+Mono:wght@400;500&display=swap');

@tailwind base;
@tailwind components;
@tailwind utilities;

@layer base {
  :root {
    color-scheme: light;
  }
  .dark {
    color-scheme: dark;
  }
  html {
    -webkit-font-smoothing: antialiased;
    text-rendering: optimizeLegibility;
  }
  body {
    @apply bg-slate-50 text-slate-900 dark:bg-slate-950 dark:text-slate-100;
    font-feature-settings: 059669">'cv02', 059669">'cv03', 059669">'cv04', 059669">'cv11';
  }
  * {
    @apply border-slate-200 dark:border-slate-800;
  }
}

@layer components {
  .card {
    @apply rounded-xl border border-slate-200 bg-white shadow-card dark:border-slate-800 dark:bg-slate-900;
  }
  .card-hover {
    @apply transition-shadow duration-200 hover:shadow-card-hover;
  }
  .btn {
    @apply inline-flex items-center justify-center gap-2 rounded-lg px-3.5 py-2 text-sm font-medium transition-colors
    focus:outline-none focus-visible:ring-2 focus-visible:ring-primary-500 focus-visible:ring-offset-2 dark:focus-
    visible:ring-offset-slate-900 disabled:cursor-not-allowed disabled:opacity-50;
  }
  .btn-primary {
    @apply btn bg-primary-600 text-white hover:bg-primary-700 active:bg-primary-800;
  }
  .btn-secondary {
    @apply btn border border-slate-200 bg-white text-slate-700 hover:bg-slate-50 dark:border-slate-700 dark:bg-slate-
    800 dark:text-slate-200 dark:hover:bg-slate-700;
  }
  .btn-ghost {
    @apply btn text-slate-600 hover:bg-slate-100 dark:text-slate-300 dark:hover:bg-slate-800;
  }
  .btn-danger {
    @apply btn bg-danger-600 text-white hover:bg-danger-700;
  }
  .input {
    @apply w-full rounded-lg border border-slate-200 bg-white px-3 py-2 text-sm text-slate-900 placeholder:text-
    slate-400 focus:border-primary-500 focus:outline-none focus:ring-1 focus:ring-primary-500 dark:border-slate-700
    dark:bg-slate-900 dark:text-slate-100;
  }
  .label {
    @apply mb-1.5 block text-xs font-medium text-slate-600 dark:text-slate-400;
  }
  .table-th {
    @apply px-4 py-3 text-left text-xs font-semibold uppercase tracking-wider text-slate-500 dark:text-slate-400;
  }
  .table-td {
    @apply px-4 py-3.5 text-sm text-slate-700 dark:text-slate-300;
  }
  .badge {
    @apply inline-flex items-center gap-1 rounded-full px-2.5 py-0.5 text-xs font-medium;
  }
}

```

```

.sidebar-link {
  @apply flex items-center gap-3 rounded-lg px-3 py-2 text-sm font-medium transition-colors;
}
.sidebar-link-active {
  @apply bg-primary-50 text-primary-700 dark:bg-primary-600/15 dark:text-primary-300;
}
.sidebar-link-inactive {
  @apply text-slate-600 hover:bg-slate-100 hover:text-slate-900 dark:text-slate-400 dark:hover:bg-slate-800
dark:hover:text-slate-100;
}
}

059669">"color:#6b7280;font-style:italic">/* Custom scrollbar */
::-webkit-scrollbar {
  width: 10px;
  height: 10px;
}
::-webkit-scrollbar-track {
  background: transparent;
}
::-webkit-scrollbar-thumb {
  @apply rounded-full bg-slate-300 dark:bg-slate-700;
}
::-webkit-scrollbar-thumb:hover {
  @apply bg-slate-400 dark:bg-slate-600;
}

059669">"color:#6b7280;font-style:italic">/* Recharts tooltip dark mode fix */
.dark .recharts-600">default-tooltip {
  @apply !bg-slate-800 !border-slate-700 !text-slate-100;
}
.dark .recharts-tooltip-label {
  @apply !text-slate-300;
}
}

```

src/vite-env.d.ts

2 lines

```
059669">"color:#6b7280;font-style:italic">/// <reference types=059669">"vite/client" />
```